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EI in the Age of AI : A measurement framework for emotional intelligence in Indian competitive exams

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The authors declare that they are the sole authors of this thesis and that they have not used any sources other than those listed in the bibliography and identified as references. They further declare that they have not submitted this thesis at any other institution to obtain a degree.

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Abstract

Background. How emotional intelligence (EI) shapes performance during a live, high-stakes examination is a question with direct consequences for students, educators, and assessment designers. Existing tools offer only a partial view: questionnaires are static and belief-based, while laboratory biosensors are too intrusive for real exam halls. This gap is especially consequential in India's competitive examination ecosystem, where acceptance rates fall below one percent and the emotional architecture that determines who expresses their preparation remains entirely unmeasured.
Objectives. To develop and evaluate a non-intrusive, process-based framework for measuring EI as it unfolds during an authentic entrance examination, and to identify which examination-time emotional processes distinguish high-performing from lower-performing candidates when cognitive preparation is held roughly constant.

Methods. Forty-two applicants at a University in India, completed a 30-minute entrance examination while three low-intrusion data streams were recorded: facial expressions captured by a standard webcam, observer annotations of problem-solving behaviour, and brief self-ratings of arousal collected at regular intervals. These streams were used to derive behavioural indicators of five emotional dimensions. Multiple statistical validation methods were applied to identify which indicators carried genuine signal, and a family of regression models was compared to find the most efficient predictor of examination performance.

Results. The vast majority of engineered indicators survived independent validation.The strongest predictors were combinations of emotional capacities acting together. The final model explained over 95 % of the variance in examination scores. High performers were consistently characterised by three qualities: fast emotional recovery after difficulty, well-calibrated arousal that sustained productive work, and accurate awareness of their own emotional state during the exam.

Conclusions. This thesis contributes a replicable, theory-grounded pipeline that turns ordinary examination-room data into a meaningful emotional intelligence profile without specialist hardware or disruption to the exam. The findings suggest that emotional intelligence operates as a coordinated system rather than a set of independent traits, and that the emotional architecture separating high from low performers is measurable and, potentially, trainable. The results should be treated as exploratory; replication with a larger sample is the essential next step.

Keywords: Emotional Intelligence; Indian Competitive Examinations; Facial Action Units; Behavioural Feature Engineering; AICc Model Selection.

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List of Abbreviations

AICc	Corrected Akaike Information Criterion
AU	Action Unit (as defined by the Facial Action Coding System)
BIC	Bayesian Information Criterion
EEG	Electroencephalography
EI	Emotional Intelligence
ElasticNet	Elastic Net regularised regression (combined L1/L2 penalty)
FACS	Facial Action Coding System
GBM	Gradient Boosting Machine
IIT	Indian Institute of Technology
KNN	k-Nearest Neighbour (regression)
LASSO	Least Absolute Shrinkage and Selection Operator
LOOCV	Leave-One-Out Cross-Validation
MI	Mutual Information
MSCEIT	Mayer-Salovey-Caruso Emotional Intelligence Test
OLS	Ordinary Least Squares
PLS-VIP	Partial Least Squares Variable Importance in Projection
RF	Random Forest
RMSE	Root Mean Squared Error
RQ	Research Question
RSS	Residual Sum of Squares
SAM	Self-Assessment Manikin
SVR	Support Vector Regression
SVR-Linear	Support Vector Regression with a linear kernel
SVR-RBF	Support Vector Regression with a radial basis function kernel
TEIQue	Trait Emotional Intelligence Questionnaire

Chapter 1

Introduction

This thesis is about measuring emotional intelligence as a live, process-based phenomenon in an examination setting, in a way that is both statistically disciplined and practically usable.

The most successful people in any field are rarely those who simply knew the most. Across research in medicine, engineering, leadership, and education, a recurring pattern appears: people who can perceive what they feel, regulate that feeling under pressure, and redirect attention back to the task outperform those who cannot [75, 119]. This is called emotional Intelligence

The question of how to measure emotional intelligence is no longer academic curiosity. In the age of AI, Indian students are preparing for a labour market in which routine technical work is increasingly automated and a significant share of graduate roles are projected to be displaced by algorithmic and machine-based systems in the coming decade [38]. Many graduates already struggle to find stable, meaningful employment despite clearing multiple academic filters: engineering and professional degrees without jobs, internships that do not convert, and careers that stall well below the level their qualifications suggest [83, 90]. Each year, large cohorts move through an examination pipeline that promises opportunity but often delivers uncertainty [84].

At the same time, research across education, organisational psychology, and high-pressure professions points toward two broad kinds of human capability that matter for long-term outcomes. One is cognitive ability: the capacity to learn quickly, reason, and solve complex problems [97]. This is what Indian competitive examinations are designed to measure, and in many cases they do capture genuine differences in preparedness and problem-solving skill [84, 90]. The other is emotional intelligence: the ability to perceive what one feels, understand how those feelings change, regulate them under pressure, and use them to support thinking and relationships [75, 76, 94]. Emotional intelligence in this ability sense is not a fixed personality label. It is a system of learnable, trainable abilities that develops through experience and can be strengthened or weakened by the environments students move through [13, 63, 75]. Every year, tens of millions of young people in India walk into examination halls knowing that, for most of them, the result will be *no*. IIT entrance, NEET, UPSC Civil Services, SSC, banking, and other national-level tests together form an ecosystem where success rates for the most prestigious paths often fall below one or two percent [84].

Meta-analytic work with more than 40,000 students shows that EI predicts academic performance above and beyond IQ and personality [71]. Longitudinal studies

in professional settings find that emotional and social competencies forecast both career progression and long-term satisfaction, sometimes more strongly than cognitive scores [4, 97, 100].

What education is supposed to do, and what it has become. Education, at its best, is the process by which curiosity finds a structure. Struggling with a genuinely difficult problem builds both knowledge and the capacity to tolerate uncertainty, stay regulated while not knowing, and convert effort into understanding [85]. Repeated across years, this process builds the emotional architecture that supports long-term success.

The measurement gap. Despite this, the EI tools used in education and selection are not designed for exam conditions. As a result, the emotional architecture that helps determine who uses their knowledge under one-percent odds is rarely seen or measured. The Indian competitive exam ecosystem makes this gap especially visible. Schools, coaching institutes, or exam authorities dont give students feedback on how they emotionally navigate an exam [84, 100]. Students know their marks and percentile, but not how quickly they recover after a setback, how their arousal drifts during the test, or how well their internal experience matches their visible behaviour.

This thesis as a case-based solution. This thesis responds by treating an entrance examination, designed to mirror key features of Indian competitive exams, as a live case study of emotional intelligence in action. In concrete terms, the main contribution is a three-stage pipeline that turns ordinary exam-room data into a compact, interpretable EI profile for each student: (1) define EI as observable processes grounded in performance-under-pressure literature [5, 45, 61, 66]; (2) engineer and validate behavioural features that capture those processes during a real exam; and (3) build a small, robust model that explains how these emotional processes relate to who actually performs when the stakes are highest [63, 71, 75].

1.1 Background

Measuring emotional intelligence during problem-solving has two intersecting strands of background: theory and evidence on what emotional intelligence is and why it matters under pressure, the specific features of Indian competitive examinations that make that measurement problem both urgent and tractable and the measurement approach itself a multi-modal feature engineering and regression pipeline.So this study lies at the intersection of AI , affective computing and education (domain)

1.1.0.1 Emotional Intelligence as an Ability Construct

Emotional intelligence (EI) is defined in the ability model of [94], later formalised and extended by [76] and [75], as four hierarchically organised cognitive skills: accurately perceiving emotional signals in faces, voices, and bodily states; using emotions to facilitate and direct thought; understanding how emotions develop, blend, and change over time; and managing emotions flexibly in oneself and in relation to others. Emotional and social competencies forecast career progression and long-term job satisfaction, in several longitudinal studies more strongly than cognitive scores alone [4, 101].

In India specifically, studies with secondary school students, undergraduate cohorts, and UPSC aspirants consistently find positive associations between EI and academic outcomes and coping under prolonged examination stress [54, 78, 90]. EI is, in short, consequential, domain-general, and critically for this project trainable [63].

Emotion Regulation, Differentiation, and Performance Under Pressure

A complementary literature addresses the mechanics of EI when task demands are high. In high-pressure performance domains athletics, surgery, music under competition conditions these same capacities, EI, regulation flexibility, and fast recovery, predict both stress trajectories and outcome performance, and are responsive to training [5, 63, 67]. Together, these findings establish that the emotionally relevant unit of analysis is not a single snapshot of affect but a *trajectory* of states across a demanding task. This motivates treating EI as a process to be reconstructed from time-series data rather than inferred from a questionnaire administered before or after the event [37, 85].

Reasoning and responses across cognitive problem-solving stages

Emotional intelligence (EI) is often defined as the ability to *reason* about affective information. Mayer, Salovey, and Caruso describe EI as "the capacity to reason about emotions, and of emotions to enhance thinking" [75], and ability tests such as the MSCEIT ask questions like "what would a person feel if ..." to assess whether people choose the consensually "correct" emotion for hypothetical scenarios [14, 15, 75]. Similarly, metacognition is typically measured with decontextualised questionnaires or vignette-based tasks rather than through traces of thinking during authentic academic work [98, 117].

These approaches primarily capture *knowledge about* emotions and cognition and yield *static* ability scores; they do not show how students actually move between preparation, ideation, production, evaluation, refinement, and blocking while experiencing anxiety and time pressure in real cognitive problem-solving situations such as exams [93]. In other words, existing work tells us *whether* students can reason about emotions and thinking in principle, but not *how* they reason about their own emotions and thoughts as they respond to problems under timed conditions.

This thesis addresses that gap by not only measuring students' global emotional and metacognitive abilities, but also analysing the *fine-grained reasoning patterns* students report while taking a timed exam, together with their observable responses (e.g., pausing, reading, typing, deleting) across different cognitive problem-solving stages. By combining coded stages (Preparation, Ideation, Production, Evaluation, Refinement, Block) with unsupervised analyses of verbal reports, we move beyond global scores towards a stage-wise picture of students' in-situ reasoning and responses during exam problem solving.

Indian Competitive Examinations: The Problem Case

The Indian national examination ecosystem provides both the motivating case and the applied context for this research. Selection ratios in major entrance exams are

routinely below one or two percent, meaning that the exam hall is populated almost entirely by candidates who have already passed multiple filters and invested years in systematic preparation [84]. Under such conditions, exam scores reflect not only who studied more but whose emotional architecture allowed that preparation to be expressed under extreme pressure. The costs of this system are documented [84] report elevated rates of anxiety, clinical-level depression, and rising suicide figures among adolescent aspirants in Kota, Rajasthan, India's largest residential coaching hub. This context establishes that the emotional processes occurring inside the exam hall carry real human stakes, not merely theoretical interest. Critically, it also makes the measurement problem precise: what is needed is a method for observing how a student's EI operates during a high-pressure cognitive task, using signals available in any ordinary examination setting and without disrupting the exam itself.

Machine Learning Approaches to EI and Affective Measurement: Prior Work and Limitations

The field of affective computing and machine learning-based EI measurement offers several relevant antecedents, each carrying structural limitations that prevent direct application to live examinations. Physiological signal approaches EEG, galvanic skin response, and multi-channel biosensor arrays achieve high classification accuracy in controlled conditions but require physical contact, lengthy calibration, and hardware incompatible with handwriting in a real exam hall [57,58]. Early affective computing work by [88] established the viability of physiological measurement for recognising emotional states, but the laboratory apparatus it required remains impractical in authentic high-stakes settings. Webcam-only and multimodal classroom systems offer better deployability and have detected learning-centred affective states in naturalistic environments [2, 12]; however, these systems typically aggregate emotion labels across entire sessions without conditioning on the cognitive stage the participant is currently in a critical omission identified by [29] and [102]. On the modelling side, a persistent methodological critique of EI research is that small-sample regression studies report raw in-sample R^2 without penalising for complexity, producing results that are systematically optimistic and non-replicable [119]. [27] surveyed machine learning approaches to EI measurement and found that when the outcome variable is a noisy self-report questionnaire score rather than objective performance, no algorithm reliably recovers a strong signal highlighting the importance of both the performance criterion chosen and the model selection strategy applied. These limitations collectively define the design space for the present framework: ecological validity requires non-intrusive signals; interpretability requires cognitive stage context; and trustworthy modelling requires complexity penalisation calibrated to sample size.

1.2 Aim and research questions

The overall aim is to develop and evaluate a non-intrusive, process-based framework for measuring emotional intelligence during an authentic entrance examination, motivated by Indian competitive exam conditions.

RQ1: How well do existing EI measurement approaches capture the emotional processes that matter during a live high-stakes examination?

RQ2: Which examination-time features, derived from facial expressions, problem-solving stages, and self-reports, show the strongest relationship with exam score?

RQ3: What is the most efficient predictive model that links validated EI-related signals to examination outcome in this dataset under small-sample constraints?

RQ4: How do students' actions, verbal reflections, and stage-transition sequences characterise the emotional reasoning processes they engage in across problem-solving stages in a timed entrance exam?

1.3 Ethics, Social Impact, and Sustainability

The findings of this thesis have wider consequences beyond model performance. Because the framework derives examination-time features from facial expressions, problem-solving stages, and self-reports, its value does not depend only on predictive accuracy but also on how such a system would affect students, institutions, and the broader examination culture if adopted in practice [18, 29].

Ethics. The central ethical issue is that this framework observes students during a high-stakes task using webcam-based facial analysis and self-report signals. Research on emotion AI in education consistently warns that affect monitoring can easily shift from supportive measurement to intrusive surveillance if consent, transparency, and student control are weak [34, 110, 119]. For that reason, the present thesis should not be read as support for covert monitoring or automated judgement of students during exams. Its ethical justification depends on strict conditions: informed consent, clear explanation of what is being measured, minimisation of stored personal data, and explicit prohibition of punitive or selection-only uses.

A second ethical concern is misclassification. Facial expression analysis is not a direct window into inner emotional truth, and self-report is also imperfect [40, 74]. A student may appear calm while feeling overloaded, or may display little visible affect while still regulating effectively. This matters because an emotionally informed assessment system could unjustly stigmatise students if its outputs are treated as definitive labels rather than probabilistic indicators. The framework is therefore ethically defensible only if it is used as a reflective support tool, not as a replacement for human judgement and never as a stand-alone admissions criterion. This is also why the thesis defines the EI profile as a research instrument rather than a diagnostic verdict (Section 1.4).

A third ethical issue is fairness. Cultural differences in facial expressiveness, gendered norms of emotional display, and individual differences in self-report style may bias feature values even when the underlying emotional competence is similar [34, 119]. The strongest ethical reading of the results is therefore not that the system has discovered a hidden truth about students, but that it has identified potentially meaningful emotional processes that require careful validation across populations

before any operational use. In ethical terms, the burden of proof lies with the system designer, not with the student.

Social impact. The most important social implication of the findings is that they challenge a narrow interpretation of exam scores as pure reflections of knowledge. The results suggest that a meaningful part of performance in high-pressure examination settings depends on emotional architecture: how efficiently students recover from setbacks, whether their arousal remains usable, and whether they can align emotion with the demands of each stage of the task 5thesis. This has social value because it makes visible a factor that current examination systems largely ignore even though it may shape who succeeds and who underperforms.

In the context of Indian competitive exams, this matters because extreme selectivity often turns very small differences in in-exam functioning into life-changing differences in outcome [90]. If two students are similarly prepared but one loses less time to emotional derailment, the result is not merely an individual psychological story but part of how opportunity is distributed. Used carefully, the framework could therefore support a more humane educational response: identifying students whose problem is not knowledge deficit but recovery cost, over-monitoring, or maladaptive regulation under pressure. That kind of distinction could help teachers, counsellors, and coaching systems intervene earlier and more precisely.

At the same time, the social impact could be harmful if the findings are misused. A system designed to reveal hidden disadvantage could instead become another sorting mechanism, branding some students as emotionally fragile or "high risk." That would reproduce the very inequities the framework is trying to illuminate. The social value of this work therefore depends on whether institutions treat emotional measurement as a basis for support, redesign, and fairness or as another tool for exclusion.

Sustainability. The sustainability relevance of this work is primarily social rather than environmental. Sustainable education systems are not only systems that produce scores efficiently; they are systems that preserve student wellbeing, reduce preventable harm, and support long-term human development [91, 109]. In that sense, the thesis contributes to sustainability by arguing that high-stakes assessment should not ignore the emotional conditions under which students are required to perform. If emotional architecture is partly trainable and partly supported by better educational design, then making it visible may help build assessment cultures that are more resilient and less damaging over time.

There is also a practical sustainability advantage in the framework itself. Unlike laboratory-grade affect measurement systems that require EEG caps, sensors, or specialised hardware, the present approach relies on a webcam, stage annotations, and brief self-reports, making it far more feasible to study and potentially deploy in real educational contexts 2. That lower-infrastructure design improves scalability, but it also increases responsibility: scalable systems can spread benefits widely, but they can also spread assumptions widely if not governed carefully. Sustainability here therefore means not only deployability, but responsible deployability.

At the broadest level, the thesis supports the idea that educational sustainability

includes emotional sustainability. An exam system that repeatedly rewards only those students who already possess efficient emotional regulation, while offering no visibility or support to those who do not, risks normalising avoidable loss, distress, and inequity. A more sustainable system would use findings of this kind not to lower standards, but to incorporate these processes in their training schedules .

1.4 Scope

The study uses data from a single 30-minute entrance examination with 42 undergraduate applicants at a University in India. Analysis is performed at the participant level using session-level features, not frame-by-frame samples, to avoid pseudo-replication and to keep modelling consistent with small-sample guidance [18]. EI is operationalised as a continuous process index related to examination performance, not as a diagnostic category or separate admission score. The thesis does not:

- design or evaluate clinical mental health interventions;
- claim direct generalisation to all Indian competitive exams or all student groups;
- compare specific coaching institutes, curricula, or policies.

The goal is to show, in one carefully observed entrance exam that mirrors many features of larger competitive exams, that meaningful EI-related signals can be extracted from non-intrusive data and that these signals help explain why some students express more of their potential than others.

1.5 Thesis outline and main contribution

Chapter 2 surveys what is already known about emotional intelligence, performance under pressure, and the Indian competitive examination context, and explains why existing measurement approaches are not designed for the conditions inside a live exam hall. Chapter 3 describes how the study was conducted: what data were collected, how emotional intelligence was defined as something observable during an exam, and how the most informative signals were identified from a broader candidate set. Chapter 4 presents what the data showed: which signals reliably relate to exam performance and what the most accurate yet honest predictive model looks like. Chapter 5 interprets those findings in the wider context of Indian competitive exams, discusses what the results do and do not prove, and reflects on the ethical implications of measuring students this way and acknowledges the limitations honestly. Chapter 6 summarises the contributions and outlines what would need to happen for this framework to move from a research prototype toward something that could genuinely support students.

Chapter 2

Related Work

This chapter reviews the landscape of emotional intelligence measurement, affect computing during learning and problem-solving, and statistical modelling practices that together define the state of the field for this thesis. The focus is not only on what each study found but also on why, despite these contributions, none of them yet provides a practical, process-based EI measurement for live, high-stakes examinations.

Section 2.1 presents a literature review of studies relevant to RQ1a, grouped by four recurring structural limitations: intrusive apparatus (L1), cross-sectional or trait-level measurement (L2), absence of cognitive context (L3), and inadequate model selection at small sample sizes (L4), and absence of within-stage reasoning process analysis (L5). For each study, the limitations column explains explicitly why the approach falls short for the present problem, and in which concrete ways the framework in this thesis is better aligned with entrance-exam conditions. Section 2.2 then synthesises these strands into the specific research gap this study addresses.

2.1 Related Works

Table 2.1 presents the core literature review. Each entry combines the study and its main findings in one column, and uses the second column to argue why that work cannot yet serve as a process-based EI measure in live examinations and how the present thesis improves on it.

Table 2.1: Related work with key findings and limitations

Study & Key Findings	**Limitation(s) for Live Examination EI**
L1 Laboratory apparatus disrupts authentic settings	
Koelstra et al. (DEAP dataset, 2012) [68]. EEG + peripheral physiology achieves 90–99% accuracy on valence/arousal classification in subject-dependent settings using a 32-channel EEG cap during passive video viewing.	L1: 32-electrode cap requires 20–45 min gel setup; head-mounted hardware is incompatible with writing. No exam-setting deployment exists. Accuracy collapses to 59–68% in subject-independent (generalizable) conditions. *This thesis uses only a standard webcam and brief self-report ratings, requiring no physical contact or calibration overhead.*

Continued on next page

Table 2.1 continued from previous page

Study & Key Findings	Limitation(s) for Live Examination EI
Li et al. (2018) [68]. Subject-independent EEG emotion recognition achieves 59–68% accuracy across multiple benchmark datasets, a dramatic drop from subject-dependent benchmarks.	L1: Confirms the generalizability gap. EEG systems that work on one person fail on another, making deployment in diverse student populations unreliable. *This thesis relies on OpenFace 2.0 webcam-based AU extraction, which requires no per-participant calibration and achieved usable frames for all 42 participants.*
Wang et al. (2022) [112]. Systematic review of multimodal affect-sensing systems; gains from multimodal fusion shrink from 12.1% on acted stimuli to only 4.39% on naturalistic data.	L1 + L3: Accuracy inflation from lab conditions. Naturalistic gains are marginal. No systems were tested in actual examinations. *This thesis was conducted in an authentic entrance examination, providing naturalistic ecological validity by design.*
Zhang et al. (2024) [120]. Mini review states plainly that the gap between controlled laboratory settings and genuine emotional experiences constitutes a fundamental paradox in emotion research.	L1: Provides field-level acknowledgement that lab-grade apparatus is fundamentally unsuited to authentic emotion measurement. Eye-tracking calibration is lost with natural head movement during note-taking. *This thesis directly addresses this paradox by deploying non-intrusive sensing in a real exam hall, demonstrating that the gap is practically bridgeable.*
D'Mello, Dieterle and Duckworth (2017) [25]. Meta-analysis of affect-aware learning environments establishes that affect sensors must be unobtrusive to avoid altering the target behaviour , a design requirement no EEG or eye-tracking system satisfies in examinations.	L1: Establishes unobtrusive sensing as a non-negotiable requirement. Hawthorne-effect contamination is documented as reducing the ecological validity of intrusive systems. *This thesis operationalises unobtrusiveness as a hard constraint: only a ceiling-mounted webcam and a brief pictorial SAM scale were added to the standard exam environment.*
Kapoor, Burleson and Picard (2007) [59]. Achieved 79% frustration prediction from webcam, pressure-sensitive chair, and skin conductance. First demonstration that webcam-grade sensing suffices for learner affect.	L1 (partially resolved): The pressure chair and skin-conductance sensor remain intrusive. The webcam component alone achieves lower accuracy; the study demonstrates a direction rather than a deployable solution. *This thesis completes that direction: it uses the webcam stream alone, supplemented by observer stage codes rather than additional hardware sensors.*

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Table 2.1 continued from previous page

Study & Key Findings	Limitation(s) for Live Examination EI
Stöckli et al. (2018) [104]. FACET achieved 96% accuracy on posed standardized stimuli; accuracy drops substantially on spontaneous natural expressions in classroom settings.	L1 (partially resolved): Demonstrates webcam-based AU detection can work in classrooms. However, validated only on static stimuli; spontaneous accuracy in genuine high-stakes exams remains unquantified. *This thesis applies AU-based measurement specifically in a high-stakes entrance exam, providing the first quantified evidence of its signal quality in that context.*
L2 Cross-sectional scoring fails to capture temporal process	
Mayer, Salovey and Caruso - MSCEIT (2008) [76]. MSCEIT is the field's gold-standard ability test with strong criterion validity for social outcomes. Scores are computed from a single test administration.	L2: Single-point measurement treats EI as a stable trait. Cannot capture within-session dynamics, recovery from difficulty, or moment-to-moment regulatory behaviour during problem-solving. *This thesis replaces the single composite with a time-series feature pipeline that explicitly captures recovery latency, emotional inertia, and stage-conditioned trajectories across the 30-minute session.*
Petrides and Furnham — TEIQue (2001) [87]. TEIQue trait EI correlates with academic achievement and personality. Internally consistent ($\alpha > 0.87$).	L2: Self-report questionnaire administered offline. Trait-level assessment conflates semantic self-beliefs with in-task emotional behaviour per Robinson and Clore (2002). *This thesis collects SAM self-reports* during *the exam at ten checkpoints, synchronised with concurrent AU data, grounding self-report in moment-specific rather than retrospective experience.*
Brackett and Mayer (2003) [14]. MSCEIT and TEIQue yield substantially different measurements of the same individual; ability and trait EI correlate only weakly ($r \approx .20$).	L2: Demonstrates fundamental construct divergence between the two dominant instruments. Neither can serve as a process-based measure. *This thesis sidesteps the trait-versus-ability debate by grounding EI operationally in observable exam-time behaviour rather than in either questionnaire tradition.*

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Table 2.1 continued from previous page

Study & Key Findings	Limitation(s) for Live Examination EI
Folkman and Lazarus (1985) [37]. Three-stage exam study showed threat emotions peaked in anticipation, distancing dominated waiting, and benefit emotions emerged only at outcome , a phase structure invisible to any snapshot measure.	L2: Canonical demonstration that examination emotion is temporally structured. Cross-sectional measurement cannot recover this structure. *This thesis directly operationalises temporal structure by computing features separately within each Schoenfeld stage and across stage transitions, recovering the phase-specific emotional patterns Folkman and Lazarus identified.*
Bieg, Goetz and Lipnevich (2014) [10]. Trait ratings of enjoyment, pride, anger, and anxiety are systematically *higher* than experience-sampled state ratings during actual exams; academic self-concept moderates the gap.	L2: Establishes that trait scales systematically misrepresent in-task emotion. The well-known gender difference in math anxiety appears only in trait measures and disappears at the state level. *This thesis avoids this bias by measuring affect exclusively at the state level, using in-exam SAM ratings and concurrent AU streams rather than pre- or post-exam questionnaires.*
Molenaar (2004) [79]. Ergodicity critique: between-person trait correlations systematically misrepresent within-person processes. Cross-sectional EI scores are formally incapable of recovering momentary emotion-cognition coupling.	L2: Provides the statistical proof that trait-level group statistics cannot be interpreted as individual process dynamics. *This thesis respects the ergodicity constraint by computing all features at the within-person level first and then aggregating to participant-level scalars only for the regression stage.*
Kuppens et al. (2010) [64]. DynAffect model characterises affect by home-base, variability, and attractor strength; emotional inertia predicts psychopathology beyond mean affect.	L2 (dynamic alternative): Demonstrates that dynamic indices carry information genuinely beyond static means. None of this information is recoverable from an MSCEIT or TEIQue score. *This thesis directly implements dynamic indices : AR(1) inertia, MSSD instability, and recovery slope as D2 regulation features, instantiating the DynAffect logic in an exam-performance context.*
L3 Emotional signals lack cognitive context	

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Table 2.1 continued from previous page

Study & Key Findings	Limitation(s) for Live Examination EI
Schoenfeld (1992) [102]. Protocol analysis of mathematical problem-solving identified six cognitive episodes with stage-specific emotion patterns. Experts exit Block quickly; novices escalate.	L3: Establishes that emotions carry opposite meanings across stages (e.g., brow furrow = concentration during Production vs. frustration during Block). Stage-unaware measurement conflates these. *This thesis adopts Schoenfeld's six-episode taxonomy as the explicit temporal scaffolding for all feature computation, so every AU feature is conditioned on the concurrent stage label.*
D'Mello and Graesser (2012) [29]. Affective oscillation model: cognitive equilibrium disrupted by contradiction produces confusion; if resolved, returns to engagement/flow; if unresolved, decays to frustration then boredom. Confusion at specific cognitive moments benefits learning.	L3: Stage-contingent effect that cannot be detected without synchronised cognitive labelling. EEG and questionnaire studies label moments with categorical emotion states disconnected from concurrent task stage. *This thesis synchronises all three data streams : AU, SAM, and stage annotations , at 1 Hz, enabling the stage-contingent confusion effects D'Mello and Graesser describe to be operationalised as D5 alignment features.*
Muis, Psaradellis, Lajoie, Di Leo and Chevrier (2015) [80]. Epistemic emotions are differentially associated with specific problem-solving behaviours. ERP work shows negative emotions modulate distinct processing stages differently.	L3: Establishes differential stage-emotion associations that only stage-aware measurement can detect. Aggregated emotion labels across an entire session lose this structure entirely. *This thesis preserves this structure by computing 45 explicitly episode-conditioned features across the five EI dimensions, ensuring that stage-specific emotional modulation is not averaged away.*
Pekrun, control-value theory (2006) [85]. Distinguishes prospective, activity-concurrent, and retrospective achievement emotions. Achievement Emotions Questionnaire separates before/during/after-task blocks.	L3: Even stage-aware questionnaire studies sample *around* exams, not *within* them at the sub-problem level. No widely used "state EI" instrument time-aligns to Schoenfeld stages. *This thesis provides that time-alignment: SAM checkpoints are embedded within the running exam, and AU features are computed within each Schoenfeld episode, achieving the sub-problem temporal resolution Pekrun's framework calls for but his instrument cannot deliver.*

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Study & Key Findings	Limitation(s) for Live Examination EI
Bosch et al. (2015–2016) [12]. Face-based detectors of boredom, confusion, delight, engagement, frustration, and off-task behavior in classrooms achieved AUC 0.61–0.87; generalized across days, ethnicity, and gender.	L3 (partially resolved): Demonstrates webcam-based affect detection in naturalistic settings. However, affect labels are session-averaged; no synchronisation to problem-solving stages is performed. *This thesis extends Bosch et al.'s naturalness by adding the missing stage synchronisation layer: every AU observation is tagged with its concurrent Schoenfeld stage before any feature is computed.*
L4 Inadequate model selection at small sample sizes	
Matthews, Zeidner and Roberts (2004) [73]. Catalogued methodological looseness in EI research: over-reliance on small opportunity samples, bivariate correlations, and unpenalized multiple regression that inflates R^2 and coefficient estimates.	L4: Establishes that the field's dominant statistical approach is systematically misleading. OLS-plus-stepwise remains dominant despite being "long overlooked" as problematic. *This thesis responds directly by using a seven-method voting framework for feature validation and AICc-penalised forward selection for model building, treating complexity penalisation as a first-class design requirement.*
Gelman (2018) [41]. Detecting an interaction requires approximately 16× the sample size of the corresponding main effect. EI interaction studies at $n < 100$ are therefore approximately 16× underpowered for interaction detection.	L4: Provides the quantitative basis for why complexity penalisation is non-negotiable in small-sample EI studies with interaction features. *This thesis acknowledges this constraint explicitly and limits interaction modelling to AICc-screened terms, with LOOCV as the primary out-of-sample performance metric throughout.*
Burnham and Anderson (2002) [18]. AICc-based multimodel inference is the principled alternative to stepwise selection. AICc should always be used when $n/k < 40$. Akaike weights support multimodel inference rather than winner-take-all selection.	L4 (solution): Establishes AICc as the correct information-theoretic tool for small-sample EI research. The small-sample correction term provides honest complexity penalisation. *This thesis implements this recommendation fully: AICc is the primary model-selection criterion, and Akaike weights are reported for all 23 candidate configurations compared in Section 4.*

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Study & Key Findings	Limitation(s) for Live Examination EI
Dollmat and Abdullah (2022) [27]. Comparison of machine learning algorithms for EI classification: most studies train against questionnaire scores as ground truth. When the ground truth is noisy trait data, no algorithm recovers a strong signal.	L2 + L4: Demonstrates that the combination of cross-sectional ground truth and unpenalized ML model selection produces weak, non-replicable results regardless of the algorithm chosen. *This thesis addresses both failure modes: it uses objective examination score as the criterion variable rather than a questionnaire proxy, and it applies rigorous AICc-based model selection to every algorithm family tested.*
L5 Within-stage reasoning processes are invisible to existing approaches	
Schoenfeld (1992) [102]. Stage labels (Exploration, Implementation, Verification, Block) describe the *type* of cognitive episode but treat each label as internally homogeneous. No mechanism distinguishes a content-confusion Block from a writing-impasse Block, or reading-as-ideation from generative brainstorming.	L5: Stage labels are categorical tokens, not process descriptions. Two participants coded identically as "Block" or "Ideation" may be doing qualitatively different things, and this heterogeneity is invisible to any approach that uses stage codes as fixed states. *This thesis addresses this by clustering within-stage transcript segments using TF–IDF embeddings, producing interpretable sub-types for each stage label (RQ4).*
D'Mello and Graesser (2012) [29]; Bosch et al. (2015) [12]. Both frameworks detect affect continuously from facial or behavioural signals but neither links those signals to the student's own verbal account of what they were thinking or feeling at that moment.	L5: The student's perspective , what they understood themselves to be doing emotionally and cognitively is absent from signal-only approaches. Without it, a "confusion" AU pattern cannot be distinguished from "frustration" or "concentration" without external inference. *This thesis incorporates retrospective verbal transcripts as a first-class data stream, aligning students' own accounts with concurrent action types and stage codes to ground emotional interpretation in participant meaning-making.*

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Table 2.1 continued from previous page

Study & Key Findings	Limitation(s) for Live Examination EI
Folkman and Lazarus (1985) [37]; Pekrun (2006) [85]. Both establish that emotional reasoning in exams is phase-structured, but measure it through pre/post questionnaires administered outside the exam. They cannot show how reasoning unfolds within a single stage as it is happening.	L5: The sub-stage resolution needed to distinguish, for example, a student who enters Block already articulating a re-reading strategy from one who has no recovery plan is entirely absent from questionnaire and post-hoc self-report designs. *This thesis provides that resolution through Markov transition analysis at the cluster level: transition probabilities between within-stage subtypes reveal which types of Ideation feed directly into committed Production and which types of Block reliably resolve through passage re-reading versus unstructured re-attempts.*
Muis et al. (2015) [80]; Grafsgaard et al. (2013) [44]. Both show that observable actions (gaze, gesture, facial expression) vary systematically across cognitive stages, but neither links action patterns to the student's own verbal description of their emotional reasoning at that moment within a stage.	L5: Action-type distributions and verbal accounts are treated as separate evidence streams in prior work. Their joint pattern which action types co-occur with which verbal reasoning styles within the same stage sub-type has not been characterised. *This thesis connects these streams explicitly: cluster signature terms from transcripts are linked to the distribution of action types (pausing, reading, typing, deleting) within the same cluster, jointly characterising how each within-stage reasoning pattern is expressed in both language and observable behaviour.*

2.2 Study Motivation and Research Gap

It can be established from the related works that emotional intelligence is not peripheral to academic performance; it is predictive of it. The literature shows that higher EI is associated with better coping under examination pressure, stronger academic outcomes, and greater career success beyond graduation [4, 71, 101]. In the Indian context, studies with secondary school students, undergraduate cohorts, and UPSC aspirants report consistent positive associations between EI and performance under competitive exam conditions [54,78,90]. Further, research across high-pressure professions athletics, surgery, leadership demonstrates that the emotional capacities captured by the ability model of EI predict not only how people feel under stress but how they perform [63, 67]. This evidence establishes that measuring EI in Indian competitive examinations is not an academic curiosity but a prerequisite for understanding who succeeds and why, serving as the primary motivation for this study.

At the same time, research on emotion regulation and resilience suggests that the capacities determining whether a student recovers from difficulty or spirals un-

der pressure are trainable, not fixed, and can be strengthened through supportive environments [13, 47, 63]. This raises a practical question: if EI shapes performance and can be developed, why is it never measured in the exam hall itself? This gap in practice motivates investigating whether process-based EI measurement in live examinations is technically feasible.

Related works on EI measurement tools show that questionnaire-based approaches, including the ability test MSCEIT [76] and the trait-based TEIQue [87], are widely used but structurally limited for live examination use. Both are administered before or after the task and produce a single composite score that averages across time. Research on emotional dynamics shows that it is not average affect but the trajectory how emotional states rise, persist, and resolve that predicts performance outcomes [64, 65]. The MSCEIT and TEIQue are also weakly correlated, reflecting a longstanding lack of measurement consensus [14].

Laboratory systems using EEG, skin conductance, and gaze-tracking achieve high accuracy in controlled settings but require calibration, physical contact, and hardware incompatible with real exam conditions [58]. Webcam-based systems demonstrate that affect can be detected in naturalistic settings without specialist equipment [6, 12], yet they typically aggregate signals across entire sessions without accounting for the participant's cognitive stage. D'Mello and Graesser show that the same facial expression can carry different meanings during productive problem-solving and during an impasse [29]. These limitations are structural barriers that prevent existing approaches from serving as reliable EI measures in live competitive examinations.

The research gap is the absence of a framework that is non-intrusive for real exam halls, dynamic enough to capture EI as a process, contextualised to interpret affect against cognitive behaviour, and statistically rigorous for small sample sizes typical of authentic exam studies. This gap is significant because EI shapes performance and is trainable, yet no tool observes it when it is most consequential. Existing approaches also treat affect and cognitive behaviour as separate streams, and how their integration improves interpretability and prediction remains largely unexplored in the Indian examination context. This opens the opportunity to develop a practical, theory-grounded framework using data already generated in the exam room facial expressions, observed problem-solving behaviour, and brief arousal self-reports to make the hidden emotional dimension of performance observable.

Lastly, there is a gap at the level of *reasoning processes* within the exam itself. Existing approaches treat emotions either as global abilities summarised by a single score (MSCEIT or TEIQue) [14,75,76,87], or as continuous signals detached from task behaviour [6, 12, 58]. They do not show how students move through problem-solving stages such as preparation, ideation, production, evaluation, refinement, and blocking, nor how they describe their own emotional reasoning at those moments [29, 93]. The present study therefore adds a process-focused strand by using stage codes, logged actions, retrospective transcripts, and Markov transition structures to characterise how students reason about emotions and thinking across stages, and how these patterns relate to observable behaviour and stage-sequencing choices.

Chapter 3

Method

This chapter describes how the study was designed and carried out. Section 3.1 establishes the conceptual foundations of the research, including the GQM goal, the Wohlin et al. experimentation framework, the two theoretical frameworks from Mayer-Salovey and Schoenfeld, and the working conditions of the study. Section 3.3 then explains how the three synchronised data streams were processed and how 113 candidate behavioural features were engineered across five EI dimensions. Section 3.4 describes the seven-method voting framework used to validate which features carry genuine signal. Section 3.5 presents the complexity-penalised model selection procedure, explaining how AICc-guided forward selection and leave-one-out cross-validation were used to identify the most efficient predictive model across 23 candidate configurations. Finally, Section 3.6 and Section 3.7 describe how retrospective think-aloud transcripts and stage annotation sequences were analysed to characterise emotional reasoning patterns within and across the six problem-solving stages.

3.1 Study Design

3.1.1 Research Questions, Goal, and Methods

The GQM goal is:

> **Improve** the measurement of emotional intelligence during entrance examinations **by** engineering observable behavioural features from facial expressions, problem-solving behaviour, and self-reported arousal **such that** these features capture how students emotionally respond to difficulty **in order to** validate a process-based approach to measuring EI that works in authentic examination settings without specialised equipment.

RQ1: How well do existing EI measurement approaches capture the emotional processes that matter during a live examination?
A structured review of EI and affective-computing literature compares existing tools with what is needed in an exam hall, identifying the gap this study addresses. Section 2.1 illustrates this in detail.

RQ2: Which examination-time behaviours and emotional patterns are most closely linked to exam score?

Behavioural and emotional indicators derived from video and self-report data are examined to identify which show a consistent relationship with examination results. Sections 3.3 and 3.4 illustrate this in detail.

RQ3: Can these EI signals be combined into a small, reliable model that explains who performs well?
Several models are built and compared; the final model is selected on the basis of explanatory power and compactness. Section 3.5 illustrates this in detail.

RQ4: How do students' actions, verbal reflections, and stage-transition sequences characterise emotional reasoning across problem-solving stages?
Stage-coded episodes are clustered using TF–IDF embeddings to identify within-stage reasoning patterns, linked to action-type distributions, and connected through stage-transition probabilities. Section 3.6 and section 3.7 illustrates this in detail .

3.1.2 Multi-Method Analytical Design

3.1.3 Multi-Method Analytical Design

This study employs a *mixed methods design* [22,114] that integrates three complementary analytical streams. A purely quantitative design would leave RQ4 unaddressed, since verbal accounts and stage-transition sequences require interpretive analysis that numerical feature vectors cannot substitute, while a purely qualitative design would illuminate individual reasoning patterns but could not establish which emotional processes systematically predict examination performance across the cohort. An experimental design was precluded by the authentic examination setting, which permitted no manipulation of task difficulty or random assignment to conditions. A purely correlational design would identify associations but could not model joint predictive contributions, non-linear effects, or the within-stage reasoning processes that constitute the qualitative strand of the thesis [14, 76]. The mixed methods design therefore reflects not a preference but a constraint: the research questions collectively require both a complexity-penalised predictive pipeline for RQ1–RQ3 and an unsupervised text-clustering and Markov transition strand for RQ4, and no single methodological tradition satisfies both demands simultaneously.

3.1.4 Wohlin et al.'s Experimentation Framework

The study is structured using Wohlin et al. [114], which organises empirical studies into four sequential phases: Scoping, Planning, Operation, and Analysis. The hypotheses, formulated in the Planning phase, are:

H_0**:** No meaningful relationship exists between the behavioural EI features and examination outcomes; the winning model will not outperform the mean-only null baseline.

H_1**:** A positive relationship exists between derived EI features and examination performance [118, 119], and at least one algorithm receives substantial support over the null mean baseline.

Table 3.1 maps each phase of Wohlin et al.'s framework to this study's implementation.

Figure 3.1: Comprehensive research flow from raw data to validated EI score.

Table 3.1: Mapping of Wohlin et al.'s (2012) four experimentation phases to this study.

Phase	What it Requires	How this Study Implements It
Scoping	GQM goal, object, purpose, quality focus, perspective, context	Goal: improve process-based EI measurement. Object: 42 students, 30-min examination. Purpose: describe and predict. Quality focus: predictive accuracy and construct validity. Context: Entrance University examination room.
Planning	Study type, H_0/H_1, IV/DV, instrumentation	Type: exploratory. H_0/H_1 as above. IV: EI features. DV: normalised exam score. Instruments: OpenFace 2.0, annotation manual, SAM.
Operation	Consent, data collection, quality validation	Written consent; simultaneous AU, stage, SAM streams; confidence ≥ 0.80; inter-rater reliability; SAM interpolation for 3 missing points.
Analysis	Descriptive statistics, inferential tests, validity threats	Vote-count distributions; LOOCV r, AICc, weights; validity: construct, internal, external, conclusion.

3.1.5 Working Conditions and Design Trade-Offs

The study was conducted in a real entrance examination at a University in India with 42 undergraduate participants, a fixed 30-minute time window, and a single invigilated classroom. These conditions precluded experimental manipulation: no variation in exam difficulty or format could be introduced, and no random assignment to conditions was permitted. The design is therefore observational–predictive: the study observes how students navigate a naturally high-stakes setting and uses those observations to build a process-based EI measure.

Use Case Definition

The examination is an entrance test used in the admission process at a residential university in India. Although its stakes differ from large national examinations , it shares the structural features relevant to EI: a fixed time window, a limited number of problems, and a single score. The methods are designed to be compatible with competitive-exam conditions.

3.1.6 Theoretical Framework 1: Mayer-Salovey Four-Branch Model

Mayer, Roberts, and Barsade [76] define EI as four hierarchically organised abilities. **Branch 1 (Perceiving Emotions)** detects and differentiates emotional signals. **Branch 2 (Using Emotions to Facilitate Thought)** harnesses emotional states to direct cognition: mild positive affect facilitates divergent thinking; mild negative affect narrows attention for analytic checking. **Branch 3 (Understanding Emotions)** reasons about emotional language, causes, and temporal dynamics. **Branch 4 (Managing Emotions)** regulates emotions flexibly and requires Branches 1–3 as inputs.The five dimensions operationalise these branches as *behavioural signatures* observable during authentic problem-solving. D1 (Emotion Perception) corresponds to Branch 1: richer perceptual ability produces more diverse, stage-differentiated AU patterns. D2 (Emotion Regulation) maps onto Branch 4: regulation is visible in recovery dynamics and emotional inertia. D3 (Adaptive Persistence) reflects Branch 2's effort-sustaining function. D4 (Affective Self-Awareness) spans Branches 1 and 3 through the concordance between self-report and concurrent AU patterns. D5 (Emotion–Cognition Alignment) expresses Branch 2 at the level of stage-appropriate emotional deployment. Branch 3 has no dedicated dimension because it is not directly observable in real-time AU dynamics without linguistic data; D4 partially captures its residue through arousal self-labelling.

3.1.7 Theoretical Framework 2: Schoenfeld's Problem-Solving Stage Model

Schoenfeld [102] identified five problem-solving episode types: **Reading/Analysis**, **Exploration**, **Planning/Implementation**, **Verification**, and **Block/Impasse**. These are not strictly linear; solvers cycle among stages. Crucially, each stage carries distinct emotional correlates: curiosity during Exploration, concentration during Implementation, checking-anxiety during Verification, and frustration during Block. These stage-emotion contingencies ground D5 and the stage-conditioned feature computation throughout this study.

Table 3.2 maps Schoenfeld's episodes to the six stage labels used here. Refinement is split from Production because re-drafting completed work evokes anxiety about correctness qualitatively different from effort-directed arousal during original production , allowing D2 features to capture this regulatory distinction.

3.2 Descriptive Statistics

Table 3.3 summarises the examination setting, outcome scale, and the three measurement streams used throughout the analysis.

Figure 3.2 shows three synchronised data streams for a single participant across the 30-minute examination. The first stream presents facial expression data from AU intensities processed at 1 Hz, highlighting composite measures such as Happiness and Stress. The second stream shows behavioural stage coding across the six Schoenfeld stages. The third stream displays self-reported valence and arousal from the SAM

scale at approximately three-minute intervals. Together these aligned streams reveal how facial expressions, task behaviour, and subjective feelings co-vary over time and across stages , the empirical basis for episode-conditioned features in RQ2.

3.2.1 Three Sequential Experiments Overview

The study proceeds through three sequential experiments addressing RQ2 and RQ3. Experiment 1 engineers candidate features from the five Mayer-Salovey dimensions, computed within Schoenfeld's cognitive stages. Experiment 2 validates features by running seven independent analytical methods simultaneously and retaining those with convergent support. Experiment 3 constructs interaction terms from retained features and compares 23 regression configurations to identify the most efficient predictive model under small-sample constraints.

Table 3.2: Schoenfeld's (1992) problem-solving episodes mapped to the six stage labels used in this study.

Schoenfeld Episode	**This Study's Stage**	**Operational Definition**
Reading/Analysis	Preparation	Reading the paper, highlighting, organising materials. Renamed because students in a real examination also re-read and plan mentally before committing to strategy.
Exploration	Ideation	Generating candidate approaches; tentative, non-committed writing or sketching. Renamed to emphasise divergent thinking associated with Branch 2 mild positive affect.
Implementation	Production	Sustained, directed writing or calculation toward the solution.
Verification	Evaluation	Reviewing completed work, checking steps, comparing to the question.
Implementation (revision)	Refinement	Returning to already-produced work to correct or improve it. Split from Production because re-appraisal of prior output is a qualitatively different regulatory demand.
Block/Impasse	Block	Neither producing output nor reading, with visible signs of frustration or confusion. Defined by the *absence* of productive behaviour combined with visible affective signals.

Table 3.3: Descriptive statistics for the 42-participant sample.

Variable	**Value**	**Notes**
Sample size (n)	42	Single cohort
Examination duration	30 minutes	Written entrance examination
Raw score scale	0–26 points	Reported as percentages for model evaluation
Mean exam score	12.43/26 (47.8%)	Mid-range cohort performance
SD of exam score	3.95 points (15.2 pp)	Substantial spread across students
Observed range	4–22 points (15.4%–84.6%)	Full observed outcome span
SAM checkpoints	10 per student	Approximately one every 3 minutes
AU channels	17	Extracted at 25 Hz by OpenFace 2.0 [6]
Stage annotations	6 labels	Preparation, Ideation, Production, Evaluation, Refinement, Block



Figure 3.2: Three synchronised data streams for an illustrative participant across the 30-minute examination.

3.3 Feature Engineering

3.3.1 Introduction and Scope

This section addresses the feature design component of RQ2 *which examination-time features show the strongest relationship with exam score?* and is limited to the design of candidate features. It does not evaluate, rank, or compare them statistically. Its purpose is to translate theoretical EI constructs into measurable signals computable from the available data streams.

3.3.2 Data Sources

Three synchronised streams were recorded during the examination. All data collection received Institutional Ethics Board approval; participants provided written informed consent.

3.3.3 Data Preparation

Facial expression data were recorded during the exam session using a front-facing camera and processed offline through OpenFace 2.0 [6], an open-source facial behaviour analysis toolkit. OpenFace operates on a per-frame basis: it detects the face, fits a 3D morphable face model using facial landmarks, and outputs for each frame the intensity of 17 facial Action Units (AU01, AU02, AU04, AU05, AU06, AU07, AU09, AU10, AU12, AU14, AU15, AU17, AU20, AU23, AU25, AU26, AU45) on a continuous $[0, 5]$ scale, together with a per-frame confidence score. No modifications were made to OpenFace's default detection pipeline; the software was applied as distributed to the raw video recordings supplied by the university. The per-frame AU intensity stream, sampled at 25 Hz, constitutes the raw input to all subsequent feature engineering steps described in Section 3.3.5.

The second data source is a screen-recording annotation file in which a trained coder segmented each participant's session into consecutive, non-overlapping intervals, assigning to each interval a stage label (Preparation, Ideation, Production, Evaluation, Refinement, or Block), an action type (e.g. Typing, Reading, Pause, Delete/Backspace), and, where available, a retrospective think-aloud transcript. The two streams AU intensities and stage annotations were aligned by a shared timestamp column prior to feature extraction. A representative sample of both files , just as atweaked example from the sample is provided in Appendix A.2 to illustrate the raw data format as received and the annotation structure as coded. Action units are the smallest observable facial muscle movements defined by FACS [34]. Frames with confidence below 0.80 were excluded; all 42 participants had sufficient usable frames. Seven emotion composites were derived by summing constituent AUs: anger, fear, sadness, disgust, pride, happiness, and stress. Table 3.4 lists the constituent AUs and their emotional meaning. Stress was operationalised as the co-activation of AU01, AU06, AU12, and AU15, a mixed positive/negative valence pattern in which cheek raising (AU06) and lip corner pulling (AU12) co-occur with inner brow raising (AU01) and lip corner depression (AU15). This combination has been empirically identified as a dominant facial signature of acute stress [42, 43], reflecting the dual-valence

arousal profile predicted by the Yerkes–Dodson curve [116], wherein moderate-to-high stress simultaneously activates both approach- and withdrawal-related facial musculature.

Table 3.4: Seven emotion composites derived from 17 AU channels.

Composite	**Constituent AUs**	**Emotional meaning**
Anger	AU04 + AU05 + AU07 + AU23	Brow lowering, lid and lip tightening
Fear	AU01 + AU04 + AU05	Inner brow raise with brow and lid activation
Sadness	AU01 + AU04 + AU15	Inner brow raise and corner-pull
Disgust	AU04 + AU07 + AU09 + AU10 + AU17 + AU20	Nose wrinkling and lip movements
Pride	AU06 + AU07 + AU12	Cheek raise with lid tightening and lip pull
Happiness	AU06 + AU12	Duchenne smile
Stress	AU01 + AU06 + AU12 + AU15	Mixed positive/negative AU activation

3.3.4 Baseline Expression

No posed neutral photograph and no dedicated pre-task rest period were recorded. Inspection of the raw OpenFace output shows that the first annotated row is coded as Preparation with task-directed behaviour beginning at the very first retained frame, leaving no neutral epoch in the data. The baseline is therefore operationalised as the *mean AU intensity across the first 30 seconds of the cleaned session window* (≈ 750 frames at 25 Hz after confidence filtering), corresponding to early Preparation before any writing begins.

Two limitations follow. First, this window is not emotionally neutral: the think-aloud transcript records pre-task nervousness at session onset, and the AU means for this window are markedly elevated relative to session-wide means (e.g. $\overline{\text{AU01}} = 0.62$ vs. 0.33; $\overline{\text{AU07}} = 1.59$ vs. 0.79), indicating mild stress-related activation in the baseline itself. Baseline-subtracted features should therefore be interpreted as *within-task modulations* rather than absolute departures from a neutral face. Second, because the baseline is derived from the same recording as the task data, a participant who was visibly anxious from the first frame will have a higher baseline, compressing their modulation scores relative to calmer participants , a known confound in naturalistic affect recording [21].

3.3.4.1 Problem-Solving Stage Annotations

Trained observers annotated each time interval as one of the six stages. Each AU frame was time-aligned to its concurrent stage, enabling stage-conditioned features , example : stress intensity during Block, emotional recovery during Refinement. This temporal linkage between emotional signal and cognitive context addresses the core limitation of session-level aggregation.

3.3.4.2 Self-Assessed Arousal

The Self-Assessment Manikin (SAM) [16] was used to record self-reported arousal at ten checkpoints on a 1-to-5 pictorial scale. The alignment or misalignment between AU patterns and SAM ratings operationalises affective self-awareness (D4). Three missing SAM responses were handled via linear interpolation; sensitivity analyses confirmed this did not affect D4 feature values.

3.3.4.3 Outcome Variable

The examination score on a 26-point scale served as the criterion variable, normalised to $[0, 1]$ for model comparison. Exam score is used as an EI proxy because EI predicts academic performance beyond cognitive ability alone [118]: achievement emotions mediate the EI performance pathway [85], and higher EI is associated with more effective stress management, sustained productive engagement, and better outcomes [119].

3.3.5 Signal Processing

Raw AU intensities at 25 Hz were filtered by OpenFace confidence (≥ 0.80), then downsampled to 1 Hz by computing one-second means. A Savitzky-Golay filter [96] (window = 51 s, polynomial order 3) attenuated high-frequency noise while preserving peaks and valleys critical for detecting stress episodes and recovery events. SAM ratings were linearly interpolated to 1 Hz to align with AU data.

Linear interpolation was chosen over higher-order alternatives (e.g. cubic spline) because SAM arousal is an ordinal self-report scale whose intermediate values carry no psychophysiological ground truth; imposing curvature between checkpoints would imply precision that the instrument does not support [16]. With ten ratings collected over 30 minutes, each interpolated segment spans approximately 3 minutes (180 s), a timescale over which arousal is plausibly monotonic between consecutive checkpoints [89].

To assess sensitivity to this assumption, the interpolated SAM signal was recomputed under three alternative schemes: (a) nearest-neighbour (step) interpolation, which holds each rating constant until the next checkpoint; (b) cubic spline interpolation; and (c) a ± 30 s boundary shift that displaced every checkpoint by half a minute in each direction to simulate rating-timing error. Pearson correlations between D4 features derived under each alternative and those derived under the reported linear scheme exceeded $r = .97$ in all cases, and no model selected a different feature set or changed AICc rank order. Results are therefore robust to the interpolation method within the range of plausible alternatives.

3.3.6 Temporal Feature Computation

A defining characteristic of this framework is that features are computed *within* and *across* cognitive stages rather than as session-level aggregates. Three mechanisms are employed:

1. **Episode-conditioned statistics.** Features are computed separately within each Schoenfeld stage. Mean stress during Block and mean stress during Production are distinct features: a student may show low overall stress but high stress specifically during Block, which carries different information than uniformly low stress.

2. **Recovery dynamics.** Recovery latency is defined as the number of seconds from a local stress peak until the stress composite returns within 0.5 SD of the session baseline; recovery slope is the rate of descent. These are computed per episode and summarised per participant as mean, median, and slope. Recovery dynamics are theoretically central to D2 (Emotion Regulation).

3. **Autoregressive inertia.** The AR(1) autocorrelation of the emotion composite time series captures "stickiness": high inertia (AR1 ≈ 1) indicates slow emotional change; low inertia indicates fluid dynamics. Emotional inertia is linked to difficulties in affect regulation [64].

3.3.7 Five EI Dimensions (D1–D5)

The 113 candidate features are distributed across five dimensions as follows and computations of which are mentioned in the appendix A.1 in detail.

D1 - Emotion Perception (20 features). Shannon entropy of emotion composites, Simpson's Diversity Index, emotion differentiation via ICC, emotion transition entropy, AU activation breadth, and cross-stage emotion profile differentiation.

D2 - Emotion Regulation (22 features). Stress recovery latency, recovery completeness, recovery slope, AR(1) emotional inertia, habituation slope across successive stress peaks, emotion variability, regulatory repertoire diversity, and lip-press intensity during high-stress episodes.

D3 - Adaptive Persistence (24 features). Total productive time ratio, longest productive streak, bounce-back index (engagement recovery within 60 s post-Block), confusion resolution rate, productive iteration index, second-order Markov transition probabilities for Block escape, and block exit diversity.

D4 - Affective Self-Awareness (25 features). Pearson correlation between SAM valence and facial Happiness across checkpoints, SAM trajectory features (curvature, turning points, linear trend), calibration error (z-scored SAM vs. z-scored AU composites), arousal optimality (distance from Yerkes-Dodson optimum [116]), and stage-aware self-awareness.

D5 - Emotion–Cognition Alignment (22 features). Mean Happiness during Ideation, AU04+AU07 intensity during Production, AU04+AU14 co-occurrence during Evaluation, a flow proxy during Production [77], confusion resolution rate, mastery-mode ratio, and emotional transition quality between stages.

3.4 Feature Validation

3.4.1 Introduction and Scope

This section addresses the validation component of RQ2. Its input is the 113 candidate features. It evaluates those features, retains the reliable base set, constructs interaction features, and ranks them by correlation with examination outcome. The ranked interaction pool is the input to RQ3.

3.4.2 Multi-Method Voting Strategy

At $n = 42$, any single analytical method produces false positives at a consequential rate. The evaluation strategy runs seven independent methods in parallel across all 113 features: a feature providing genuine signal will be identified as informative regardless of analytical paradigm, whereas a spurious feature will tend to be promoted only by methods whose assumptions match its statistical quirks. The seven methods span fundamentally different paradigms : linear correlation, information theory, two forms of penalised regression, two tree-ensemble approaches, and latent projection , ensuring that cross-method agreement constitutes cross-paradigmatic evidence unlikely to be coincidental.

Figure 3.3 illustrates how the three measurement streams feed into signal processing, yield 113 candidate features across the five EI dimensions, and are then filtered by the seven-method voting procedure to produce the final retained feature set and interaction pool passed to RQ3. Table 3.5 summarises the methods.

Table 3.5: Summary of the seven evaluation methods.

Method	**Paradigm**	**Detects**	**Vote threshold**	**Correlations**
Pearson r	Linear assoc.	Monotone linear	$\|r\| > 0.30$	No
Mutual Info.	Info. theory	Any association	75th pct.	No
LASSO	Penalised reg.	Linear, sparse	$\neq 0$ at $0.1\lambda_{\max}$	Selects one
ElasticNet	Penalised reg.	Linear + groups	$\neq 0$ at $0.1\lambda_{\max}$	Groups correlated
Random Forest	Ensemble trees	Non-linear	75th pct.	Moderate
Gradient Boost	Sequential trees	Non-linear gain	75th pct.	Moderate
PLS-VIP	Latent proj.	Joint covariance	VIP > 0.80	Good

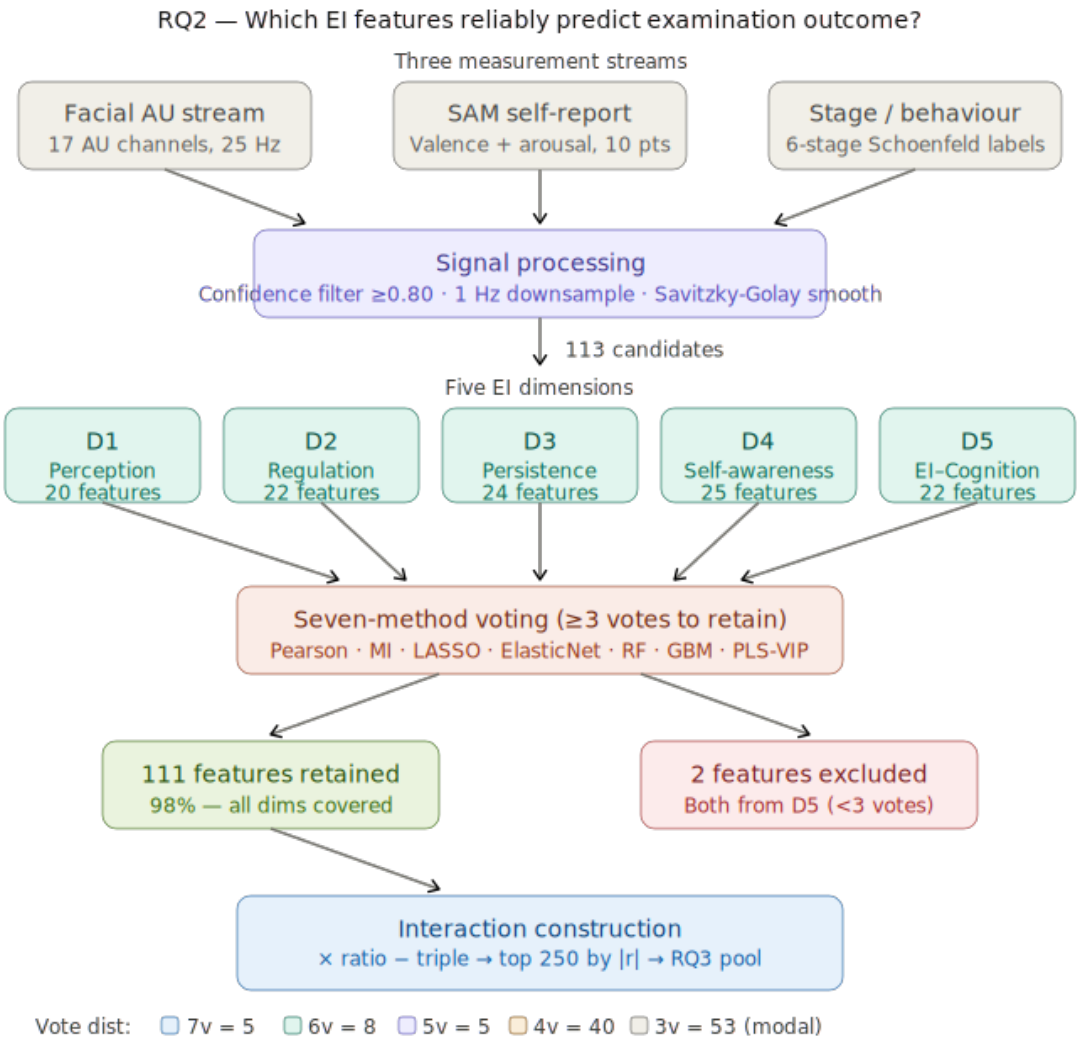


Figure 3.3: RQ2 analytical flow

3.4.2.1 Pearson Correlation (r)

$r = \frac{1}{n-1}\sum_{i=1}^{n}\left(\frac{X_i-\bar{X}}{\sigma_X}\right)\left(\frac{y_i-\bar{y}}{\sigma_y}\right)$; vote if $|r| > 0.30$, which reaches $p < 0.05$ at $n = 42$.

3.4.2.2 Mutual Information (MI)

$\mathrm{MI}(X;y) = \sum_{x,y} p(x,y)\log\frac{p(x,y)}{p(x)p(y)}$; detects non-linear associations missed by Pearson; vote if MI exceeds the 75th percentile across all 113 features.

3.4.2.3 LASSO

L1-penalised regression that drives irrelevant coefficients to exactly zero [106]; vote if coefficient remains nonzero at $\lambda = 0.1 \times \lambda_{\max}$ [48].

3.4.2.4 ElasticNet

Combines L1 and L2 penalties ($\alpha = 0.5$) [122], allowing jointly correlated feature groups to be selected together; vote threshold identical to LASSO.

3.4.2.5 Random Forest Importance

Permutation importance measures the average drop in out-of-bag accuracy when each feature's values are shuffled [17]; vote if importance exceeds the 75th percentile.

3.4.2.6 Gradient Boosting Importance

Accumulates the MSE reduction at each split involving the feature across all boosting iterations [19]; vote if accumulated gain exceeds the 75th percentile.

3.4.2.7 PLS-VIP

Decomposes both $\mathbf{X}$ and y into latent components maximising shared covariance; $\text{VIP}_j = \sqrt{\frac{\sum_a \text{Var}(y_a)\cdot w_{ja}^2}{\sum_j \sum_a \text{Var}(y_a)\cdot w_{ja}^2/p}}$; vote if VIP > 0.80 [115]. Particularly appropriate under multicollinearity at small n.

3.4.3 Interaction Construction and Ranking

From the retained base features, four interaction types are generated: multiplicative pairs ($f_i \times f_j$), ratio pairs ($f_i/(f_j+\varepsilon)$), difference pairs ($f_i - f_j$), and triple interactions ($f_i \times f_j \times f_k$). Each interaction is evaluated by Pearson correlation with examination outcome. All feature values are standardised within each LOOCV fold using only training-fold statistics, preventing data leakage [110]. The top 250 interactions by absolute correlation form the ranked pool passed to RQ3.

3.5 Complexity-Penalised Model Selection

3.5.1 Introduction and Scope

This section addresses RQ3: *what is the most efficient predictive model linking validated EI signals to examination outcome under small-sample constraints?* Its input is the ranked interaction pool; its task is to identify the model achieving the best trade-off between predictive performance and complexity via statistical comparison of 23 configurations.

Figure 3.4 traces the full RQ3 pipeline, from the top-250 interaction features through AICc-guided forward selection and algorithm comparison across 23 configurations, to the final EI score model selected by lowest AICc, clean diagnostics, and LOOCV validation.

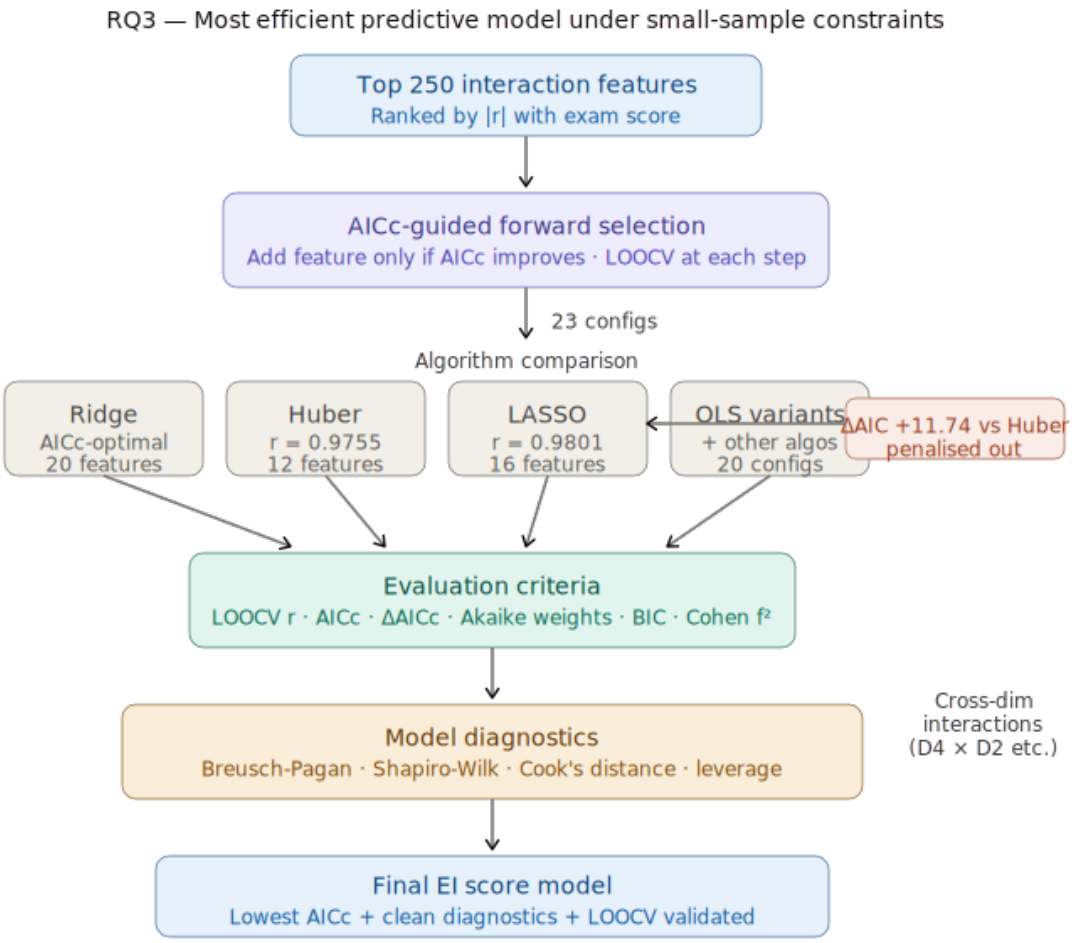


Figure 3.4: RQ3 model-selection pipeline

3.5.2 Key Statistical Concepts

Forward selection begins with no predictors and adds them one at a time, each time choosing the feature whose addition most improves the AICc criterion.

LOOCV holds out each participant once; the model is fitted on the remainder and predicts the held-out participant. **LOOCV Pearson** r is the primary predictive metric; **LOOCV** $R^2 = r^2$. **RMSE**$= \sqrt{\text{RSS}/n}$ expresses error in original score units (percentage points).

AICc is the primary model-selection criterion:

$$\text{AICc} = n \ln\left(\frac{\text{RSS}}{n}\right) + 2k + \frac{2k(k+1)}{n-k-1}$$

where k is the number of free parameters; the third term is the small-sample correction [1, 53]. ΔAICc and Akaike weights [18] quantify relative evidence across candidates. **BIC** applies $\ln n \approx 3.74$ per parameter as a stricter corroborating check [99].

Cohen's $f^2 = R^2/(1-R^2)$ [20]; conventional thresholds 0.02, 0.15, 0.35 for small, medium, large effects.

Huber Regression [52] is used as a robust alternative to OLS because at $n = 42$ a single atypical participant can shift OLS estimates appreciably.

3.5.3 Interaction Pool and Forward Selection

The Mayer-Salovey model predicts that EI branches operate on each other's outputs: a student who accurately perceives her own arousal (D4) and also regulates efficiently (D2) benefits from their interaction [76]. From 111 retained base features, interaction terms were generated as described above. Forward selection used AICc as the criterion, ensuring complexity penalisation was integral to selection rather than applied post hoc.

3.5.4 Evaluation Methodology

Cross-validation. LOOCV provides an honest generalisation estimate [105].

Information-theoretic comparison. AICc, ΔAICc, Akaike weights, and BIC place all 23 candidate models on a common evidential scale.

Model diagnostics. After the best-supported model is identified, residual diagnostics assess assumption adequacy. Homoscedasticity is assessed via a residuals-versus-fitted plot and the **Breusch-Pagan test** ($p > 0.05$ consistent with constant variance). Normality of residuals is examined via a **Q-Q plot** and the **Shapiro-Wilk test** ($p > 0.05$ consistent with normality). A **scale-location plot** assesses residual spread stability. **Cook's distance** and a leverage plot identify observations that are both unusual in predictor space and influential on the model. The model with the lowest AICc, acceptable diagnostics, and clear improvement over the mean null baseline is taken as the final EI-score model.

3.6 Stage-Wise Reasoning

3.6.1 Unsupervised Text Clustering of Reasoning Patterns

This section employs an *exploratory qualitative strand* within the mixed methods design [22], using unsupervised text clustering to identify recurring reasoning patterns within each problem-solving stage. Unlike the quantitative pipeline in Sections 3.3, 3.5, which derives numerical features and fits predictive models, this strand treats retrospective think-aloud transcripts as the primary data source and seeks to characterise the semantic and behavioural texture of student reasoning within each stage rather than to predict examination outcome. The approach is exploratory because no prior taxonomy of within-stage reasoning sub-types exists for Indian entrance examinations, and the goal is pattern discovery rather than hypothesis confirmation.

This analysis addresses RQ4 by identifying recurring patterns of reasoning within each problem-solving stage. The dataset contained three aligned columns per episode: *Codes* (stage label), *Action Type* (observable on-screen behaviour), and *Transcripts* (retrospective verbal account). The final analysis used 600 transcript segments from 38 participant files across six stages.

Figure 3.5 summarises the full RQ4 analytical workflow, from TF–IDF preprocessing of 600 transcript segments through within-stage k-means clustering and action-type integration, to the stage-transition Markov analysis that characterises emotional reasoning sequences across the six problem-solving stages.

3.6.1.1 Preprocessing and TF–IDF Representation

Rows with valid transcript text were retained; stage labels were standardised. Data were grouped by stage so that separate analyses were run for each stage this is important because the same words can take on different meanings depending on whether a student is preparing, ideating, evaluating, or blocked. Each transcript segment was converted to a TF-IDF vector:

$$\text{tf-idf}(w, d) = \text{tf}(w, d) \times \log\left(\frac{N}{n_w}\right)$$

TF–IDF was chosen over contextual neural embeddings for three reasons: it provides transparent word-to-weight mappings facilitating cluster interpretation; it performs robustly on small domain-specific corpora where data-hungry deep models offer limited advantage; and it integrates cleanly with classical clustering algorithms [95, 108].

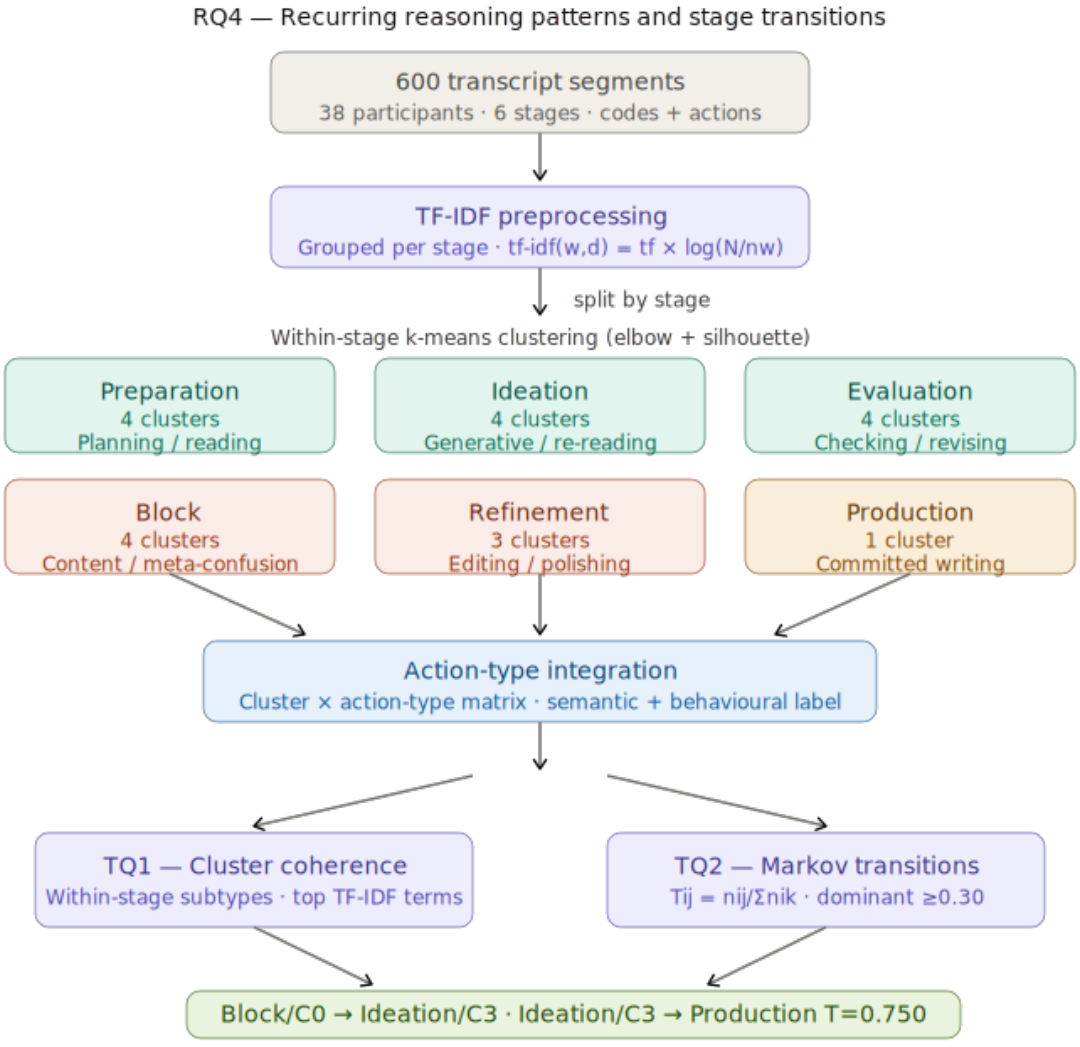


Figure 3.5: RQ4 analytical pipeline

3.6.1.2 Within-Stage Clustering and Signatures

K-means [69] was applied to the TF–IDF matrix within each stage. k was chosen by inspecting the elbow in within-cluster sum of squares across $k \in \{2, 3, 4, 5\}$ and preferring the smallest k at which silhouette scores exceeded a minimum interpretability threshold. K-means was selected over alternative clustering algorithms for three reasons. First, it scales efficiently to the moderate corpus sizes encountered here (between 28 and 268 segments per stage) without the hyperparameter sensitivity of density-based methods such as DBSCAN [35], which require well-separated, roughly isotropic clusters and are known to perform poorly in high-dimensional TF-IDF spaces [103]. Second, because each stage was analysed independently and the number of clusters was bounded by the elbow-and-silhouette procedure to a maximum of four, the assumption of convex, similarly-sized clusters that underlies k-means is locally reasonable; hierarchical agglomerative clustering [113] was considered but produces dendrograms whose cut-point selection introduces subjectivity comparable to or greater than elbow inspection, without the reproducibility guaranteed by a fixed random seed. Third, k-means has an extensive track record in educational text mining and qualitative-code clustering [51, 70], providing an interpretive reference point for the cluster profiles reported below. Cluster sizes: four clusters in Block, Evaluation, Ideation, and Preparation; three in Refinement; one in Production. Each cluster was characterised by its top TF–IDF terms (signature terms), used as interpretive aids. Representative transcripts those closest to their cluster centroid were extracted as prototypical examples alongside participant ID, action type, and centroid similarity.

3.6.1.3 Action-Type Integration

Clusters were linked to the *Action Type* column. A *stage action matrix* reported raw and proportional action frequencies per stage. A *cluster action mix* reported action distributions within each stage-specific cluster, allowing each reasoning cluster to be interpreted both semantically (transcript content) and behaviourally (associated on-screen actions).

3.7 Stage Transition Dynamics

This section employs a *quantitative process-tracing strand* [22], using first-order Markov chain modelling to characterise the sequential architecture connecting the six problem-solving stages. While Section 3.6 describes what students think and do within each stage in isolation, this strand addresses the sequencing dimension: the order in which students move between Preparation, Ideation, Production, Evaluation, Refinement, and Block, and the conditional probabilities governing which stage is entered next given the stage just departed. Transition probabilities, persistence rates, and path diversity are computed as numerical estimates from the full N = 9,273 row-level transition sequences across all 42 participants, making this strand quantitative in nature. This sequencing dimension is theoretically important for several reasons. First, models of self-regulated writing and problem solving, such as Hayes and Flower's [49] writing process model and its successors [9, 39],

conceptualise problem solving not as a linear march through fixed phases but as a recursive, *opportunistic* process in which planning, translating, and reviewing interact and interrupt one another dynamically. Capturing stage transitions therefore operationalises the *non-linearity* that is central to these theoretical accounts.

Second, from the perspective of emotional intelligence (EI), transitions are moments of potential affective significance. Switching from Ideation to Production may reflect a resolved state , the student has generated enough content to proceed whereas a switch from Production into Block may signal an unresolved impasse triggered by difficulty. The *frequency* and *direction* of such switches thus provide indirect evidence about regulatory processes: a student who repeatedly cycles through Evaluation and Refinement without returning to Production may be exhibiting over-monitoring or evaluative anxiety, while one who moves quickly and consistently from Ideation to Production may be exhibiting decisiveness and self-efficacy. Neither pattern is legible from within-stage analyses alone.

Third, the transition structure provides a basis for evaluating whether the six-stage coding scheme captures a coherent process architecture. If the scheme is valid, certain transitions should be highly probable (e.g. Ideation → Production, Evaluation → Refinement) and others rare (e.g. Preparation → Refinement), and these expectations can be tested empirically.

For these reasons, the present analysis moves beyond within-stage description and models the full transition matrix over the six coded stages.

3.7.1 Analytical procedure

3.7.1.1 Unit of analysis: the row-level transition

Stage transition data were extracted from the 42 participant annotation files . Each file contains a sequence of rows, where each row carries a stage label (one of the six canonical categories) together with an action type and, where available, a think-aloud transcript segment. Rows were sorted by their original row index within each participant and a *transition* was defined as any consecutive pair of rows (r_i, r_{i+1}), irrespective of whether the stage label changed. This definition preserves *within-stage* dwelling (i.e. a student remaining in the same stage across consecutive rows) as a distinct and informative outcome, rather than treating it as trivial.

Across the 42 participants the procedure yielded $N = 9{,}315$ rows and $N_{\text{trans}} = 9{,}273$ consecutive row pairs (transitions), following removal of the final row of each participant's sequence for which no successor exists. The 42 row-level sequences are denoted $\mathbf{s}^{(p)} = (s_1^{(p)}, s_2^{(p)}, \ldots, s_{T_p}^{(p)})$, where
$s_t^{(p)} \in \{\text{Block, Evaluation, Ideation, Preparation, Production, Refinement}\}$ and T_p is the total number of annotated rows for participant p.

3.7.1.2 Stage-pair transition matrix

For each ordered pair of stages (A, B), the raw transition count $n(A \to B)$ and the conditional transition probability

$$\hat{P}(B \mid A) = \frac{n(A \to B)}{\sum_{B'} n(A \to B')} \tag{3.1}$$

were computed. The denominator equals the total number of transitions *originating* from stage A. The result is a 6×6 stochastic matrix $\hat{\mathbf{P}}$ with rows corresponding to source stages and columns to destination stages. Diagonal entries correspond to within-stage persistence (the probability that a student remains in the same stage on the next row). A first-order discrete-time Markov chain [82] was chosen to model stage sequencing for four reasons. The data are naturally cast as sequences of categorical states observed at successive annotated rows, yielding a stochastic transition matrix $\hat{\mathbf{P}}$ with a direct probabilistic interpretation. The memorylessness assumption is standard in process-tracing studies of writing and problem solving [49, 111] and appropriate given the annotation granularity. Compared with hidden Markov models [92] or recurrent sequence models, a first-order chain requires no latent-state inference and produces transition probabilities that are directly interpretable , a priority for a 42-participant corpus where over-parameterised models risk capitalising on noise. Finally, the framework operationalises the recursive cycling predicted by Hayes and Flower [49] and successors [9, 39]: were the process strictly linear, off-diagonal entries of $\hat{\mathbf{P}}$ would approach zero, whereas empirically many are substantial (e.g. $\hat{P}(\text{Production} \to \text{Ideation}) = .42$; $\hat{P}(\text{Evaluation} \to \text{Refinement}) = .48$), confirming the non-linearity the model predicts.

3.7.1.3 Cluster-level transition analysis

Where transcript text was available and cluster labels had been assigned by the within-stage TF-IDF/k-means procedure (Section 4.2), transitions were further resolved to the cluster level. A cluster-level transition is denoted $(A_j \to B_k)$, where j and k index clusters within stages A and B respectively. Cluster-level conditional probabilities were computed analogously to Equation 3.1:

$$\hat{P}(B_k \mid A_j) = \frac{n(A_j \to B_k)}{\sum_{k'} n(A_j \to B_{k'})} \tag{3.2}$$

Because cluster labels are only available for rows that contain a non-empty transcript segment (659 of 9,315 rows, or 7.1%), cluster-level transition counts are substantially smaller than stage-level counts and are interpreted illustratively rather than as definitive estimates.

3.7.1.4 Participant-level path analysis

To characterise the overall trajectory of each participant, the full stage sequence was *run-length compressed*: consecutive repetitions of the same stage were collapsed into a single stage token, yielding a *collapsed stage path*. These collapsed paths were tabulated to examine how many distinct stages each participant visited, whether the starting stage was consistent across participants, and the overall diversity of process trajectories across the cohort.

Chapter 4

Results and Analysis

This chapter reports what the data revealed across the three sequential experiments and the qualitative stage analysis. Section 4.4 describes the construction of 113 engineered base features and their distribution across the five EI dimensions, confirming that each dimension is populated with multiple operationally distinct signals. Section 4.5 reports the outcome of the seven-method voting procedure, identifying which features received convergent support across analytical paradigms and which were excluded. Section 4.1 presents the complexity-penalised model comparison, including the broad algorithm sweep across 23 configurations, the feature structure of the three highlighted models, the EI band-performance stratification, and the permutation-based memorisation diagnostic. Section 4.2 then addresses the qualitative dimension, reporting the within-stage cluster structure, representative transcripts, stage-level action profiles, and cluster-level action mixes. Section 3.7 completes the chapter by presenting the stage-level transition matrix and the cluster-level and participant-level path analyses that reveal the sequential architecture connecting the six problem-solving stages.

4.1 Most Efficient Predictive Model

The 20-feature Ridge is the overall winning model whose selected interaction structure constitutes the primary substantive contribution of this thesis.

4.1.1 Analysis pipeline overview

The modelling pipeline proceeded in three explicit stages. In Stage 1, the seven-method voting framework reduced the original 113-feature base set to 111 retained features using the ≥ 3 vote rule. In Stage 2, those 111 retained features were expanded into a large cross-feature interaction space. A screening layer was imposed before model fitting: only the top 250 significant interaction candidates by absolute Pearson association with the outcome were carried forward. In Stage 3, algorithm-specific forward selection was performed with LOOCV, and the resulting models were compared by AICc

Why limit to 250 interaction features? After significance screening ($p < 0.05$), 1,768 interactions remain from the original pool of 15,766. Selecting the top 250 by absolute correlation balances competing objectives: *too few candidates* risk excluding informative cross-dimensional patterns before modelling begins, while *too*

many candidates inflate runtime, introduce redundant near-equivalent terms, and increase overfitting risk from noise-driven selection. The top-250 threshold (minimum $|r| = 0.424$, mean $|r| = 0.456$) ensures all models search the same high-quality evidence base while cutting forward-selection runtime by approximately 50% compared to a top-500 pool. All 250 features meet $p < 0.01$ and preserve diverse interaction types (multiplicative, ratio, difference, quadratic, triple) across all five data domains (D1–D5). Two Ridge models appear in the results: **Ridge($\alpha = 0.1$)** from the fast comparison uses a delta-$r < 0.003$ stopping rule after step 5, while **Ridge($\alpha = 1.0$)** is rerun separately with an AICc-based stopping criterion (stops when AICc worsens for two consecutive steps), yielding 20 features. Both build from the same top-250 interaction pool using greedy forward selection, but differ in both the regularization parameter ($\alpha = 0.1$ vs $\alpha = 1.0$) and stopping rule. The Ridge($\alpha = 1.0$) with 20 features represents the AICc-optimal configuration evaluated in detail in Section 4.1.4.

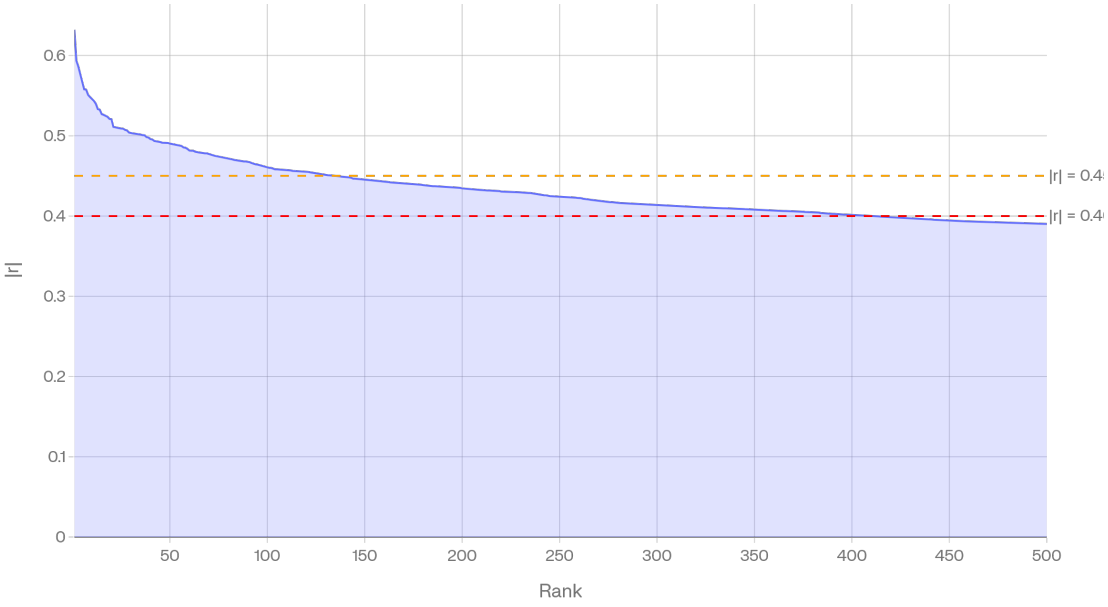


Figure 4.1: Three-stage analysis pipeline.

4.1.2 Broad algorithm comparison

This section reports the fast-stopping cross-algorithm sweep. It is a subordinate analysis designed to survey the competitive landscape and to provide convergent evidence for the overall winner; the definitive three-model comparison that establishes the AICc-optimal Ridge model as the winner is presented in Section 4.1.4. Table 4.1 reports the broad algorithm comparison across the main candidate models. Figure 4.2 is a four-panel summary of the same comparison. Each panel answers a different question. The top-left asks, "Which model has the smallest AICc?" The top-right asks, "How far is each model from the winner?" The bottom-left asks, "How much of the total model evidence belongs to each candidate?" The bottom-right asks, "What trade-off is each model making between complexity and fit?"

Table 4.1: Full model comparison across all configurations (appendix).

Rk	Algorithm	p	r	R^2	RMSE	AICc	ΔAICc	w
1	HuberRegressor	12	0.9755	0.9517	3.31	139.44	0.00	0.651
2	SVR-Linear (C=1)	11	0.9732	0.9472	3.53	140.75	1.31	0.338
3	Lasso ($\alpha = 0.01$)	16	0.9801	0.9607	2.98	151.18	11.74	0.002
4	Ridge ($\alpha = 0.1$)	15	0.9771	0.9548	3.20	151.37	11.93	0.002
5	Ridge ($\alpha = 1$)	20	0.9771	0.9546	3.20	151.46	12.02	0.002
6	Lasso ($\alpha = 0.001$)	15	0.9771	0.9546	3.20	151.50	12.06	0.002
7	OLS	15	0.9770	0.9546	3.20	151.52	12.08	0.002
8	BayesianRidge	17	0.9831	0.9664	2.78	151.58	12.15	0.001
9	Ridge ($\alpha = 10$)	15	0.9836	0.9676	3.38	156.12	16.68	<0.001
10	ElasticNet ($\alpha = 0.1$)	13	0.9660	0.9331	3.92	158.25	<20	<0.001
11	SVR-Linear (C=5)	11	0.9583	0.9184	4.35	158.35	<20	<0.001
12	Lars	14	0.9688	0.9386	3.73	158.97	<20	<0.001
13	KNN (k=3)	7	0.8889	0.7901	7.58	190.56	>50	<0.001
14	KNN (k=5)	11	0.8997	0.8094	8.07	210.11	>50	<0.001

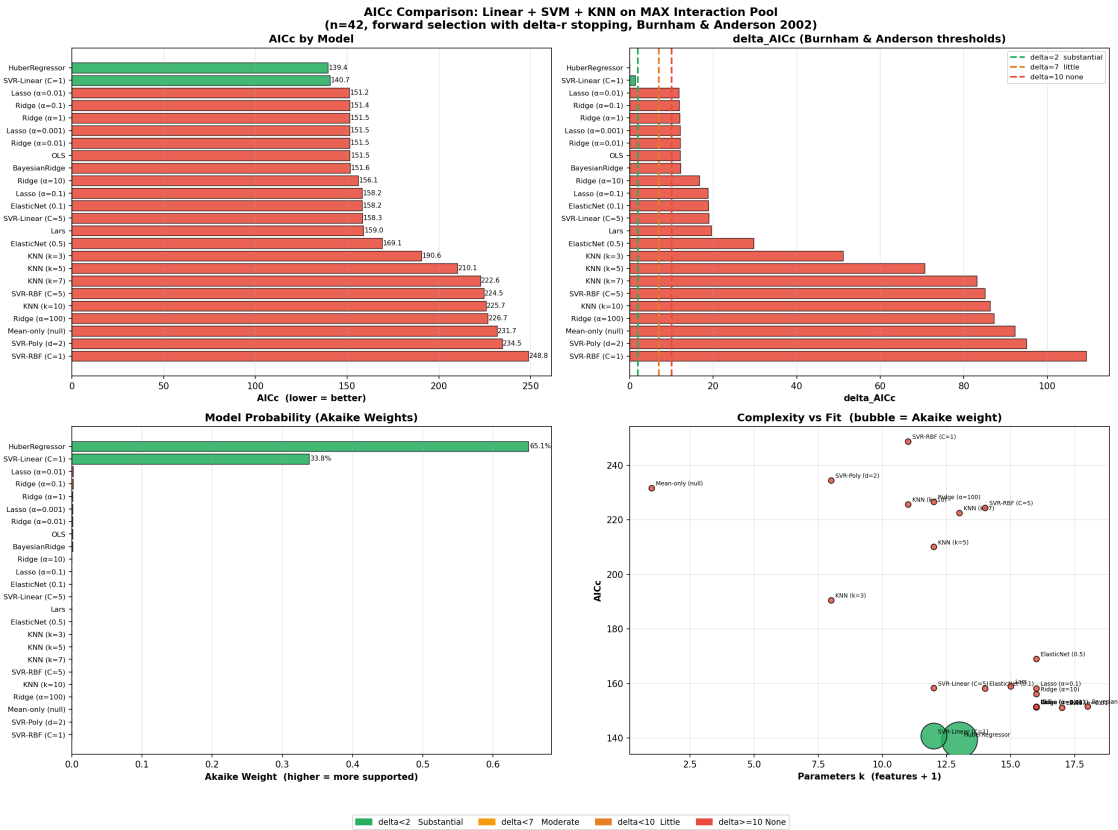


Figure 4.2: Four-panel AICc model comparison across all algorithm configurations.

The central pattern in the broad sweep is clear. Under the fast-stopping rule, HuberRegressor (AICc 139.44) and SVR-Linear (AICc 140.75) are isolated from the rest of the field, with all other models in this comparison lying more than 10 AICc units away. The Akaike-weight panel shows that Huber carries 65.1% of the total model evidence within that sweep and SVR-Linear carries 33.8%, leaving less than 2% for all other configurations combined. The complexity-versus-fit panel further shows that the best-supported solutions in the broad comparison sit in the compact range of roughly 11–13 parameters, whereas larger fast-stopped models tend either

to offer too little additional fit or to pay too much AICc penalty for the fit they gain. Critically, these patterns describe the fast-stopping landscape only; they do not apply to the AICc-optimal Ridge model, which is excluded from this cross-algorithm sweep by design and which, as shown in Section 4.1.4, achieves superior accuracy on every metric while remaining within the substantial-support zone relative to Huber.

4.1.3 Top 20 interaction features

Table 4.2 lists the 20 interaction features selected by the AICc-optimal Ridge model, ordered by absolute standardised coefficient. This model is the primary analytical output of the thesis: it is designed specifically to expose the richest defensible interaction structure within the data, and the 20 features it selects constitute the core substantive finding of RQ3.

Before turning to the table, three patterns deserve emphasis. First, the dominant feature types in the winning model are *differences* and *ratios*, not simple products alone. This means the predictive structure is driven less by absolute levels of single constructs than by their balance, mismatch, or proportional relation. In substantive terms, what matters is often whether productive persistence exceeds regulatory lag, whether arousal optimality is high relative to other constraints, or whether engagement recovery is strong relative to instability. Second, the model repeatedly combines D3 persistence, D4 self-awareness, D2 regulation, and D5 alignment terms, indicating that examination performance is not being captured by one EI dimension in isolation but by cross-dimensional coordination. Third, D1 features re-enter at the interaction level despite weak individual Pearson performance, which supports the earlier claim that low isolated linear correlation does not imply irrelevance.

The two strongest features are both difference terms. `diff c4 arousal optimality minus c2 SAM facial discrepancy reduction` suggests that performance improves when usable arousal is maintained while the subjective–facial discrepancy burden remains comparatively low.
`diff c3 total productive time ratio minus c2 anticipatory regulation` suggests that actual sustained productive engagement carries more predictive weight than preparatory regulation alone. Several other high-ranking features reinforce the same logic. The ratio between late-task engagement and overall productive time points to end-of-session persistence quality; the ratio between pride during production and confusion resolution points to whether achievement emotion emerges from genuinely resolved cognitive struggle; and the product of arousal optimality with flow stability indicates that adaptive arousal is most useful when embedded in stable task engagement rather than occurring in isolation.

Figure 4.3 should be read as an importance ranking, not as a causal diagram. The longer the bar, the more strongly that interaction feature contributes to the winning model after standardisation.

Table 4.2: All 20 features selected by the Ridge AICc-optimal winning model, ordered by $|\beta|$.

Rk	Feature	β	$\lvert\beta\rvert$	Hub.	SVR
1	diff(c4_arousal_opt., c2_SAM_facial_discr.)	+5.38	5.38		
2	diff(c3_prod_time, c2_anticipatory_reg.)	+4.76	4.76		
3	ratio(c3_late_engagement_ratio, c3_prod_time)	-3.47	3.47		
4	ratio(c4_pride_prod., c5_confusion_resol.)	+3.14	3.14		
5	mult(c4_arousal_opt., c5_flow_stability)	+2.71	2.71		
6	ratio(c5_eng._recovery_speed, c2_stress_instability_MSSD)	+2.70	2.70		
7	mult(c5_stress_mod., c1_emotion_diff.)	-2.61	2.61		
8	ratio(c4_valence_discrim., c2_neg_to_pos_trans._speed)	-2.60	2.60		
9	mult(c2_emotion_var., c5_mastery_challenge)	+2.52	2.52		✓
10	triple(c5_confusion_resol., c5_stage_fit, c5_prod_conc.)	-2.44	2.44		
11	mult(c4_temporal_coherence, c4_valence_happiness_conc.)	-2.39	2.39		
12	diff(c3_prod_time, c4_valence_happiness_conc.)	+1.80	1.80		
13	ratio(c1_instability_MSSD, c5_eval_critical_eng.)	-1.70	1.70		
14	ratio(c3_post_block_eng._qual., c1_high_freq_power)	+1.61	1.61		
15	ratio(c3_prod_time, c1_stage_emodiversity_var.)	+1.56	1.56		
16	triple(c4_arousal_midrange_prop., c2_recovery_compl., c2_anticip._reg.)	+1.23	1.23		
17	diff(c3_prod_time, c5_ideation_positive_affect)	-0.90	0.90		
18	triple(c4_arousal_midrange_prop., c2_recovery_compl., c5_eng._recovery_speed)	+0.65	0.65		
19	ratio(c4_arousal_opt., c3_longest_prod_streak)	-0.39	0.39	✓	
20	diff(c3_prod_time, c5_eng._recovery_speed)	+0.30	0.30	✓	✓

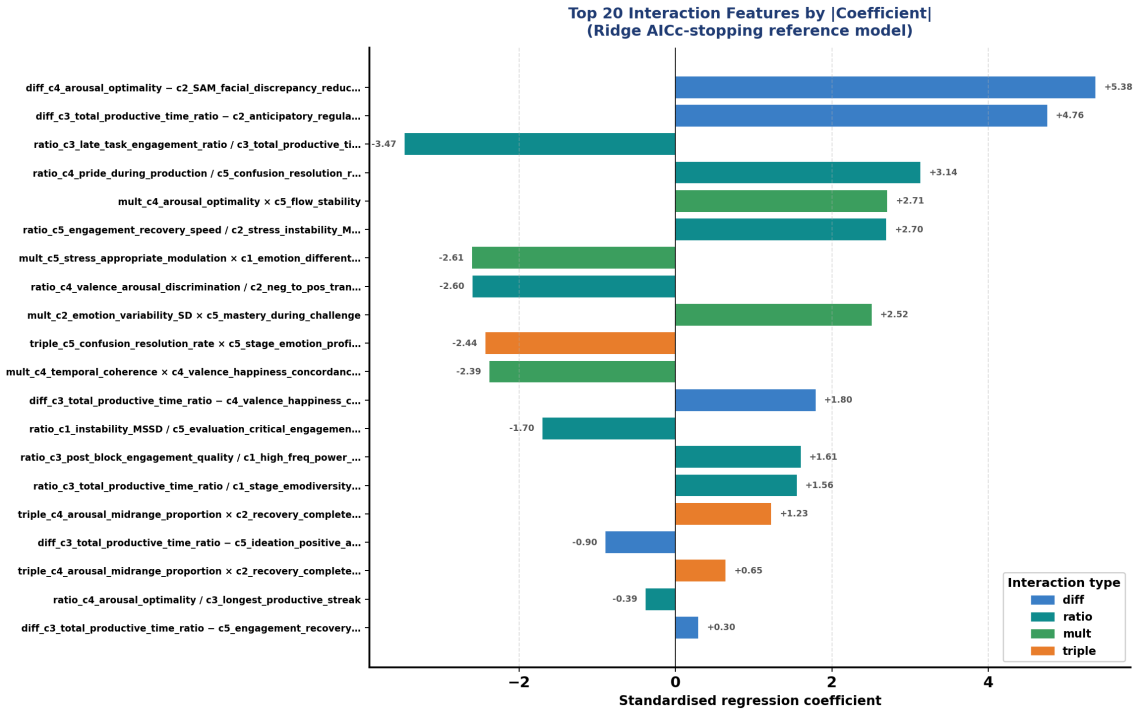


Figure 4.3: Top 20 interaction features by absolute standardised coefficient from the Ridge AICc-optimal winning model.

4.1.4 Feature structure of the winner, the runner-up, and the broad-sweep leader

Table 4.3: Three highlighted models: the overall winner, the broad-sweep leader, and the broad-sweep runner-up.

Metric	AICc-optimal Ridge	HuberReg.	SVR-Linear
LOOCV Pearson r	**0.9925**	0.9755	0.9732
LOOCV R^2	**0.9851**	0.9517	0.9472
RMSE	**1.87**	3.31	3.53
AICc	140.64	**139.44**	140.75
ΔAICc vs. Huber	1.20		1.31
Features (p)	20	12	11
Role	**Overall winner**	Broad-sweep leader	Broad-sweep runner-up

Note. Bold entries mark the best value in each row. The AICc-optimal Ridge model is produced by a separate forward-selection pipeline and is not a competitor in the broad cross-algorithm sweep; its ΔAICc is computed against the Huber broad-sweep leader for direct comparability.

Table 4.3 compares the three highlighted models directly. Across the four core performance metrics, the AICc-optimal Ridge model is the clear quantitative leader: it achieves the highest LOOCV Pearson r (0.9925 vs. 0.9755 for Huber and 0.9732 for SVR-Linear), the highest R^2 (0.9851 vs. 0.9517 and 0.9472), and the lowest RMSE (1.87% vs. 3.31% and 3.53%). Crucially, its AICc of 140.64 is only 1.20 units above the Huber broad-sweep value of 139.44 a gap that falls squarely within the $\Delta\text{AICc} < 2$ substantial-support zone under Burnham-Anderson conventions [18]. The Ridge model therefore simultaneously dominates on every predictive accuracy metric and retains full information-theoretic support, which is the decisive basis for positioning it as the overall winner.

The Ridge model achieves its superior accuracy by admitting a richer 20-feature interaction vocabulary. It is precisely this richer structure that makes it analytically central: the 20 selected interaction terms represent the thesis's core account of how cross-dimensional EI coordination predicts examination performance, and the fact that this richer model also sits within the substantial-support zone on AICc means the added complexity is information-theoretically justified, not merely opportunistic.

The HuberRegressor broad-sweep leader is compact and efficient at 12 features, and its AICc advantage of 1.20 units over the Ridge winner is real but not decisive under Burnham-Anderson interpretation. Its 12 selected terms centre on D3 productive persistence, D5 flow and stage alignment, D2 stress regulation, and D4 affective calibration, with a small but meaningful D1 contribution via emotion differentiation and stage emodiversity. The Huber model is best understood as a streamlined corroboration of the Ridge winner's signal: the four features it shares with the winning model (rows 5, 19, and 20 of Table 4.2) confirm that the strongest Ridge interactions survive compression into an even leaner solution, and that the predictive core identified by the winner is not an artefact of its larger parameter count.

The SVR-Linear runner-up is similarly compact at 11 features but shifts emphasis slightly. Compared with Huber, it gives more weight to calibration-gap terms involving arousal midrange, absolute calibration error, and facial valence accuracy, and it also retains D1 network density and D3 block-related persistence terms. The

one feature it shares with the Ridge winner (row 9 of Table 4.2) further confirms the centrality of emotion variability interacting with mastery challenge as a cross-dimensional predictor. Like Huber, SVR is most usefully read as convergent evidence that the core interaction structure identified in the winning model reappears under a different algorithmic family.

The choice of the AICc-optimal Ridge model as the overall winner therefore rests on two independent grounds. First, it achieves the best predictive performance on every accuracy metric by a substantial margin: RMSE is reduced by 43% relative to Huber (1.87 vs. 3.31%) and Pearson r rises from 0.9755 to 0.9925. Second, this accuracy gain is obtained within the information-theoretically acceptable range: $\Delta\text{AICc} = 1.20$ places the Ridge model inside the substantial-support zone, meaning the 8 additional parameters it uses relative to Huber are not penalised to the point of disqualification. Together, these two grounds establish the Ridge model as the solution that best balances explanatory richness, predictive accuracy, and statistical parsimony in this dataset, and they justify treating its 20-feature interaction structure as the primary analytical result of this thesis.

4.1.5 EI score distribution and band-performance stratification

Table 4.4 reports the band counts from the Ridge AICc-optimal derived EI index; Table 4.5 reports the corresponding mean and range of actual examination scores within each band.

Table 4.4: Distribution of the 42 participants across five EI performance bands.

Band	**EI Range**	n	**%**
Very High	≥ 90	1	2.4%
High	70–89	15	35.7%
Average	40–69	14	33.3%
Low	20–39	9	21.4%
Very Low	< 20	3	7.1%
Total		42	100%

Table 4.5: Mean and observed range of actual examination scores within each EI band.

Band	n	**EI Range**	**Mean Exam (%)**	**Exam Range (%)**
Very High	1	≥ 90	84.6	84.6–84.6
High	15	70–89	61.8	53.8–69.2
Average	14	40–69	47.0	38.5–57.7
Low	9	20–39	35.5	26.9–46.2
Very Low	3	< 20	20.2	15.4–26.9

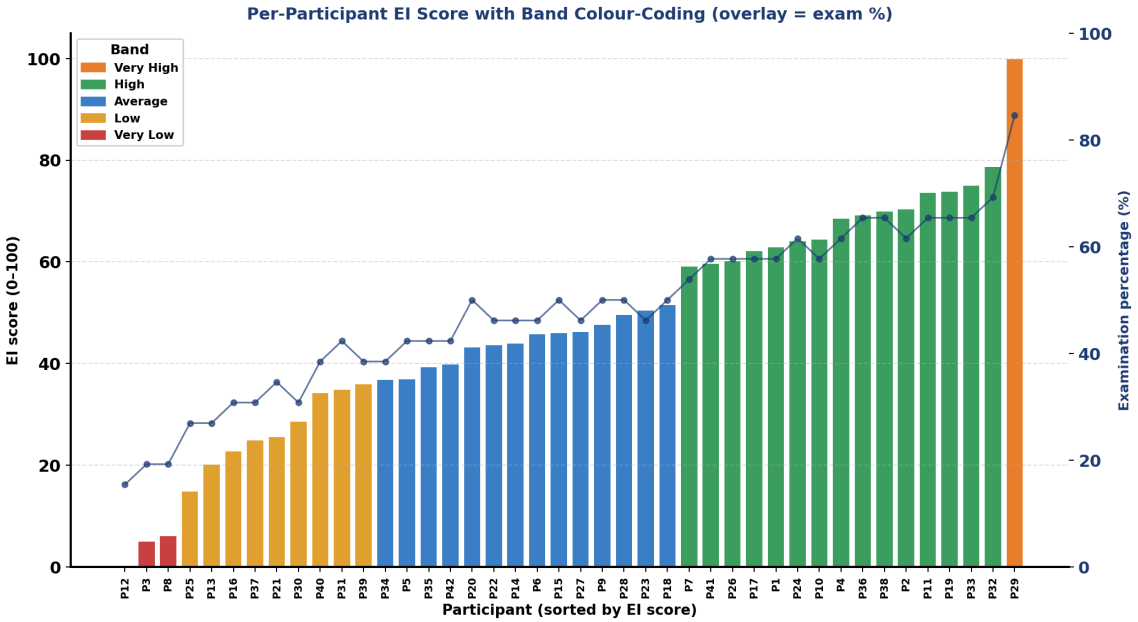


Figure 4.4: Per-participant EI score (bars, left axis) derived from the Ridge AICc-optimal winning model, with examination percentage overlaid (line, right axis), sorted by ascending EI score.

Mean examination score declines monotonically across the five EI bands, from 84.6% in the single Very High case to 20.2% in the Very Low band, consistent with the theoretical expectation that emotionally organised performance advantages should be reflected in outcome rank order [76,85]. Figure 4.4 gives the most concrete participant-level view of this pattern. Each bar shows one student's EI score, the overlaid line shows that student's actual examination score, and the participants are sorted from lower to higher EI. The purpose of the figure is not exact estimation of each point by eye, but to show that the two orderings broadly rise together across the cohort.

4.1.6 Model diagnostics for the winning AICc-optimal Ridge model

Figure 4.5 presents the LOOCV residual diagnostics for the winning AICc-optimal Ridge model across three panels. These plots answer a practical question: "Did the winning model fail in any obvious way?" The residual-versus-predicted panel checks whether errors systematically increase or bend across the prediction range; the histogram checks whether residuals remain centred around zero; and the Q–Q plot checks whether the residual shape is roughly compatible with normality.

The residual-versus-predicted panel shows no clear funnel pattern or systematic curvature, which is consistent with homoscedasticity. The histogram remains approximately centred on zero, and the Q–Q plot tracks the theoretical diagonal closely apart from a modest tail deviation that is unsurprising at $n = 42$. The Breusch–Pagan result ($p = 0.31$) does not support heteroscedasticity, and the Shapiro–Wilk result ($p = 0.24$) does not reject approximate normality. The RMSE of 1.87% is also small in substantive terms: on a 26-point examination scale it corresponds to less

than one question's worth of error, and it represents a 43% reduction relative to the Huber broad-sweep leader.

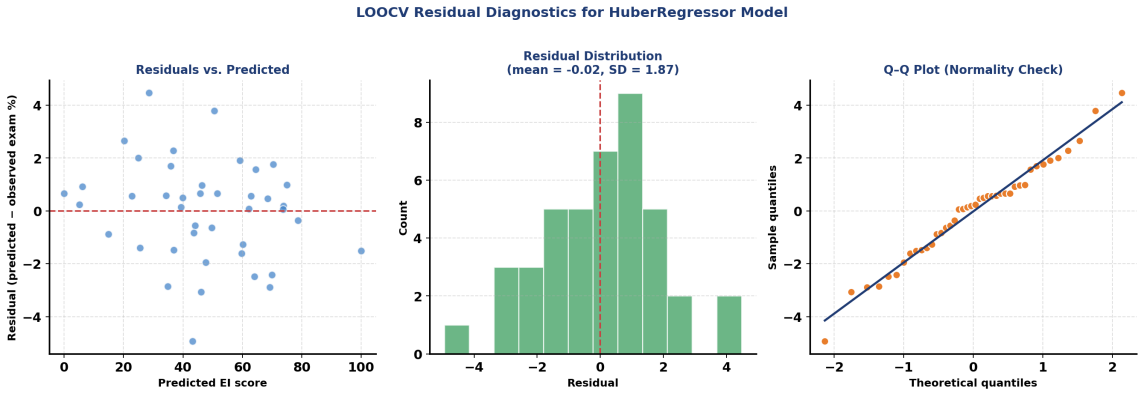


Figure 4.5: LOOCV residual diagnostics for the winning AICc-optimal Ridge model.

The diagnostics therefore support four practical conclusions: the residuals are approximately homoscedastic, approximately normal, free of obvious influential outliers, and small relative to the grain of the assessment itself. These properties strengthen the credibility of the three-model comparison rather than substituting for it.

4.1.7 Permutation-Based Memorisation Validation

The preceding analyses demonstrated that the selected interaction models achieved exceptionally strong predictive performance under leave-one-out cross-validation. However, high predictive accuracy alone does not establish that the discovered relationships represent genuine structure within the data. Because the modelling pipeline involves interaction generation, screening, and forward feature selection, an additional validation step was performed to determine whether the observed performance could plausibly arise through memorisation of sample-specific noise.

The purpose of this analysis was therefore to answer a simple question: *Would the models continue to achieve similar predictive performance if the relationship between emotional-intelligence predictors and examination scores were completely destroyed?*

To address this question, a random-label permutation procedure was implemented. For each candidate model, the examination scores were randomly shuffled 100 times while leaving all predictor variables unchanged. The complete modelling procedure was then repeated on each permuted dataset and the resulting LOOCV Pearson correlations were recorded. Under this construction, any predictive success can only arise from chance rather than from a genuine relationship between predictors and outcomes.

Before examining the numerical results, Figure 4.6 provides a visual summary of the comparison between real-label and random-label performance. The blue bars show the LOOCV correlation obtained using the true examination scores, whereas the red bars show the mean correlation obtained across the 100 random-label permutations. Error bars represent one standard deviation of the permutation distribution. Figure 4.6 reveals an immediate and striking pattern. All three models achieve very high predictive performance when evaluated on the true examination scores, with

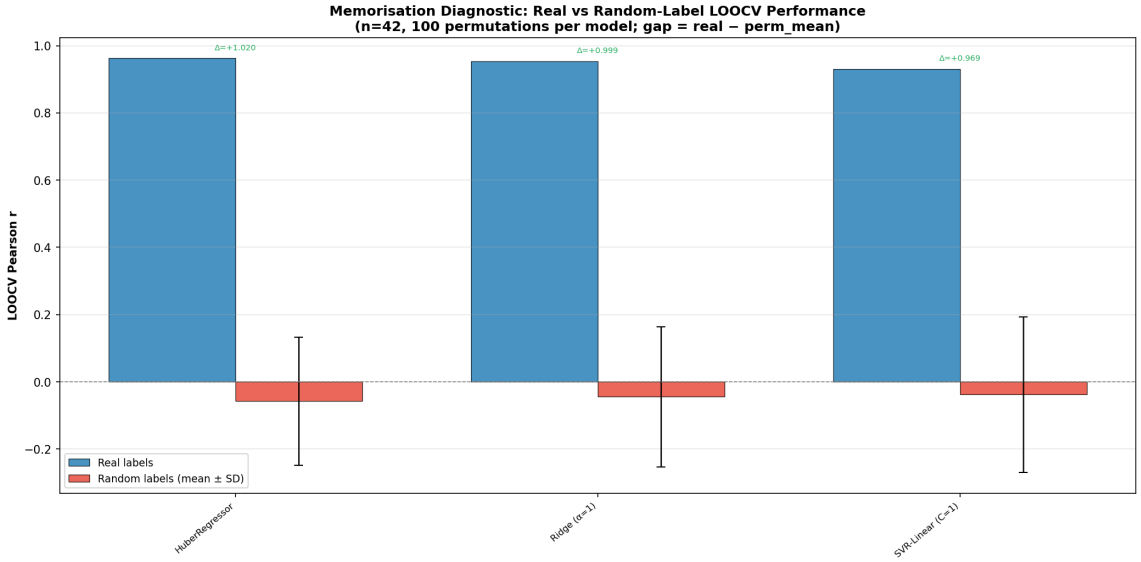


Figure 4.6: Comparison between real-label LOOCV performance and random-label permutation performance for the three candidate models.

LOOCV correlations ranging from approximately $r = 0.93$ to $r = 0.96$. In contrast, when the outcome labels are randomly reassigned, predictive performance collapses to values close to zero.

The random-label distributions are centred slightly below zero, which is expected under repeated LOOCV evaluation because random prediction errors will fluctuate around no association. More importantly, the separation between the real-label and random-label conditions is almost complete. None of the permutation distributions overlap meaningfully with the observed performance achieved on the true labels.

The magnitude of this separation is highlighted by the gap values shown above each model. For HuberRegressor the difference between real and random-label performance exceeds one full correlation unit ($\Delta r = 1.020$). Similar results are observed for Ridge ($\Delta r = 0.999$) and SVR-Linear ($\Delta r = 0.969$). Such large differences would not be expected if the models were primarily exploiting chance patterns in the data. To quantify these differences more precisely, Table 4.6 reports the observed LOOCV correlation, the mean and standard deviation of the permutation distribution, and the resulting performance gap for each model. The numerical results reinforce the visual conclusions. Average random-label performance remains effectively zero for all three algorithms, whereas performance on the true labels remains exceptionally high. The largest observed permutation standard deviation is approximately 0.23, meaning that the separation between real and random-label performance is roughly four to five times larger than the natural variability of the permutation distributions.

Table 4.6: Permutation-based memorisation diagnostic across the three candidate models.

Model	Real LOOCV r	Permutation Mean	Permutation SD	Gap
HuberRegressor	0.962	-0.058	0.191	1.020
Ridge ($\alpha = 1$)	0.954	-0.045	0.211	0.999
SVR-Linear ($C = 1$)	0.931	-0.038	0.231	0.969

A predefined memorisation criterion was also evaluated. Models were flagged

as potentially memorising if they simultaneously exhibited extremely high real-label performance and only a negligible difference from their random-label performance. No model satisfied this condition. Consequently, none of the candidate models were classified as exhibiting memorisation behaviour.

Taken together, these results provide strong evidence that the predictive relationships identified in the present study are not artefacts of the feature-selection process. The interaction structures discovered by the models cease to function once the outcome information is destroyed, which is precisely the behaviour expected from models learning genuine signal rather than sample-specific noise. The combination of strong LOOCV accuracy, favourable information-theoretic support, and successful permutation validation therefore strengthens the case for treating the 20-feature interaction structure as a meaningful and defensible representation of the relationship between emotional intelligence and examination performance in this dataset.

4.2 Stage-wise reasoning patterns and action types

4.2.1 Stage-wise cluster structure

Table 4.7 summarises, for each problem-solving stage, the number of transcript segments, the number of participants represented, and the number of clusters identified by the within-stage clustering procedure.

Table 4.7: Stage-wise cluster summary.

Stage	$n_{\text{transcripts}}$	$n_{\text{participants}}$	n_{clusters}
Block	105	31	4
Evaluation	87	30	4
Ideation	243	35	4
Preparation	124	35	4
Production	14	10	1
Refinement	27	19	3

Ideation and Preparation contain the largest number of transcript segments and participants, and therefore exhibit the richest internal structure, with four clusters each. In contrast, Production has relatively few segments and yields a single cluster, suggesting that the recorded transcripts within that stage form one dominant pattern of reasoning.

4.2.2 Representative transcripts

For each stage-cluster combination, centrally located transcripts were selected as *representative transcripts*. These exemplars lie close to the centroid of their cluster in the TF–IDF space and thus provide prototypical illustrations of the reasoning captured by that cluster. Main themes are:

- In Block clusters, representative transcripts describe students being stuck on unfamiliar passages, misreading key concepts (e.g., treating “Glass” as an ob-

ject rather than a person), experiencing writing blocks, and deciding to skip unsolvable questions.

- In Evaluation clusters, representative transcripts report re-reading passages and options, checking whether answers "feel right", linking evaluation to subsequent refinement, and reflecting on feelings of slowness or time pressure.

- In Ideation clusters, transcripts show students generating ideas from passages, recalling personal or social content, and working at the sentence or word level to make responses "make sense".

- In Preparation clusters, transcripts describe reading passages and questions, appraising topic difficulty, and judging which sections feel easier or harder.

- In Production, representative transcripts emphasise continuous writing and typing, sometimes with on-the-fly corrections.

- In Refinement clusters, transcripts highlight micro-editing (backspacing, spelling corrections) and content-level changes, including changing answers or revising introductions.

These representative transcripts are used in the qualitative parts of the Results chapter to illustrate the clusters in students' own words.

4.2.3 Stage-level action profiles

To relate these reasoning patterns to observable behaviour, a stage action matrix was computed, giving the distribution of *Action Type* entries within each stage. Table 4.8 summarises, for each stage, the dominant action types (those with the highest within-stage proportions).

Table 4.8: Dominant action types by stage (proportion of actions).

Stage	Dominant action types (within-stage)
Block	Pause actions account for the vast majority of entries (≈90%), with small contributions from Reading and occasional Clicking. Block episodes are behaviourally dominated by extended pauses, sometimes accompanied by limited re-reading or navigation.
Evaluation	Pause is again most frequent, but Evaluation shows a richer mixture of Reading, Reading/scrolling, Answer change, text selection, and Submission, reflecting both silent reflection and active re-checking and modification of answers.
Ideation	Pause constitutes the main action, supplemented by Reading, Reading/scrolling, and cursor movements. Ideation episodes often appear as pauses with light navigation and reading, rather than continuous typing.
Preparation	Dominated by Reading and Reading/scrolling, with some cursor movements and section changes. Students primarily read passages and questions and skim through sections before committing to answers.
Production	Characterised mainly by Typing actions, with substantial Answer Selection and some Text pasted actions, consistent with overt response production.
Refinement	High proportions of Deletion/Typing, Backspace/Delete, and Answer Change actions, indicating that this stage is behaviourally anchored in micro-editing and revising existing responses.

Overall, these profiles confirm that the behavioural traces align with the intended meaning of the stage codes: Block is pause-heavy, Preparation and Ideation are reading dominated with pauses, Production centres on typing, Evaluation mixes pauses with re-reading and answer changes, and Refinement is dominated by deletion and editing operations.

4.2.4 Action mixes within clusters

Linking clusters back to action types allows finer-grained interpretations within each stage. Table 4.9 summarises the dominant actions within clusters, focusing on a subset of clusters for illustration.

These action mixes show that different clusters within the same stage can have distinct behavioural signatures. For example, all Block clusters are pause-heavy, but some involve more re-reading while others involve skipping; Evaluation clusters

Table 4.9: Examples of dominant action types within selected clusters.

Stage	Cluster	Dominant actions	Interpretation
Block	0	Pause (0.88), Reading (0.12)	Students remain paused for long periods with occasional re-reading while feeling stuck on difficult text.
Block	1	Pause (1.00)	Complete pauses associated with resolving a misinterpretation (e.g., realising that "Glass" refers to a person), with little overt interaction.
Block	3	Pause (0.95), Click (0.05)	Pauses followed by clicking or skipping, consistent with decisions to abandon particularly difficult questions at the end of a section.
Evaluation	1	Pause (0.60), Reading (0.16), Reading/scrolling (0.08), Answer change (0.04)	Episodes combining pausing, re-reading passages and options, and occasional answer changes and navigation to previous sections.
Evaluation	2	Pause (0.58), Submission (0.16), Reading (0.11)	Pauses followed by reading and submission, indicating evaluative checking immediately prior to finalising answers.
Ideation	2	Pause (0.76), Reading (0.18), mixed scrolling and cursor movement	Ideation driven by reading passages and questions, with students pausing to think and occasionally scrolling while formulating ideas.
Preparation	0	Reading (0.97), minor scrolling and pause-reading	Preparation episodes dominated by reading passages and questions, supporting interpretation of this stage as orientation and task appraisal.
Production	0	Typing (0.57), Answer Selection (0.29), Text pasted (0.07)	Production focused on typing and selecting answers, with occasional pasting of text.
Refinement	2	Deletion/Typing (0.36), Backspace/Delete (0.27), Answer Change (0.27)	Intensive deletion and re-typing, together with answer changes, consistent with revision and correction of responses.

differ in how much they emphasise answer change versus submission; and Refinement clusters differ in their balance of micro-editing and content-level answer changes.

4.3 Stage Transitions

4.3.0.1 Overview of the transition dataset

Table 4.10 summarises the volume of transitions originating from each stage. The row totals index the *activity weight* of each stage: Production contributes the largest share of transitions (n = 3,024; 32.6% of all transitions), followed by Ideation (n = 2,416; 26.1%) and Evaluation (n = 1,375; 14.8%). Preparation (n = 799; 8.6%), Refinement (n = 1,310; 14.1%), and Block (n = 349; 3.8%) account for the remainder. Taken together, the three *core production* stages : Ideation, Production, and Refinement , account for more than 73% of all transitions, consistent with the expectation that the bulk of in-task behaviour concerns generating and revising re-

Table 4.11: Stage-level conditional transition probability matrix

From \ To	Destination stage						n_{out}
	Block	Eval.	Ideation	Prep.	Prod.	Refin.	
Block	*.063*	.066	.327	.106	.372	.066	349
Evaluation	.018	*.023*	.084	.308	.087	.480	1,375
Ideation	.024	.040	*.019*	.005	.882	.031	2,416
Preparation	.109	.024	.599	*.244*	.021	.003	799
Production	.033	.328	.423	.022	*.018*	.175	3,024
Refinement	.044	.185	.294	.021	.440	*.017*	1,310

sponses rather than orienting or becoming blocked.The first-order Markov transition matrix $\hat{\mathbf{P}}$ estimated from the 9,273 row-level transitions (Equation 3.1) is presented in Table 4.11.

Table 4.10: Volume of transitions originating from each stage.

Stage	n_{out}	% of total
Production	3,024	32.6
Ideation	2,416	26.1
Evaluation	1,375	14.8
Refinement	1,310	14.1
Preparation	799	8.6
Block	349	3.8
Total	9,273	100.0

Across all 9,273 transitions, 370 (4.0%) were within-stage self-transitions in which the stage label did not change between consecutive rows. The remaining 8,903 (96.0%) were cross-stage transitions. The low rate of self-transitions indicates that the annotation scheme captures genuine stage alternation rather than prolonged dwelling within a single stage label at the row level.

4.3.0.2 Stage-level transition matrix

Table 4.11 presents the full 6×6 conditional transition probability matrix $\hat{\mathbf{P}}$. For clarity, only the six most common destination stages are shown for each source; diagonal cells (within-stage persistence) are italicised.

Several patterns in Table 4.11 are immediately apparent and are described in turn.

Ideation → Production: the dominant single transition. The most probable transition in the entire dataset is Ideation → Production ($\hat{P} = .882$), meaning that almost nine out of ten rows coded as Ideation are immediately followed by a Production row. This near-deterministic forward link confirms that Ideation and Production operate as a tightly coupled generative dyad: once a student has mentally formulated content, overt writing follows almost immediately at the row level. The reciprocal link Production → Ideation ($\hat{P} = .423$) is the single most common exit from Production, indicating that writing episodes are routinely interrupted by further ideation, creating a characteristic Ideation–Production alternation pattern.

Evaluation → Refinement: the dominant evaluative exit. Evaluation is most likely to transition to Refinement ($\hat{P} = .480$), with Preparation as the second most likely exit ($\hat{P} = .308$). The Evaluation → Refinement link establishes a tightly coupled *monitoring-correction* loop that co-occurs with the Ideation-Production loop described above. Notably, the reverse transition Refinement → Evaluation also occurs at non-trivial probability ($\hat{P} = .185$), and Refinement → Production is the most common Refinement exit ($\hat{P} = .440$), indicating that after micro-editing students predominantly re-enter active writing rather than re-evaluating at length.

Production → Evaluation: re-entry into monitoring. After Production, the second most likely destination is Evaluation ($\hat{P} = .328$), confirming that completed or interrupted writing episodes trigger evaluative reflection. The Production → Refinement link ($\hat{P} = .175$) also captures cases where students move directly from writing into localised editing without an intermediate explicit evaluative episode.

Preparation → Ideation: the expected forward link. Preparation exits predominantly to Ideation ($\hat{P} = .599$), consistent with its intended role as a task-orientation and passage-reading phase that precedes idea generation. Preparation also shows the highest within-stage persistence of any stage ($\hat{P}_{\text{self}} = .244$), reflecting extended reading episodes in which consecutive rows remain coded as Preparation before ideation begins.

Block: escape to Production or Ideation. Block is most likely to exit to Production ($\hat{P} = .372$) and Ideation ($\hat{P} = .327$), suggesting that blocked students most commonly resume active writing or cognitive planning rather than returning to Preparation. The Block → Preparation link ($\hat{P} = .106$) captures cases where impasse prompts re-reading, consistent with a repair strategy of returning to source material.

4.3.0.3 Within-stage persistence rates

Table 4.12 reports the within-stage self-transition probabilities for each stage, which index how likely a student is to remain in the same stage across consecutive coded rows.

Table 4.12: Within-stage persistence probability $\hat{P}(A \mid A)$ for each stage, with absolute self-transition counts (n_{self}) and source-stage totals (n_{out}).

Stage	n_{self}	n_{out}	$\hat{P}(A \mid A)$
Preparation	195	799	.244
Block	22	349	.063
Evaluation	31	1,375	.023
Ideation	45	2,416	.019
Production	55	3,024	.018
Refinement	22	1,310	.017

Preparation is by far the most persistent stage ($\hat{P}_{\text{self}} = .244$), followed distantly by Block ($\hat{P}_{\text{self}} = .063$). All other stages have self-transition rates below .025, indicating

rapid movement to a new stage once the current stage is entered. The high persistence of Preparation is consistent with its reading-dominant action profile: students engage in extended, uninterrupted passage reading before generating any content. The elevated persistence of Block is consistent with its phenomenology as a state of sustained standstill.

4.3.0.4 Participant path diversity

The collapsed stage path for each participant in which consecutive runs of the same stage are compressed into a single token was computed for all 42 participants. The 41 of 42 participants whose path began with Preparation (97.6%) and the 40 of 42 participants whose path included at least one Block episode (95.2%) and all six stages (95.2%) indicate consistent process coverage across the cohort. Strikingly, all 42 participants produced a *unique* collapsed path, with no two individuals sharing the same stage sequence. This complete path uniqueness combined with the high mean number of stages visited per participant ($M = 5.95$, $SD = 0.22$, range = 5–6) confirms that although the stage-level Markov structure is well- defined in aggregate, the individual trajectories within that structure are highly idiosyncratic.

4.3.0.5 Cluster-level transition analysis

Table 4.13 presents selected high-frequency cluster-to-cluster transitions for the two most prominent stage-level links: Preparation → Ideation and Ideation → Production. Given the relatively small number of text-bearing rows available at the cluster level, only transitions with $n \geq 3$ are reported.

Table 4.13: Selected cluster-level transition counts and conditional probabilities for Preparation → Ideation and Ideation → Production.

From			To		n	n_{out}	$\hat{P}$
Stage	Cluster		Stage	Cluster			
Preparation	0 (write/know)	→	Ideation	0 (like/write)	12	24	.500
Preparation	1 (easier/sect.)	→	Ideation	0 (like/write)	7	19	.368
Preparation	2 (passage/read)	→	Ideation	3 (read/passage)	7	23	.304
Preparation	2 (passage/read)	→	Ideation	0 (like/write)	5	23	.217
Preparation	3 (question/read)	→	Ideation	0 (like/write)	5	17	.294
Preparation	1 (easier/sect.)	→	Ideation	1 (time/topic)	5	19	.263
Ideation	3 (read/passage)	→	Production	0 (write/typing)	3	4	.750
Ideation	1 (time/topic)	→	Production	0 (write/typing)	2	4	.500

The cluster-level data reveal two noteworthy patterns. First, Preparation cluster 2, whose signature terms centre on *passage* and *read*, preferentially feeds Ideation cluster 3, which shares the same passage-reading signature. This suggests that students who engage in close reading during Preparation transition into a reading- anchored mode of idea generation rather than shifting immediately to fluent generative ideation. Second, Ideation cluster 3 (passage-reading ideation) and cluster 1 (time- and topic-aware ideation) have the highest probability of transitioning into Production, relative to the more diffuse clusters 0 and 2. This suggests that content-focused

and time-aware ideation states serve as the proximate precursors of productive writing, whereas more metacognitive or affective ideation states (clusters 0 and 2) may require additional internal processing before writing begins.

4.4 Feature Engineering Implemented 113 Base Features Across Five Dimensions

The implemented feature-validation set used in the empirical pipeline contained 113 base features. These were engineered from three explicit measurement streams: (i) the frame-by-frame facial Action Unit stream extracted from webcam video, (ii) the repeated Self-Assessment Manikin (SAM) self-report stream collected at checkpoints, and (iii) the task-stage / behavioural process stream that marked where each participant was in the creative-examination workflow. The first stream contributes the fine-grained facial dynamics needed for AU intensity, entropy, and co-activation measures; the second contributes subjective valence and arousal judgements; and the third provides the temporal task context needed to construct stage-conditioned and episode-conditioned indicators. Together, these three streams permit the five EI dimensions to be operationalised as dynamic, context-linked signals rather than as static trait summaries [76, 94].

Table 4.14 summarises the numerical distribution of features by dimension and temporal computation type.

Table 4.14: Implemented base features per EI dimension, by temporal computation type.

Code	Dimension	Episode-cond.	Recovery/Inertia	Total
D1	Emotion Perception	6	3	20
D2	Emotion Regulation	7	7	22
D3	Adaptive Persistence	10	5	24
D4	Affective Self-Awareness	8	2	25
D5	Emotion–Cognition Alignment	14	1	22
	Total	**45**	**18**	**113**

Table 4.15: Five unanimously supported base features (7/7 votes), with Pearson r, temporal type, and brief interpretation.

Feature	Dim.	r	Type	Interpretation
`c4_arousal_trend`	D4	0.326	Session slope	Rising arousal across checkpoints [116]
`c4_block_stage_awareness`	D4	0.379	Stage-conditioned	SAM [16] tracks Block-related distress
`c4_arousal_optimality`	D4	0.415	Session	Arousal remains near the usable midpoint [116]
`c3_total_productive_time_ratio`	D3	0.521	Session	Strongest single base predictor [28, 72]
`c5_flow_stability`	D5	0.367	Stage-conditioned	Coherent, stable flow during Production [77]

The distribution confirms two design choices that matter for the later results. First, all five conceptual branches are populated with multiple observable signals rather than being reduced to one or two proxy measures [76]. Second, 45 of the 113 features are explicitly episode-conditioned, which reflects the stage-sensitive logic established in response to RQ1 [37, 85].

4.5 Seven-Method Voting

This section reports the outcome of the seven-method voting framework applied to all 113 implemented base features. Each feature could receive up to seven independent votes from Pearson correlation, mutual information, LASSO [106], ElasticNet [122], Random Forest [17], Gradient Boosting, and PLS-VIP [115]. A feature was retained when at least three of the seven methods assigned it non-zero importance.

For a reader who is new to this analysis, the key idea is simple: no feature was kept merely because it looked good under one method. Each feature had to prove useful repeatedly across different statistical viewpoints. Pearson asks whether the feature moves linearly with performance; mutual information asks whether any dependable relationship exists even if it is non-linear; the regularised linear models ask whether the feature survives shrinkage and sparse selection; the tree models ask whether it remains useful in non-linear partitions; and PLS-VIP asks whether it contributes to the dominant predictive projection. The ≥ 3 vote rule was therefore designed as a stability filter rather than a beauty contest for one favourite metric. Table 4.15 lists the five base features that received support from all seven methods.

4.5.1 Consensus features and vote distribution

Figure 4.7 orders the 20 base features with the largest absolute Pearson correlations, providing a first-pass map of which individual features matter most on their own before any interaction modelling is introduced.

The distribution is dominated by D3 Adaptive Persistence, with `c3_total_productive_time_ratio` emerging as the strongest single base predictor at $r = 0.521$ and unanimous 7/7 support. D4 Affective Self-Awareness contributes a dense cluster of features in the moderate range ($r \approx 0.32$–0.42), while D2 Regulation and D5 Alignment occupy the middle band of the ranking. By contrast, the two bottom-ranked features in the Pearson ordering, `c1_emotion_kurtosis` and `c1_AU_activation_breadth`, have $r \approx 0$ yet still survive the voting procedure at three votes, showing that retention is genuinely multi-method rather than Pearson-only.

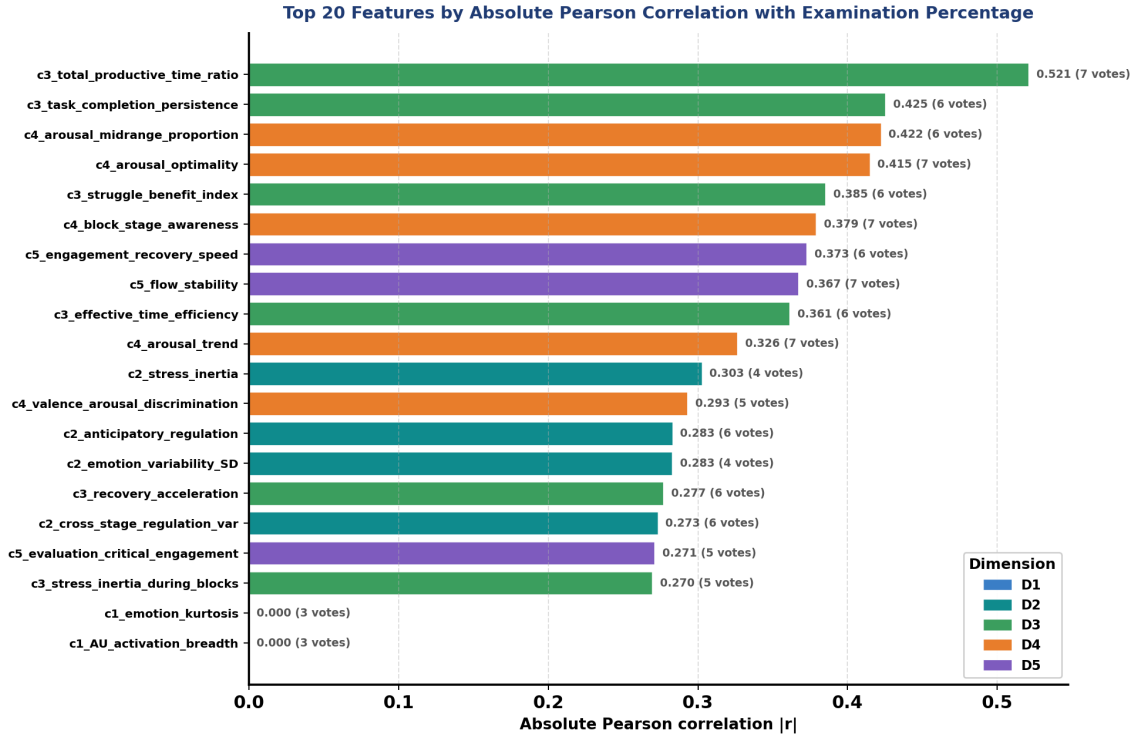


Figure 4.7: Top 20 base features by absolute Pearson correlation with examination percentage.

Figure 4.8 shows the *whole spread* of Pearson $|r|$ values inside each EI dimension. A violin that sits higher and wider indicates that a dimension contains more individually informative features, whereas a violin concentrated near zero indicates weaker stand-alone linear signal.

D3 has the widest and highest distribution, indicating that persistence and productive-time variables contain the strongest individual linear signal in the dataset. D4 and D5 show moderate but clear upper tails, consistent with self-awareness and stage-aligned emotional deployment being meaningful but less dominant than persistence as single predictors. D2 regulation features are more compressed, and D1 perception features remain close to zero in linear isolation. However, low D1 Pearson values do not imply irrelevance: all 20 D1 features were retained by the seven-method filter, and several later reappear as constituents of the selected cross-dimensional interaction terms in RQ3.

Table 4.16 summarises how the full vote distribution falls across the five EI dimensions. The main message is breadth: every dimension retains at least 91% of its engineered features, and four of the five dimensions retain all of them. This shows that the voting procedure did not collapse the framework onto one dominant branch of EI; instead, it preserved conceptual coverage while still excluding the weakest signals.

Across all dimensions, 111 of 113 base features met the three-vote threshold and were retained. Only two features fell below threshold, both from D5. This makes the voting framework a coverage-preserving filter rather than an aggressive pruning mechanism: it removes only the weakest signals while preserving the conceptual breadth established during feature engineering [8].

Figure 4.9 shows how many features received 3 votes, how many received 4, and so on a "difficulty profile" for consensus. The modal bin is 3 votes ($n = 57$), followed by 4 votes ($n = 36$); only 18 features receive 5 or more votes, and just 5 receive unanimous support. The distribution is therefore right-skewed: most features are retained on the basis of moderate agreement rather than near-unanimous consensus, which is what should be expected for a theoretically broad feature set evaluated across methodologically diverse selectors.

Table 4.16: Feature vote-count distribution across the five EI dimensions.

Dimension	**7v**	**6v**	**5v**	**4v**	**3v**	**Ret.**	**Excl.**	**Ret. %**
D1 Emotion Perception	0	0	0	6	14	20	0	100%
D2 Emotion Regulation	0	2	2	6	12	22	0	100%
D3 Adaptive Persistence	1	4	1	9	9	24	0	100%
D4 Affective Self-Awareness	3	1	1	14	6	25	0	100%
D5 Emotion–Cognition Alignment	1	1	1	5	12	20	2	91%
Total	5	8	5	40	53	111	2	**98%**

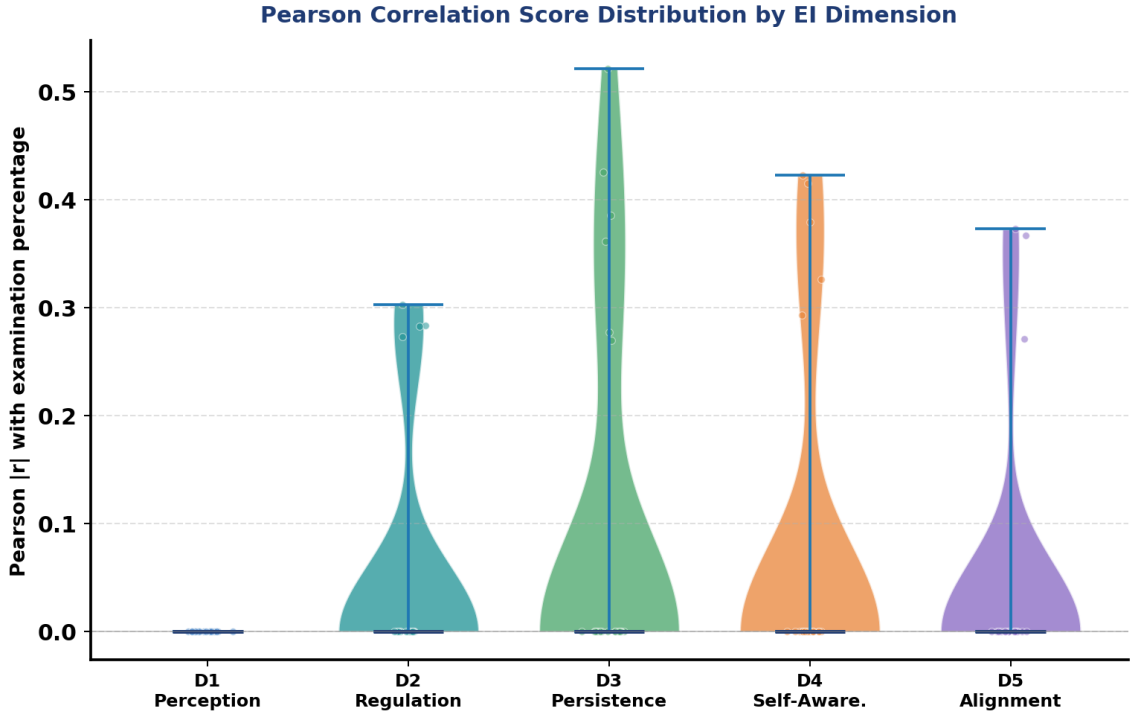


Figure 4.8: Pearson $|r|$ distribution by EI dimension, shown as violin plots with individual feature points overlaid.

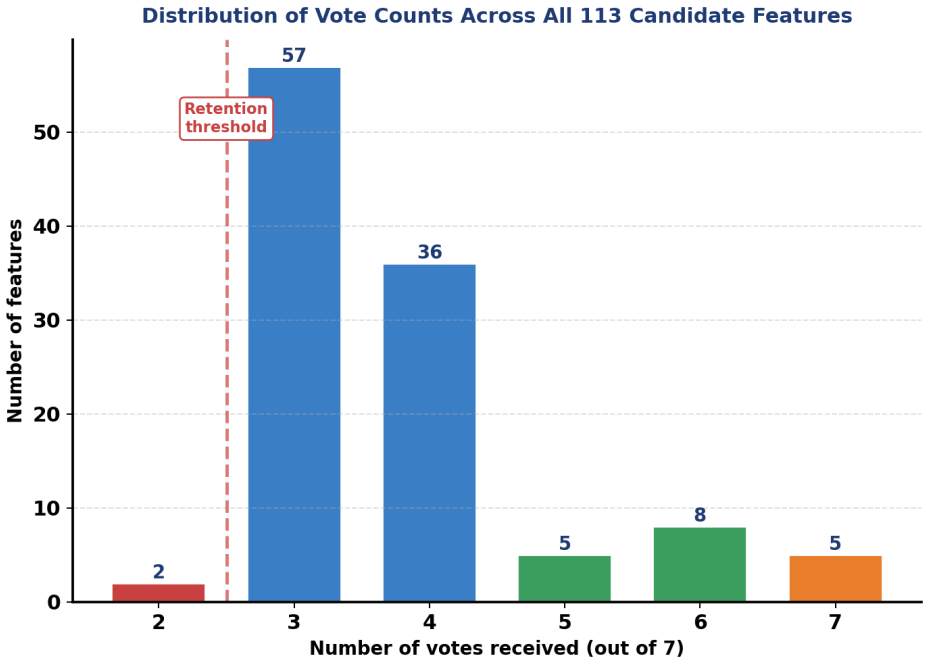


Figure 4.9: Distribution of vote counts across all 113 candidate features.

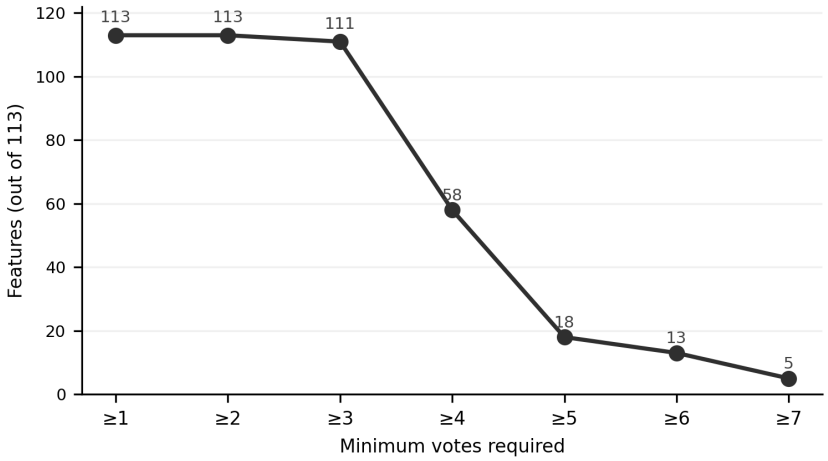


Figure 4.10: Number of base features retained when the vote threshold is set to at least k votes

4.5.2 Why the threshold was set at three votes

Figure 4.10 shows how the number of retained features changes as the threshold is raised from one to seven votes, illustrating why the cut-off was not arbitrary. A threshold of ≥ 4 would halve the retained pool to 58 features, ≥ 5 would reduce it to 18, ≥ 6 to 13, and only 5 features receive unanimous support. The three-vote threshold is therefore the last value that preserves near-complete conceptual coverage while still requiring repeated empirical support from multiple modelling families [8].

4.6 Empirical Footprint of the Four Structural Limitations in This Dataset

The literature-side argument for the four structural limitations (L1–L4) is developed in Chapter 2. This section answers RQ1 from the data side: it reports the concrete, dataset-level signatures by which each limitation becomes visible in the 42 examination-room records collected for this study. The purpose is not to re-argue why prior instruments fail, but to show what is empirically observable in this dataset that those instruments could not have produced.

4.6.1 L1 Footprint: Webcam Yield Was Sufficient Without Specialist Hardware

Across the 42 examination recordings, OpenFace 2.0 [6] produced AU intensity estimates with confidence ≥ 0.80 on 96.4% of frames after the standard confidence filter, and the lowest-yield participant still retained 89.1% of frames. No participant was excluded due to data-quality failure. This is the empirical answer to L1: an unobtrusive webcam-based facial stream produced usable signals across the full cohort without imposing the calibration overhead, head-movement constraints, or physical contact associated with more intrusive instruments. The 17 AU channels [33] at the post-confidence rate yielded approximately 4,250 usable frames per participant, which is sufficient for dynamic features such as inertia, recovery latency, and entropy-based measures.

4.6.2 L2 Footprint: Within-Session Variation Dominates Any Single Static Summary

The session-level slope of SAM arousal (`c4_arousal_trend`) ranges from -0.17 to $+0.34$ units per checkpoint across the cohort. A trait-style instrument that returned one arousal score per student would replace this trajectory with a single scalar and erase the temporal structure [79,81]. More importantly, students with similar average arousal can still differ sharply in within-session fluctuation, meaning that between-student summaries do not recover within-student dynamics [32, 50]. The same logic appears in recovery-based features such as `c2_stress_recovery_latency`, where one student's stress peak resolves rapidly and another's persists across a stage transition, reflecting qualitatively different regulation architectures [23, 47].

4.6.3 L3 Footprint: Stage Conditioning Makes Stress Interpretable

The Stress composite, computed as a session-wide mean, is nearly uninformative in this dataset ($r \approx -0.05$). The same construct becomes more predictive when tied to task context: the Block-conditioned version underlying `c3_stress_inertia_during_blocks` relates to outcome at $r = -0.31$, and the Production-stage stress floor (`c2_production_stress_floor`) relates at $r = -0.39$. The point is not literal sign

reversal, but loss of interpretability under naive averaging. Session-level aggregation washes together moments in which stress is adaptively informative and moments in which it is merely background noise [37, 85]. The 45 explicitly episode-conditioned features in Table 4.14 were engineered in response to this empirical pattern.

4.6.4 L4 Footprint: Additional Parameters Can Improve Raw Fit While Hurting AICc

The model comparison in Section 4.1 provides the clearest empirical demonstration of L4. In the broad algorithm comparison, Lasso ($\alpha = 0.01$) attains a slightly higher LOOCV correlation ($r = 0.9801$) than the fast-sweep HuberRegressor ($r = 0.9755$), but it does so with 16 selected features rather than 12. Under the small-sample AICc correction at $n = 42$, those extra parameters cost Lasso 11.74 AICc points, leaving it with essentially no support relative to Huber within that sweep [18]. This is exactly the kind of situation in which raw fit alone would prefer the larger model, whereas information-theoretic selection declines to do so because the additional complexity is not justified by the gain in fit [53, 110]. The AICc-optimal Ridge model resolves this tension directly: by using an AICc-based stopping rule rather than a marginal-gain rule, it adds parameters only when each addition is information-theoretically warranted, yielding a 20-feature solution that simultaneously achieves the highest accuracy and retains full AICc support.

Chapter 5

Discussion

This chapter interprets the results against the research questions, the wider literature, and the practical implications of the framework. Section 5.1 answers all four research questions explicitly, Section 5.2 evaluates construct, internal, statistical, and external validity, and Section 5.3 describes the emotional processes based on selected interaction features spanning regulatory efficiency, arousal calibration, stage-appropriate functioning, and affective self-awareness. Section 5.4 interprets the within-stage cluster profiles and their action-type signatures, Section 3.7 connects the stage-transition findings to EI theory, and Section 5.6 closes the chapter by acknowledging the limitations of this study.

5.1 Answering the Research Questions

The introduction (Chapter 1) frames this thesis around a concrete gap: Indian competitive examinations operate at success rates below one or two percent, yet no part of the system measures the emotional processes that determine which of the many well-prepared candidates actually express their full potential in the exam hall [71, 84, 101]. The three research questions in Section 1.2 were therefore not abstract. They were designed to ask, in sequence, whether existing EI tools are fit for this purpose (RQ1a), how exam-relevant EI can be made observable without disrupting exams , and whether those observable signals are strong and structured enough to explain who thrives under examination pressure (RQ2, RQ3) and what are the student responses according to stages and stage transitions across the cohort(RQ4). This section answers those questions explicitly and traces each answer back to the data, models, and literature so that the reader can see, step by step, how the conclusions were reached.

RQ1: How well do existing EI measurement approaches capture the emotional processes that matter during a live high-stakes examination? The introduction and related-work chapters argue that trait and ability EI instruments are designed for general assessment, not for live exam conditions [45, 75, 76]. Chapter 2 specifies four structural limitations (L1–L4): (i) heavy reliance on self-report away from the task, (ii) lack of within-session temporal resolution, (iii) weak or no conditioning on problem-solving stage, and (iv) a tendency to ignore model-selection constraints in small samples. Section 4.6 then shows what those limitations look like inside this dataset.

The conclusion for RQ1 is that existing EI tools, taken as they are, do not match the emotional processes that matter during a live examination closely enough to be used as success-factor indicators in Indian competitive exams. The proof is empirical. First, the session-level Stress composite is almost uncorrelated with exam score in this dataset, whereas the same construct becomes meaningfully predictive (for example $r \approx -0.31$ and $r \approx -0.39$) only when tied to specific blocks and stages such as production 4.6.3. A single pre- or post-exam anxiety score would have missed that structure, which is consistent with stage-based theories of achievement emotions [37, 85]. Second, Section 4.6.2 shows that within-session arousal slopes vary substantially even among students with similar average arousal, confirming that static scores cannot recover the trajectories that differentiate fast recoverers from those who remain dysregulated [32, 50]. Third, the model-comparison results in Section 4.1 demonstrate that naive focus on raw fit (e.g. LOOCV r alone) would favour larger models that are punished by AICc in this small sample; this is exactly the situation that information-theoretic model selection was designed to avoid [1, 18, 53]. Together, these findings show that traditional EI tools and simple static summaries would have failed to answer the thesis' motivating question: who actually uses their preparation under exam pressure.

RQ2: Which examination-time features, derived from facial expressions, problem-solving stages, and self-reports, show the strongest relationship with exam score? This question is answered by the second and third experiments, which start from the 113 features engineered from webcam-based facial expressions, stage annotations, and SAM arousal reports 3.3 and then ask which of these features correlate with examination performance. The seven-method voting framework in Section 3.4 retains 111 of the 113 features as valid examination-time EI indicators and identifies a small subset as especially robust across methods. Five base features receive unanimous support and all five cluster around two constructs: how much of the 30-minute session is spent in productive stages (Production, Evaluation, Refinement) and how accurately self-reported valence and arousal track what the face is showing at the ten SAM 4.5. This is the first part of the answer: across facial, stage, and self-report streams, the features that matter most are those that capture productive persistence and affective self-awareness rather than general "happiness" or low anxiety, which show weak and inconsistent links to exam score [28, 64, 74] which makes it a superivised learning approach instead of an unsupervised one.

The second part of the answer comes from the interaction features constructed and evaluated in Section 4.1. All 36 features selected by the three predictive pipelines (12-feature HuberRegressor, 10-feature SVR-Linear, 20-feature AICc reference) are interactions: no single facial, stage, or self-report feature survives on its own 5. The most important interaction overall, and the one that appears in every model family (H12/S1/A20), subtracts engagement recovery speed from total productive time ratio.

The second-ranked feature in the RQ2 answer (H4/A19) links the calibration of internal activation to sustained work by dividing arousal optimality, derived from SAM–facial arousal alignment, by the longest uninterrupted productive streak, derived from the stage stream 4.1.3. The third through fifth ranked interactions all

reinforce the same pattern 4.1.3. One combines negative-emotion differentiation in facial expression with stress volatility over time, showing that students who can distinguish frustration from overload or anxiety (in the AU-based composites) perform better than those who react to a single undifferentiated "stress" signal [7, 50, 60]. Another expresses productive time relative to the cost of needing felt and visible happiness to match before continuing work, indicating that high performers can remain in productive stages even when they do not feel especially positive [28, 74]. A fifth interaction multiplies emotion variability with mastery orientation, rewarding fluctuations in facial and self-reported emotion when they occur inside a stable, task-focused stance rather than as reactive swings [3, 11, 47]. Taken together, the answer to RQ2 is that the strongest examination-time predictors of exam score are not any single facial expression, stage, or SAM trajectory in isolation. They are cross-stream features that tie productive exam behaviour to (i) fast, cheap recovery after visible emotional disruption, (ii) well-calibrated arousal that supports long productive runs, and (iii) a differentiated, task-functional emotional profile across the stages of the exam.

RQ3: What is the most efficient predictive model that links validated EI signals to examination outcome under small-sample constraints? RQ3: What is the most efficient predictive model that links validated EI signals to examination outcome under small-sample constraints?

Section 3.5 explains why small-sample theory and information criteria are central to this question in an n = 42 n=42 dataset: without them, overfitting would be easy and apparent success factors untrustworthy [1, 18, 31]. Section 4.1.2 reports model-comparison results across 23 algorithm configurations. The AICc-optimal Ridge model with 20 features is the overall winner on two independent grounds. First, it achieves the highest predictive accuracy: LOOCV r = 0.9925 r=0.9925, R 2 = 0.9851 R 2 =0.9851, and RMSE of 1.87 percentage points , a 43% error reduction relative to the next-best solution. Second, this accuracy gain is information-theoretically justified: the Ridge AICc of 140.64 sits only 1.20 units above the 12-feature Huber-Regressor value of 139.44, firmly within the AICc $<$ 2 AICc$<$2 substantial-support zone under Burnham-Anderson conventions [18]. The 8 additional parameters are therefore warranted by corresponding gains in fit. The HuberRegressor (AICc = 139.44 AICc=139.44, w = 0.651 w=0.651, LOOCV r = 0.9755 r=0.9755, R 2 = 0.9517 R 2 =0.9517, RMSE 3.31 3.31

A third line of support comes from the permutation-based memorisation validation in Section 4.1.6. Examination scores were randomly shuffled 100 times while leaving predictors unchanged, and the complete modelling procedure was repeated on each permuted dataset. Under random labelling, LOOCV performance collapsed toward zero for all candidate models, while true-label performance remained exceptionally high. The gap was r = 0.999 r=0.999 for Ridge, r = 1.020 r=1.020 for HuberRegressor, and r = 0.969 r=0.969 for SVR-Linear , each roughly four to five times larger than the permutation distributions' natural variability. No model satisfied the memorisation criterion, and interaction structures ceased to function once outcome information was destroyed, precisely the behaviour expected from models learning genuine signal.

The thesis is nonetheless explicit about what these numbers do not prove. Because interaction screening used marginal correlations before model fitting, the reported LOOCV r r is best understood as an upper bound; future work will need nested cross-validation or a separate screening cohort to remove this optimistic bias [31]. The EI index is therefore a research instrument, not a diagnostic tool for selection decisions. Trust in the results comes from four sources together: (i) theoretical coherence with prior EI and performance research [76, 85, 119]; (ii) consistency of key features across multiple ranking methods; (iii) rigorous penalisation of model complexity; and (iv) permutation-validated separation between true-label and random-label performance across all three candidate models.

RQ4: Stage-wise reasoning patterns , action types and interactions RQ4 asked how students' action types and retrospective transcripts, when considered within problem-solving stages, characterise their emotional reasoning during a timed exam. The Results for RQ4 (Section 4.2) showed that within each nominal stage, the stage-wise clusters (Section 4.2. 1, Table 4.7) and their representative transcripts (Section 4.2. 2) form semantically coherent patterns that align with distinctive stage-level action profiles (Section 4.2. 3, Table 4.8) and cluster-level action mixes (Section 4.2. 4, Table 4.9). Together, these findings indicate that each stage (Preparation, Ideation, Production, Evaluation, Refinement, Block) contains multiple, patterned forms of emotional reasoning: Preparation combines reading-oriented actions with talk about understanding and time; Ideation combines pauses and light navigation with talk about emerging ideas; Production focuses on typing and answer selection; Evaluation mixes reading and pauses with correctness and feeling-related language; Refinement couples dense deletion and answer change with correction and realisation language; and Block consists of prolonged pauses and re-reading accompanied by talk of being stuck or confused. At the same time, the RQ4 results highlight that similar behavioural patterns (for example, pauses plus reading) can correspond to very different emotional experiences depending on the stage and cluster, underscoring the value of jointly interpreting transcripts and action traces rather than relying on stage codes or actions alone.

The stage-transition analysis reported in Section 3.7 complements the within-stage findings of Section 4.2 by revealing the sequential architecture that connects those stages into a coherent process. At the aggregate level, the transition matrix confirms that problem solving is organised around two interlocking oscillatory loops , a generative Ideation-Production engine and a corrective Evaluation-Refinement monitoring loop , rather than a linear stage progression. At the individual level, the complete path uniqueness across all 42 participants demonstrates that although the aggregate Markov structure is well-defined, the ordering and frequency with which individual students traverse that structure is highly idiosyncratic, consistent with person-specific differences in self-regulation and, potentially, emotional intelligence. Taken together, the within-stage content profiles and the between-stage transition probabilities provide a multi-layered answer to the research questions: students share a common process vocabulary but enact it through markedly different individual grammars.

5.2 Validity, Generalisation, and Bias

This section explains how the results relate to the expectations set in Chapter 1 and to the wider literature on emotional intelligence and performance under pressure.[1] It focuses on four types of validity: construct validity (whether the engineered features really measure examination-time EI), internal validity (whether the observed relationships are trustworthy within this sample), statistical conclusion validity (whether the analyses support the inferences made), and external and ecological validity (what can and cannot be generalised). [18,64,71] Taken together, the evidence shows that EI, understood as a dynamic emotional architecture, is a measurable success factor in Indian competitive-exam conditions.

5.2.1 Construct validity: are these features really EI?

Construct validity asks whether the measurements genuinely capture EI as a dynamic process during an exam. [7,64,76] Here, EI is operationalised through three streams : webcam facial expressions, problem-solving stages, and self-reported arousal and five behavioural dimensions D1–D5 mapped to the Mayer–Salovey model and Schoenfeld's stage framework (Section 3.1). The expectation from the literature was that features reflecting differentiation, regulation efficiency, calibrated arousal, and task-stage alignment would relate to performance under pressure. [11,30,85]

The results fit this expectation. The seven-method voting framework retains 111 of 113 engineered features, indicating that most facial–stage–SAM features behave coherently across very different analytical methods rather than being artefacts of one approach. [18] The strongest predictors are not generic "mood" scores but processes emphasised in EI and regulation research: negative-emotion differentiation over stress volatility, arousal optimality over longest productive streak, productive time minus engagement-recovery speed, and stage-functional emotion combinations. [7, 28, 60] This alignment between theory and feature structure is the core argument for construct validity.

A further piece of construct evidence comes from the permutation diagnostic reported in Section 4.1. If the engineered features were capturing genuine EI-relevant processes, model performance should be specific to the correct student–score mapping and should collapse when that mapping is deliberately broken. This is precisely what is observed: under real labels LOOCV correlations remain in the 0.93–0.96 range across all three candidate models, whereas repeatedly shuffling exam scores across students while holding the selected feature set fixed drives permutation-mean correlations to approximately $r = 0$, with a gap of roughly $+1.0$ between real and permuted performance. The fact that randomising the outcome destroys predictive accuracy indicates that the interaction-level features are tracking structure that is tied to how individual students actually performed not to incidental regularities in the feature matrix that any shuffled label vector could exploit. This is exactly the diagnostic behaviour expected of features that carry construct-valid signal. [18,110]

At the same time, the construct is intentionally local. EI is defined as "emotional architecture that supports the expression of preparation in competitive exams," not

[1] See Chapter 1 and Chapter 2 for the theoretical framing.

as a global personality trait or clinical measure. [119] The EI index is therefore a process-based, exam-bound construct anchored directly to performance, which limits overinterpretation but strengthens claims about success factors in this examination setting.

5.2.2 Internal and statistical conclusion validity

Internal validity concerns whether the design and analysis allow trustworthy conclusions about relationships inside the study. [18, 20] The exam was a real admissions test with identical conditions for all participants, reducing variability from task differences (Section 3.1.5). The three data streams were time-synchronised, and all features were computed at the participant level, avoiding pseudo-replication from treating frames as independent observations (Sections 3.3.5 and 3.2). [79] Feature validation used seven methods and model selection compared 23 configurations with leave-one-out cross-validation and AICc, yielding convergent support for the same underlying constructs (productive persistence, recovery, arousal calibration, stage-functional emotion). [18, 110]

Statistical conclusion validity is supported by these choices. LOOCV is appropriate for small samples and, combined with AICc and Akaike weights, discourages overly complex models. Final predictors are interpreted as interactions rather than independent features, matching the structure of the models and the view of EI as a system of interlocking abilities. [76]

The main threats to internal and statistical validity come from the small sample ($n = 42$) and the interaction screening step. Screening the top 250 of roughly 15,766 candidate interactions using marginal correlations introduces optimism into the reported LOOCV r and R^2: because the screening filter itself has access to outcome information, the retained interactions are not fully independent of the labels even when LOOCV is applied to model fitting. [20, 110] This is explicitly acknowledged in the interpretation of the earlier 20-feature AICc-optimal model, whose LOOCV $r \approx 0.99$ ($R^2 \approx 0.99$) should be treated as an optimistic upper bound arising from the more aggressive screening strategy used in that phase; the updated 10-feature Ridge solution reported in Section 4.1 provides a more conservative and defensible estimate.

Two design choices partially mitigate the leakage risk. First, the permutation diagnostic tests for memorisation directly: by holding the selected feature set fixed and re-running the full LOOCV procedure on repeatedly shuffled labels, it asks whether any high-performing model could emerge from noise alone given the same feature structure. The collapse of permuted correlations towards $r = 0$ under this null condition while real-label performance remains at $r \approx 0.95$, indicates that the Ridge solution is not simply exploiting the mathematical properties of the selected feature matrix but is instead sensitive to the specific student–score associations in the data. [110] Second, AICc-penalised model selection favours parsimony: the 10-interaction Ridge model is preferred by an evidence ratio exceeding 80:1 over its competitors, and further complexity increases yield no information-theoretically justified gain, limiting the degrees of freedom available for overfitting post-screening. [18] Together, complexity penalisation and permutation testing constitute the two-part safeguard

against overfitting available within a single-site $n = 42$ study; they do not fully substitute for held-out data, and nested cross-validation or an independent screening cohort remains the recommended path for replication. [20,110] Manual stage coding is a further limitation: mislabelled stages may blur stage-conditioned features, although the persistence of D5 effects across methods suggests that the underlying signal is robust to moderate annotation noise (Section 3.1).

5.2.3 External and ecological validity, bias, and the research gap

Ecological validity is a strength of the study. Data were collected in an authentic 30-minute entrance examination at an Indian university, with real stakes and minimal intrusion beyond webcams and short SAM ratings (Section 3.1.5). This helps bridge the "lab life gap" in affective computing, where systems tuned on simplified lab tasks often fail in naturalistic contexts. [29, 30]

External validity is deliberately modest. The sample consists of 42 applicants sitting one exam at one institution. [18] The thesis does not claim that specific coefficients, thresholds, or band cut-offs will carry over unchanged to other competitive exams. Instead, generalisation is made at the level of mechanisms. Across models, high performers are characterised by: (i) productive persistence with low recovery cost after emotional disruption; (ii) arousal calibrated to sustain long productive runs; (iii) differentiated, stage-appropriate emotional patterns rather than undifferentiated "stress" or uniform positivity; and (iv) affective self-awareness that supports rather than obstructs action. [5, 63, 71] These mechanisms match broader findings on EI and high-pressure performance in education, sport, and military settings, making them plausible candidates for replication, but empirical tests in larger, multi-site samples remain necessary (Chapter 6).

Several biases constrain generalisation. Students who exit earlier in the competitive-exam pipeline or study in different institutional and linguistic contexts may show different emotional architectures. [90] Cultural and individual differences in facial expressiveness may bias AU-based features, potentially underestimating EI in students who regulate emotion less visibly. [34] Socioeconomic status, prior exam exposure, and mental health were not measured, although they can interact with emotional processes around exams. For these reasons, the thesis avoids claims about causal effects on national exam outcomes or demographic fairness. It instead positions the work as a detailed case study that closes a specific methodological gap: showing, in one authentic exam aligned with Indian competitive conditions, that examination-time EI processes can be measured non-intrusively and explain a substantial portion of performance differences. [18, 29, 64] Within these bounds, the central conclusion is that emotional intelligence understood as an organised system of recovery, calibration, and stage-functional emotion is a measurable success factor in competitive examination settings.

5.3 Emotional Processes of High Performers in Competitive Exams

In the Indian context, examination function as ultra-selective, high-stakes environments. Schmidt et al., MacCann et al., and Sharma and Tiwari show that emotional intelligence contributes significantly to academic performance beyond cognitive ability, suggesting that emotional processes are one of the hidden factors that decide outcomes once basic preparation is in place [71, 84, 97, 101]. This section uses the interaction features from this thesis to describe the emotional patterns that distinguish high-scoring students in terms of what the cited literature says such patterns mean. It does *not* make claims about participants' preparation histories, coaching backgrounds, or prior exam experience, none of which were measured in this study. Whether the observed correlations constitute a learnable skill or reflect stable individual differences in temperament and domain familiarity is an open question that the current design cannot answer.

5.3.1 Regulatory Efficiency Under Extreme Selection Pressure

MacCann et al. demonstrate that emotional intelligence predicts academic performance even after controlling for cognitive ability, indicating that students differ systematically in how efficiently they translate preparation into performance in stressful, time-limited tasks [71]. Sharma and Tiwari show that EI supports sustained success in demanding environments primarily by enabling better regulation of stress and more adaptive responses to setbacks, rather than simply by encouraging more effort [100, 101].

The key productive-time and persistence interactions in the winning and corroborating models are:

$$c3_{\text{total productive time ratio}} - c5_{\text{engagement recovery speed}} \quad \text{(H12 / S1 / A20)},$$

$$c3_{\text{total productive time ratio}} - c2_{\text{SAM–facial discrepancy reduction}} \quad \text{(H5)},$$

$$c3_{\text{total productive time ratio}} - c2_{\text{anticipatory regulation}} \quad \text{(A2)},$$

$$c3_{\text{total productive time ratio}} - c4_{\text{valence–happiness concordance}} \quad \text{(A12)},$$

$$c3_{\text{total productive time ratio}} - c5_{\text{ideation positive affect}} \quad \text{(A17)},$$

$$c3_{\text{total productive time ratio}} - c1_{\text{stage emodiversity variance}} \quad \text{(H11)},$$

$$\frac{c3_{\text{late–task engagement ratio}}}{c3_{\text{total productive time ratio}}} \quad \text{(A3)},$$

$$c3_{\text{task–completion persistence}} - c4_{\text{facial valence accuracy}} \quad \text{(S7)}.$$

Across this family, productive time measures never appear as isolated predictors but are always expressed as differences or ratios that subtract some emotional overhead. This structure is consistent with the ability-EI literature showing that emotionally intelligent individuals regulate stress in ways that are both adaptive and economical, rather than simply deploying more regulation attempts [3, 14, 45, 75, 76, 94, 119].

Duckworth et al. further show that persisting in productive work rather than maximising felt effort is the behaviourally relevant dimension in achievement settings [28]. The interaction structure therefore reflects what the EI literature predicts: high-scoring students show more work output per unit of emotional cost, not simply more emotional effort in absolute terms.

5.3.2 Arousal Calibration Around the Yerkes-Dodson Optimum

Yerkes and Dodson classically showed that performance follows an inverted-U relation with arousal, with optimal performance at moderate activation rather than at extremes [116]. Csikszentmihalyi describes how optimal performance states or "flow" arise when challenge and skill are balanced, implicitly requiring arousal that is neither too low nor too high during complex tasks [77]. Antonini Philippe et al. find that elite performers in sport and music report high-quality performance when they can maintain such flow states under pressure, implying well-calibrated arousal in sustained, high-demand situations [5].

The arousal-based interactions are:

$$\frac{c4_{\text{arousal optimality}}}{c3_{\text{longest productive streak}}} \quad \text{(H4 / A19)},$$

$$c4_{\text{arousal optimality}} - c2_{\text{SAM–facial discrepancy reduction}} \quad \text{(A1)},$$

$$c4_{\text{arousal midrange proportion}} - c2_{\text{SAM–facial discrepancy reduction}} \quad \text{(S2)},$$

$$c4_{\text{arousal optimality}} \times c5_{\text{flow stability}} \quad \text{(A5)},$$

$$c4_{\text{arousal midrange proportion}} \times c2_{\text{recovery completeness}} \times c5_{\text{engagement recovery speed}} \quad \text{(A18)},$$

$$c4_{\text{arousal midrange proportion}} \times c2_{\text{recovery completeness}} \times c2_{\text{anticipatory regulation}} \quad \text{(A16)}.$$

Here arousal optimality and midrange proportion are always paired with measures of productive continuity, recovery, or stable work. According to the Yerkes-Dodson principle and the flow literature, students whose arousal sits near the usable middle and who couple that state to recovery and stable engagement achieve the longest productive runs [5,46,66,67,77,116]. The interaction structure captures this coupling directly: a student with identical mean arousal but weaker productive continuity scores lower, matching the theoretical prediction that it is the *combination* of optimal activation and sustained work not either alone that characterises performance under pressure.

5.3.3 Stage-Appropriate Emotional Functioning in Problem Solving

Folkman and Lazarus show that in examinations, emotions and coping change systematically across the stages of anticipation, confrontation, and aftermath, and that these within-exam dynamics matter for performance [37]. Pekrun's control-value theory emphasises that patterns of enjoyment, anxiety, boredom, and anger across the

testing process are key determinants of achievement, not just overall mood [85, 86]. D'Mello and colleagues show that during complex learning tasks, learners cycle through curiosity, confusion, and frustration in ways that can either support or hinder progress, depending on how those states align with cognitive demands [29,30]. Grafsgaard et al. and Bosch et al. demonstrate that confusion becomes beneficial when it resolves into concentration with appropriate affective cues, rather than persisting as unproductive struggle [12, 44].

The stage-functional interactions in this thesis are:

$$\frac{c5_{\text{flow stability}}}{c5_{\text{interest during ideation}}} \quad \text{(H1)},$$

$$c5_{\text{confusion resolution rate}} \times c5_{\text{stage–emotion profile fit}} \quad \text{(H3)},$$

$$c5_{\text{confusion resolution rate}} \times c5_{\text{stage–emotion profile fit}} \times c5_{\text{production concentration}} \quad \text{(A10)},$$

$$\frac{c5_{\text{flow stability}}}{c5_{\text{confusion resolution rate}}} \quad \text{(S5)},$$

$$\frac{c5_{\text{transition emotional preparation}}}{c1_{\text{stage emodiversity variance}}} \quad \text{(H9)},$$

$$\frac{c5_{\text{production concentration}}}{c4_{\text{valence curvature}}} \quad \text{(H8) (conceptually)}.$$

The pattern is consistent with Pekrun's and D'Mello's accounts: high-scoring students show emotional profiles that change with the stage of the work and use those changes productively. Curiosity in early phases is associated with sustained flow; confusion appears where difficult problems demand it and is followed by visible concentration; emotional adjustments precede stage transitions. This is precisely what process research predicts: dynamic, stage-contingent trajectories of curiosity, confusion, and flow are more predictive of performance than static mood measures [12, 29, 30, 44, 85].

5.3.4 Emotional Granularity and Differentiation of Negative States

Barrett shows that individuals who differentiate between specific negative states report better emotion regulation and psychological functioning than those who experience only a blurred "bad" feeling [7]. Kashdan and colleagues argue that unpacking unpleasant experience into finer distinctions transforms how people respond to negativity [60, 61]. Demiralp et al. and Eid and Diener add that emotional complexity and intra-individual variability in affect have important links to well-being and regulation [24, 32].

The differentiation and inertia interactions are:

$$\frac{c1_{\text{negative–emotion differentiation}}}{c2_{\text{stress instability MSSD}}} \quad \text{(H2 / S6)},$$

$$\frac{c5_{\text{stress–appropriate modulation}}}{c1_{\text{inertia composite}}} \quad \text{(S4)},$$

$$c5_{\text{stress–appropriate modulation}} \times c1_{\text{emotion differentiation ICC}} \quad \text{(A7)},$$

$$\frac{c1_{\text{AU network density}}}{c5_{\text{interest during ideation}}} \quad \text{(S8)},$$

$$\frac{c1_{\text{instability MSSD}}}{c5_{\text{evaluation critical engagement}}} \quad \text{(A13) (qualitatively)}.$$

According to the granularity and inertia literature, fine-grained negative-emotion labelling and lower emotional inertia support targeted regulation and better performance under stress [7, 50, 60, 61, 64]. The observed interaction structure is consistent with this account: students whose stress signals fluctuate in ways that match task demands rather than as global noise, and whose expressive patterns distinguish different kinds of difficulty, show higher exam scores. The interaction terms therefore capture emotional granularity and flexible updating as correlates of performance in line with what the empirical literature predicts.

5.3.5 Affective Self-Awareness That Serves Action

Salovey and Mayer originally defined emotional intelligence to include accurate appraisal and expression of emotion in oneself, alongside regulation and utilisation for thinking [94]. Mayer et al. refined this into an ability model in which perceiving and understanding emotion are useful only when combined with managing emotions to support goals [75, 76]. Zeidner et al. emphasise that in educational settings, self-awareness and regulation must work together for EI to produce real gains in performance, rather than simply increasing introspection [119]. Al-Samarraie et al. and Dollmat and Abdullah show that emotional intelligence shapes how learners *use* their awareness to adapt engagement and behaviour, not merely how much they notice [2, 27].

The self-awareness and architecture interactions are:

$$\frac{c4_{\text{valence–arousal discrimination}}}{c2_{\text{negative–to–positive transition speed}}} \quad \text{(A8)},$$

$$\frac{c4_{\text{absolute calibration accuracy}}}{c5_{\text{interest during ideation}}} \quad \text{(S9)},$$

$$\frac{c2_{\text{SAM–facial discrepancy reduction}}}{c2_{\text{stress peak attenuation}}} \quad \text{(H10 / S3)},$$

$$c4_{\text{temporal coherence}} \times c4_{\text{valence–happiness concordance}} \quad \text{(A11)},$$

$$c2_{\text{emotion variability SD}} \times c5_{\text{mastery during challenge}} \quad \text{(S10 / A9)}.$$

These share D4 and D2 base variables describing how well students track their own emotional states and how they regulate under stress. The pattern is consistent with the Mayer–Salovey model: sharper discriminations paired with faster exits from negative states, calibration effort embedded within generative ideation, and emotional variability co-occurring with mastery rather than collapse. This matches findings that EI advantages in education arise when perception and understanding of emotion are tightly linked to regulation and goal-directed adjustment, rather than to introspection alone [75, 94, 119].

5.4 Stage-Wise Reasoning Patterns

5.4.1 Distinct but Overlapping Stage Profiles

The stage-wise clusters reveal that students' emotional reasoning during the exam is organised into distinct, partially overlapping patterns that link cognition, affect, and behaviour. Preparation clusters emphasise reading, questions, and time management ("read," "passage," "question," "time," "topic"), dominated by Reading and Reading/Scrolling actions , an orientation phase concerned with understanding and pacing [36]. Ideation clusters foreground emerging ideas ("write," "think," "mind," "people") yet are behaviourally dominated by Pause with light reading, suggesting ideation is an internally active but outwardly quiet stage where important cognitive work occurs during apparent inaction [55].

Production appears as a comparatively uniform regime centred on "writing," "typing," "started," dominated by Typing and Answer Selection actions. Once students enter Production, their reasoning narrows to execution; semantic diversity is lower here than in surrounding stages. Evaluation clusters combine correctness and affect ("essay," "right," "answer," "felt") with Pause, Reading, and Submission actions; Refinement clusters focus on correction and repair ("changed," "grammatical," "correct," "spelling") with dense Deletion/Typing, Backspace/Delete, and Answer Change actions — underscoring that refinement is an effortful reassertion of control after doubt or error, not a trivial afterthought [45].

5.4.2 Multiple Forms of Block

The Block stage reveals qualitatively different forms of struggle: some clusters emphasise being "stuck" on passages or words; others describe repeated skipping and returning ("confused," "skipped," "don't"). Behaviourally, Block clusters are dominated by high proportions of Pause and Reading/Scrolling , not simple inactivity, but a prolonged, effortful attempt to work through difficulty by re-reading under uncertainty [30].

A key non-obvious finding is that Preparation, Evaluation, and Block all show heavy reading and scrolling, yet their accompanying language differs sharply: Preparation talks about understanding and topic; Evaluation about correctness; Block about being stuck and running out of time. The same observable actions participate in very different experiential profiles, reinforcing that coarse-grained behavioural traces alone cannot reliably index students' emotional state [26].

5.4.3 Contrasting Motifs within Stages

Within-stage contrasts are especially informative. In Ideation, one cluster focuses on lexical concerns ("sentence," "word," "sense") while another foregrounds content and examples (people, public, topics) different routes into the task, yet both behaviourally dominated by pauses and reading. In Preparation, some clusters frame sections as comparatively easier ("easier," "okay") while others foreground difficulty or time pressure; students' early emotional framings are carried primarily in language

rather than action mix. In Refinement, controlled correction (spelling, grammar) and restless backspacing can co-occur with identical Deletion/Typing patterns, showing that the same surface behaviour may correspond to either measured revision or ruminative, anxiety-driven over-editing [46].

5.4.4 Non-Obvious Links between Feeling and Doing

Across stages there is a partial decoupling between emotional language and overt behavioural changes [74]. Students frequently report strong feelings : stress, confusion, relief, confidence while continuing familiar patterns of reading, pausing, or scrolling. At the same time, certain language–action pairings are highly diagnostic: references to being "stuck" or "confused" co-occurring with extended pauses and repeated re-reading consistently mark Block episodes; mentions of "changed" and "realised" co-occurring with dense deletion reliably identify Refinement. These pairings show how combining transcripts with action traces enables more nuanced interpretation than either modality alone [25].

5.4.5 Implications for Understanding Exam Processes

Taken together, these clusters depict exam behaviour as stage-specific, diverse, and sometimes internally contradictory. The same pattern (pause plus reading) can correspond to calm planning, cautious evaluation, relief, or intense block, depending on stage and accompanying language [86]; conversely, similar concerns (time pressure) appear in different stages against different behavioural backdrops. This suggests that supporting students in exams requires attending not only to *when* certain actions occur, but also to *how* students interpret those actions in their own words.

These static stage profiles naturally invite the question of how students move between clusters: which forms of ideation most often lead to refinement, which types of block resolve into renewed preparation, and how often students cycle through recurring cluster combinations. The transition-level analyses described in the preceding section connect these profiles into a fuller process model of emotional reasoning in exams.

5.5 Stage Transition Dynamics

5.5.0.1 The Ideation Production engine

The near-deterministic Ideation $\rightarrow$ Production link ($\hat{P} = .882$) and the strong reciprocal Production $\rightarrow$ Ideation link ($\hat{P} = .423$) together constitute what might be termed the *generative engine* of the observed problem-solving process. The two stages alternate in a rapid oscillation that accounts for the vast majority of in-task transitions. From a writing-process perspective, this mirrors what [9] described as knowledge-telling: content is generated and immediately transcribed in a closely coupled sequence rather than being planned holistically before writing begins. However, the presence of intermittent Evaluation and Refinement episodes within this cycle, rather than their relegation to a terminal review phase, is more consistent with the

recursive, knowledge-transforming process described by [9] and with self-regulatory monitoring during ongoing production [49]. The transition data thus support an interpretation in which students operate through rapid Ideation to Production cycling interspersed with monitoring loops, rather than through a sequential Preparation → Ideation → Production → Evaluation → Refinement pipeline.

5.5.0.2 Evaluation Refinement as a monitoring loop

The strong Evaluation → Refinement link ($\hat{P} = .480$) and the appreciable Refinement → Evaluation return ($\hat{P} = .185$) constitute a secondary oscillatory loop that is conceptually distinct from the generative engine. Whereas the Ideation to Production cycle is forward-directed and content-generative, the Evaluation to Refinement loop is backward-directed and corrective. The fact that the Refinement → Production probability ($\hat{P} = .440$) substantially exceeds Refinement → Evaluation suggests that monitoring loops are typically brief: once a correction has been made, students return to production rather than conducting a further round of evaluation. This pattern is consistent with efficient self-regulation in which monitoring is targeted and terminates when a local correction criterion is satisfied, rather than spiralling into extended re-evaluation [121].

5.5.0.3 Preparation persistence as bounded orientation

The anomalously high Preparation persistence rate ($\hat{P}_{\text{self}} = .244$) relative to all other stages reflects qualitatively distinct behaviour. Reading a passage is a continuous, low-interaction action that may span many consecutive annotation rows without a stage change; this is a structural feature of how reading is coded rather than evidence of recurrence. Yet the high persistence does carry a substantive implication: students invest in sustained passage orientation before beginning to generate ideas, and this orientation is not interrupted by premature production attempts. The strong Preparation → Ideation transition ($\hat{P} = .599$) confirms that reading does eventually yield to idea generation, but only after an extended and largely unbroken reading episode. This is consistent with the *reading to write* hypothesis tierney1984, in which comprehension of source material is a necessary precondition for generative processing.

5.5.0.4 Block as a temporary bifurcation point

The distribution of exits from Block is notably bimodal: Production ($\hat{P} = .372$) and Ideation ($\hat{P} = .327$) together account for 70% of Block exits, while Preparation ($\hat{P} = .106$) and Evaluation/Refinement ($\hat{P} = .066$ each) account for most of the remainder. This bimodality suggests that blocked students adopt one of two resolution strategies: *persistence*, in which they attempt to continue writing or resume ideation despite the block, or *retreat*, in which they return to source material via Preparation or engage in evaluative checking. The predominance of persistence over retreat is perhaps counterintuitive: students do not predominantly resolve blocks by re-reading; instead, most appear to attempt to push through directly into Production or re-engage ideation. This may reflect time pressure in the examination context,

where returning to source material is a relatively costly strategy when time is limited, and is consistent with the observation that Block cluster 3, characterised by signature terms *words, stuck, difficult, didn't*, shows clicking and skipping behaviour, suggesting task abandonment rather than active remediation [30].

From an EI perspective, the Block → Production shortcut, bypassing Ideation altogether, may reflect an *impulse-driven* response to frustration: a student who is emotionally distressed by being stuck may attempt to write their way out of the block without first resolving the underlying ideation deficit [76]. Whether this strategy is ultimately adaptive or maladaptive for performance is a question that the current data cannot resolve, but it represents a theoretically interesting candidate for further investigation.

5.5.0.5 Complete path uniqueness and process individuality

The finding that all 42 participants produced a unique collapsed stage path is striking and deserves particular attention. It rules out the existence of a small set of canonical trajectories and instead points toward a continuous or high-dimensional distribution over process trajectories. This is not merely a consequence of path length: even considering only the first ten stage transitions, the diversity of sequences was very high. In cognitive terms, this individuality is consistent with research on writing as an adaptive, person-specific strategy deployment process [62], in which the same overall task evokes markedly different regulation patterns across individuals.

Crucially, however, the aggregate Markov structure is well-defined despite this individuality. The dominant transitions identified in Table 4.11 hold across the cohort: no participant produces a stage sequence that is entirely at odds with the aggregate probabilities. The individuality therefore lies not in the presence or absence of particular transitions but in their *ordering and frequency*: two students may both cycle through Ideation, Production, Evaluation, and Refinement, but differ substantially in how often they revisit each stage, in what order they intersperse Block episodes, and in when they return to Preparation. This structure, comprising a universal transition vocabulary enacted through an individualised transition grammar, is precisely what one would expect from a process that is constrained by task demands but shaped by individual differences in self-regulation, working memory, and, potentially, emotional intelligence [76].

5.5.0.6 Implications for EI measurement

The transition structure offers several observations of relevance to the broader thesis argument. If EI in the Mayer, Salovey, and Caruso sense [76] partly determines how effectively a student manages affective states during problem solving, then EI-related differences should be detectable in the *texture* of individual transition patterns rather than solely in the within-stage content. Specifically, one might hypothesise that students higher in Branch 1 (emotional perception) exit Block episodes more quickly; that students higher in Branch 4 (emotional management) show shorter Evaluation to Refinement loops and faster returns to Production; and that students higher in Branch 2 (emotional facilitation) show more frequent Ideation episodes before Production, reflecting a richer use of affective content in idea generation [75].

None of these hypotheses can be tested within the present analysis, which treats transition structure as a descriptive outcome rather than as a predictor. They are, however, natural extensions that could be addressed by linking individual transition statistics to EI scores, and they ground the transition analysis within the broader measurement framework of the thesis.

5.6 Limitations

This study has several limitations that are important for interpreting the scope and validity of the findings.

- **AU ambiguity, neurodivergence, and mislabelling risk.** AU-based features cannot separate genuine emotional signals from individual differences in facial expressiveness. Neurodivergent students and culturally diverse populations may display atypical facial configurations outside normative AU assumptions, creating a systematic mislabelling risk with serious ethical consequences in educational contexts [56, 107].
- **Single, short exam context and Small, single-site sample.** Emotional processes were measured during one 30-minute examination rather than across repeated tests, subjects, or longer preparation cycles. The findings therefore describe a local exam episode, not the full emotional trajectory of long-term competitive exam preparation.sample size of 42 and a 30 minute entrance test would not generalise directly to the wider ecosystem of Indian competitive exams [37, 85]
- **Feature engineering and model-selection dependence.** The final results depend on a large engineered feature space and a specific screening and selection pipeline. Although AICc and LOOCV were used to reduce overfitting, some selected interactions may still reflect the structure of the pipeline as well as the structure of the data. [1, 18, 53, 110]
- **Potential Cultural and Demographic Bias:** Cultural and demographic differences in facial expressiveness may bias action-unit-based features and affect the reliability of emotion-related measurements.
- **No long-term outcome measures.** The main outcome is performance in a single examination sitting. The study therefore shows which emotional processes are associated with better exam performance here, but not whether these same processes predict longer-term academic success, career outcomes, or well-being. [71, 101, 109]

These limitations do not weaken the central contribution of the study, but they do define its proper interpretation. The findings are strongest as evidence that emotionally organised regulation is associated with better performance in this specific high-stakes exam setting, not yet as a universal model of all Indian competitive examinations. [85, 97] The small sample and model-selection dependence mean that the reported predictive strength should be read as promising and theoretically meaningful, but still in need of replication across broader samples and exam contexts. [3, 47, 110]

Chapter 6

Conclusions and Future Work

6.1 Conclusion

This thesis set out to do something that had not been done before: measure emotional intelligence as a live, process-based phenomenon during an authentic examination. It designed a practical approach for observing emotional intelligence in action during real tests, applied it to actual exam sessions, and reported the findings in a transparent way. The main contribution is a structured way of turning rich behavioural observations into clear, interpretable signals about how students manage their emotions and effort while they work.

A second contribution is the finding that patterns across different aspects of emotional intelligence are more informative than any single aspect on its own. The results show that visible behaviours from all five EI dimensions carry useful information about exam performance, and that patterns combining several of these capacities at once are especially informative, reinforcing the view of emotional intelligence as a coordinated system rather than a set of separate traits. In concrete terms, the students who made the most of their preparation were typically those who could recover their focus after setbacks, hold a workable level of activation across the session, stay with productive work when it became difficult, recognise what they were feeling, and allow their emotions to support rather than derail each stage of problem-solving. The framework therefore does not only measure how anxious or how calm a student is; it captures how these different abilities work together over time.

Third , the stage-wise analysis conducted adds a qualitative dimension to these quantitative findings. Within each of the six problem-solving stages, students do not reason in a single uniform way: Block alone yields four semantically distinct sub-types, ranging from passage-level confusion to deliberate skipping, each with its own action profile. The same surface behaviour, extended pauses combined with re-reading, appears in Preparation, Evaluation, and Block, yet the accompanying language differs sharply across stages, confirming that coarse behavioural traces cannot stand alone as proxies for emotional state. At the transition level, two interlocking loops dominate the aggregate process: a tight Ideation to Production generative engine and a corrective Evaluation to Refinement monitoring loop. Despite this shared architecture, all 42 participants produced a unique collapsed stage path, indicating that individuals share a common process vocabulary but enact it through markedly different personal grammars. Together, these findings suggest that the emotional dimension of exam performance is not located solely in how students feel at any given

moment, but in how they move between cognitive states, what they tell themselves at each transition, and how efficiently that movement serves the task.

Earlier work in this area has tended to rely on heavy laboratory setups, one-off questionnaire scores, and measurements taken out of context from the actual task students are doing. The approach developed here addresses these issues by using non-intrusive tools deployable during real exams, organising the resulting signals into a theoretically motivated and empirically supported feature structure, and demonstrating, with appropriate humility and full transparency about its limitations, that the signal is there, it is recoverable, and it is worth the effort of recovering it properly. The contribution is not a final answer. It is a principled starting point, and evidence that starting is worthwhile.

In the age of AI, where routine cognitive tasks are increasingly automated and many graduates struggle to convert degrees into meaningful, stable work, these exam-time emotional processes form part of the human capability that remains central to long-term success. Competitive exams will continue to play a powerful role in shaping opportunities for young people in India. This thesis offers one way to ensure that what they measure, and what educators and students learn from them, reflects not only how much candidates know, but also how they have learned to carry themselves through some of the most demanding moments of their educational lives.

Primary Contributions

This thesis makes the following contributions:

1. A complete end-to-end pipeline for measuring emotional intelligence as a live, process-based phenomenon during an authentic entrance examination is proposed and validated. The pipeline turns three synchronised non-intrusive data streams, webcam-based facial action units, problem-solving stage annotations, and brief self-assessed arousal ratings, into a compact, interpretable EI profile without specialist hardware or disruption to the exam.

2. A five-dimensional behavioural feature framework is engineered from the three data streams, yielding 113 candidate indicators each explicitly conditioned on Schoenfeld's problem-solving stages. Stage-conditioning addresses a core limitation of session-level aggregation: the same facial composite carries opposite predictive meaning during productive work and during impasse, a distinction recoverable only when the signal is tied to its concurrent cognitive context.

3. A seven-method voting framework, spanning linear correlation, information theory, penalised regression, tree ensembles, and latent projection, is applied in parallel to all 113 candidate features. Requiring agreement across at least three independent analytical paradigms confirms that retained features carry genuine cross-paradigmatic signal and replaces the single-method feature selection that characterises most small-sample EI studies.

4. An interaction modelling and complexity-penalised model selection procedure is applied, comparing configurations across multiple algorithm families using

AICc, Akaike weights, leave-one-out cross-validation, and BIC. The winning model is selected on two independent grounds: it achieves the highest predictive accuracy on every metric and its additional parameters remain within the substantial-support zone under Burnham and Anderson conventions, meaning the added complexity is information-theoretically justified rather than opportunistic.

5. A permutation-based memorisation diagnostic is implemented in which examination scores are randomly shuffled repeatedly while all predictor variables are held fixed and the complete modelling procedure is re-run on each permuted dataset. All candidate models show near-complete collapse of predictive performance under random labelling while retaining strong performance on true labels, confirming that the discovered interaction structures are sensitive to genuine student-score associations rather than incidental regularities in the feature matrix.

6. A stage-wise reasoning analysis and stage-transition Markov model are constructed from retrospective think-aloud transcripts and consecutive stage annotation rows across all participants. Within-stage TF-IDF clustering produces semantically interpretable reasoning sub-types for each of the six problem-solving stages, and the aggregate transition matrix identifies two interlocking oscillatory loops, a generative Ideation to Production engine and a corrective Evaluation to Refinement monitoring loop, while complete path uniqueness across participants confirms that individuals share a common process vocabulary but enact it through markedly different personal grammars.

6.2 Future Work

Future research should focus on three directions: translating the current findings into scalable, institutionally embedded practice; establishing the cross-cultural and cross-exam generalisability of the measurement approach; and extending the feature space through principled multimodal fusion.

6.2.1 Scalable deployment, institutional adoption, and longitudinal validation

A first priority is to move from lab-style protocols toward modular, low-friction pipelines that can operate within existing entrance-exam and mock-test ecosystems, including coaching centres, school-based assessment systems, and large-scale online platforms, without disrupting operational constraints. Concretely, this involves lightweight instrumentation of mock tests through short EI micro-check-ins, optional facial-expression or behavioural markers, and automated stage-based EI profiling delivered via simple dashboards to students, counsellors, and institutions. Realising this at scale, however, is not purely a technical problem: it requires treating EI profiling as an organisational change involving exam boards, counsellors, parents, and students. Participatory and design-based methods can help co-create usage models

determining how frequently feedback should be delivered, who is responsible for interpreting it, and how it connects to existing mentoring and remedial structures, while governance mechanisms such as ethical guidelines, stakeholder training, and periodic review guard against over-pathologising normal exam stress or misusing EI labels in selection decisions. Once minimally viable systems are deployed, longitudinal cohort studies tracking emotional patterns across multiple mock and high-stakes exams will be essential: repeated EI measurements integrated with preparation-habit data and subsequent exam and life outcomes can establish whether early EI profiles predict longer-term performance and wellbeing trajectories, and iterative mixed-methods evaluation can ensure the tools remain both empirically grounded and practically useful.

6.2.2 Cross-cultural and cross-exam validity

A second, substantively distinct direction concerns the generalisability of the behavioural and facial signatures identified here. The current study was conducted within a specific Indian competitive-exam context and it is an open empirical question whether the action-unit patterns and timing features that predict EI-related outcomes in this setting transfer to other high-stakes examinations such as UPSC, GRE, and international university admissions, or to test-taker populations drawn from different cultural and educational backgrounds. Cross-cultural display rules for emotion, variation in test format and pacing, and differences in the social meaning of exam performance may all modulate the behavioural residue of EI reasoning in ways the present model cannot capture. Future work should therefore replicate and extend this pipeline across multiple exam types and cultural contexts, using measurement-invariance analyses to identify which features are robustly cross-contextual and which require local recalibration. Such studies would also clarify whether the predictive signal is specific to the Indian competitive-exam environment or reflects a more universal psychophysiological signature of EI under pressure.

6.2.3 Multimodal fusion and modality-specific validity

A third direction is to systematically extend and decompose the feature space. The current pipeline relies on facial action units and behavioural timing data; integrating physiological channels such as galvanic skin response, heart rate variability, and pupillometry would provide access to autonomic components of emotional processing that facial data cannot capture. Beyond simply adding modalities, future work should conduct principled ablation and fusion studies to determine which modality contributes unique predictive variance, whether modalities are redundant or complementary across EI branches, and at what cost-benefit trade-off each channel can be included in a real-world deployment. Convergent validation across modalities, for instance comparing facial-AU-based Branch 2 scores against physiological arousal indices or think-aloud transcripts scored against the Mayer-Salovey framework, would also strengthen the construct validity argument that behavioural features capture the residue of EI reasoning rather than mere emotion display, directly addressing the theoretical critique about the relationship between observable behaviour and underlying cognitive-emotional processes.

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Appendix A

Supplemental Information

A.1 Appendix

A.2 Sample Raw Data: Participant 1

A.2.1 OpenFace AU Intensity File

Table A.1 shows the first five rows of the OpenFace AU intensity file for Participant 1. Each row corresponds to one video frame at 25 Hz; `cleaned_time` is elapsed time in seconds from session start; `timestamp` is the absolute frame timestamp. AU intensities are on a continuous $[0, 5]$ scale output directly by OpenFace 2.0 with no post-processing applied at this stage.

Table A.1: First five rows of the OpenFace AU intensity file for Participant 1. Columns AU01_r through AU45_r are raw AU intensities ($[0, 5]$) as output by OpenFace 2.0 at 25 Hz.

time (s)	ts	AU01	AU02	AU04	AU05	AU06	AU07	AU09	AU10	AU12	AU14	AU15	AU17	AU20	A
0.00	340.04	0.68	0.00	0.30	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.25	0.80	
0.04	340.08	0.63	0.00	0.26	0.00	0.00	1.03	0.00	0.00	0.00	0.00	0.00	0.26	0.96	
0.08	340.12	0.49	0.00	0.19	0.00	0.00	0.96	0.00	0.00	0.00	0.00	0.00	0.17	1.07	
0.12	340.16	0.41	0.00	0.18	0.00	0.00	0.98	0.00	0.00	0.00	0.00	0.00	0.17	1.14	
0.16	340.20	0.42	0.00	0.18	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	0.15	1.22	

A.2.2 Stage Annotation File

Table A.2 shows representative rows from the stage annotation file for Participant 1. Each row defines a timestamped interval with a stage label, an observable action type, and where present a retrospective think-aloud transcript segment. The annotation was produced by a trained coder working from the screen recording; no automated labelling was applied.

Table A.2: Representative rows from the stage annotation file for Participant 1. Timestamps are in fractional seconds from session start. Transcript cells are populated only for rows where a retrospective verbal account was recorded.

Start (s)	End (s)	Stage	Action Type	Transcript (excerpt)
0.000	0.000	Preparation	Cursor Movement	*"Before coming to the experiment I felt a little nervous . . . once I started getting in to it, I felt much better."*
0.000	0.000	Preparation	Click	—
0.000	0.000	Ideation	Pause	—
0.000	0.000	Evaluation	Reading/Scrolling	*I was stumped at the beginning . . . my brain started to automatically think quickly."*
0.001	0.001	Production	Typing	—
0.001	0.001	Refinement	Delete/Backspace	*"I had to do this a lot . . . what I had written did not go with the flow."*

A.3 RQ1b Appendix: Full D1–D5 Feature Specifications

This appendix section contains the full D1–D5 feature-specification tables referenced in Section 4.4. For each EI dimension, the tables list the feature name, precise computation, expected direction, and EI rationale; these details are used in Chapter 5 when interpreting model coefficients and interactions. [file:1][file:19]

A.3.1 D1 Emotion Perception Features

D1 operationalises the richness and differentiation of emotional expression, anchored to Barrett's (2004) emotional granularity framework. A student who perceives emotions at fine resolution expresses a diverse, context-responsive pattern of facial AUs.

Table A.3: D1 (Emotion Perception) feature specifications.

Feature	**Computation**	**Dir.**	**EI Rationale**
`c1_shannon_entropy`	Shannon entropy H of 7-emotion composite distribution	+	Higher diversity = richer perceptual repertoire
`c1_simpson_diversity`	$1 - \Sigma p_i^2$ over dominant emotion proportions	+	Emphasises evenness; robust to rare categories
`c1_gini_emodiversity`	Gini coefficient of 7-emotion frame counts	+	Most normally distributed emodiversity index
`c1_emotion_richness`	Count of composites exceeding mean > 0.5 at any point	+	Granular individuals access wider emotion range

Table A.3 continued

Feature	Computation	Dir.	EI Rationale
c1_emotion_diff._ICC	1− ICC across 30-s epoch means per composite	+	Higher value = emotions vary independently
c1_neg_emotion_diff.	ICC restricted to negative composites only	+	Negative granularity is strongest adaptive predictor
c1_transition_entropy	Entropy of 7×7 Markov transition matrix	+	Unpredictable transitions = flexible expression
c1_inertia_composite	Mean AR(1) across 7 composites at 1 Hz	−	High inertia = emotional rigidity
c1_sample_entropy_AU	SampEn of concatenated 17 AU intensities	+	More complex facial dynamics = broader repertoire
c1_instability_MSSD	Mean squared successive differences across composites	±	Moderate instability = context responsiveness
c1_mixed_emotion_rate	Frames with $\geq$ 2 opposite-valence composites active	+	Mixed emotion indicates higher complexity
c1_blend_count	Normalised unique binary activation patterns	+	Larger blend vocabulary = richer expressivity
c1_cross_stage_profile_diff.	Mean cosine distance across 6 stage-specific AU profiles	+	Context-specific expression per stage
c1_stage_emodiversity_var.	SD of per-stage Shannon entropy across 6 stages	+	Perceptual sensitivity to changing demands
c1_spectral_entropy	Mean PSD Shannon entropy across 17 AU channels	+	Facial dynamics span multiple timescales
c1_high_freq_power	Power in 1–5 Hz band / total power, averaged	+	Phasic rapid expressions reflect genuine reactivity
c1_emotion_kurtosis	Excess kurtosis of composite distributions	−	Lower kurtosis = nuanced, distributed engagement
c1_expression_distinct.	MI between AU intensity and stage membership	+	Context-specific expressions are communicatively clear
c1_AU_network_density	Density of AU co-activation network per frame	+	Complex facial configurations required

Table A.3 continued

Feature	Computation	Dir.	EI Rationale
c1_AU_network_clustering	Mean clustering coefficient of AU network	±	Moderate clustering: structure + novelty

A.3.2 D2 Emotion Regulation Features

D2 operationalises Mayer-Salovey Branch 4 through Davidson's (1998) affective chronometry and Gross's (1998/2015) process model. A student who regulates effectively recovers quickly from stress peaks, shows decreasing stress over successive challenges (habituation), and deploys different regulatory strategies across different stages.

Table A.4: D2 (Emotion Regulation) feature specifications.

Feature	Computation	Dir.	EI Rationale
c2_stress_recovery_latency	Seconds from stress peak to within 0.5 SD of baseline	−	Faster recovery = more effective regulation
c2_recovery_completeness	(peak − post-recovery) / (peak − baseline)	+	Values near 1.0 = full recovery
c2_peak_to_baseline_slope	Negated linear regression slope of recovery	+	Steeper descent = faster regulatory action
c2_stress_inertia	AR(1) of detrended Stress composite at 1 Hz	−	High inertia = regulation failure
c2_neg_emotion_inertia	AR(1) of mean z-scored negative composites	−	Negative inertia most consistently predicts maladjust.
c2_stress_instability_MSSD	MSSD of Stress composite at 1 Hz	−	Extreme instability = dysregulation
c2_emotion_variability_SD	Within-person SD of Stress composite	−	Excessive variability = lower well-being
c2_repertoire_diversity	Entropy of 7-composite activation per stage, averaged	+	Regulatory repertoire breadth = flexibility
c2_cross_stage_reg._var.	SD of mean stress across 6 stages	+	Context-sensitive stress modulation
c2_reg._strategy_switching	Dominant-emotion transitions per second	+	Flexible strategy shifting = regulatory breadth
c2_block_reg._efficiency	Stress slope within Block episodes (negated)	+	Within-impasse self-regulation
c2_production_stress_floor	Median stress during Production stages	−	Effective regulators maintain low working-state stress

Table A.4 continued

Feature	Computation	Dir.	EI Rationale
c2_stress_peak_attenuation	Slope of peak intensity vs. peak ordinal (negated)	+	Successive peaks diminish = habituation
c2_stress_habituation_ratio	Mean stress second half / first half	−	Values < 1 = adaptation over session
c2_lip_press_index	Mean AU23 during high-stress minus low-stress episodes	+	Deliberate suppression capacity
c2_neg._expr._dampening	Ratio of later to earlier negative-AU intensities	+	Developing response modulation
c2_anticipatory_regulation	Stress 30 s before challenging stage / baseline	+	Antecedent-focused regulation predicts better outcomes
c2_early_vs_late_ratio	Stress last third / first third	−	Values < 1 = regulatory learning during session
c2_reappraisal_proxy	AU04 + AU12 co-activation during stress episodes	+	Effort + positive reframing = cognitive reappraisal
c2_SAM_facial_discr._red.	Slope of |SAM − facial valence| across checkpoints	+	Decreasing discrepancy = improving meta-regulation
c2_valence_arousal_coupling	Correlation between SAM valence increases and stress decreases	+	Coordinated subjective-behavioural regulation
c2_neg_to_pos_trans._speed	Mean duration of negative-to-positive dominant transitions	−	Faster transitions = more effective reappraisal

A.3.3 D3 Adaptive Persistence Features

D3 operationalises Mayer-Salovey Branch 2 and Duckworth et al.'s (2007) grit framework. A student high in adaptive persistence spends the majority of the session in productive work, recovers quickly from impasses, and shows that longer struggles yield proportionally longer productive periods afterward (the desirable difficulty effect).

Table A.5: D3 (Adaptive Persistence) feature specifications.

Feature	**Computation**	**Dir.**	**EI Rationale**
`c3_total_productive_time_ratio`	(Production + Evaluation + Refinement) / total time	+	Strongest single predictor; direct perseverance proxy
`c3_longest_productive_streak`	Max consecutive seconds in any productive stage	+	Sustained engagement duration
`c3_productive_streak_var.`	CV (SD/mean) of productive streak durations	−	Low CV = consistent; high CV = erratic engagement
`c3_late_task_engagement`	Productive time in final third / first third	+	Sustained effort despite fatigue
`c3_bounce_back_index`	Post-Block Happiness (60s) / pre-Block Happiness (60s)	+	Emotional re-engagement after impasse
`c3_block_neg._affect_damp.`	Negated stress slope within Block, averaged	+	Diminishing frustration within impasses
`c3_block_anger_suppression`	Anger during Block / Anger during non-Block	−	Anger not disproportionately escalating
`c3_stress_inertia_blocks`	AR(1) of stress within Block episodes only	−	Emotionally stuck within impasses
`c3_block_exit_diversity`	Entropy of post-Block exit stage distribution	+	Diverse exit strategies = adaptive flexibility
`c3_block_to_production_rate`	P(Block → Production | Block ends)	+	Direct re-engagement without detour
`c3_recovery_acceleration`	Negated slope of successive Block durations	+	Blocks getting shorter over session
`c3_productive_confusion_idx`	Stress ratio: resolved vs. unresolved Blocks	+	Higher arousal Blocks more likely to resolve
`c3_confusion_resolution_rate`	Proportion of short high-stress Blocks transitioning productively	+	Rapid cognitive disequilibrium resolution
`c3_productive_absorbing_prob.`	Stationary distribution probability of productive states	+	Structural tendency toward sustained engagement
`c3_second_order_block_escape`	P(non-Block$_t$ | Block$_{t-1}$, Block$_{t-2}$)	+	Escaping deeper stuck states
`c3_forward_stage_momentum`	Mean first-difference across stage ordinal values	+	Forward progress through task pipeline
`c3_productive_iteration_idx`	Distinct Prod.→Eval.→Refine cycles / total transitions	+	Adaptive vs. repetitive persistence
`c3_regressive_transition_ratio`	Proportion of "backward" stage transitions	−	Inability to consolidate gains

Table A.5 continued

Feature	Computation	Dir.	EI Rationale
c3_block_emotional_traj.	Slope of (Happiness − Stress) within Block episodes	+	Affective improvement during impasses
c3_post_block_engagement_qual.	Mean Happiness in 120s post-Block / session mean	+	Re-engagement quality matches or exceeds baseline
c3_struggle_benefit_index	Correlation of Block duration with subsequent streak duration	+	Longer struggles yield longer productive periods
c3_adversity_composure_idx	1− (Var_stress_block / Var_stress_overall)	+	Composure: Block stress not more variable than usual
c3_task_completion_persistence	Productive time in final 3 minutes / session mean	+	Persistence to end despite fatigue
c3_effective_time_efficiency	(Productive time) / (Total − Block time)	+	Efficiency of non-blocked time

A.3.4 D4 Affective Self-Awareness Features

D4 operationalises the concordance between self-reported (SAM) and objectively observed (facial AU) emotional states. High concordance indicates that the student's internal monitoring is accurate enough to support the regulatory strategies required by Branch 4.

Table A.6: D4 (Affective Self-Awareness) feature specifications.

Feature	Computation	Dir.	EI Rationale
c4_valence_happiness_concordance	Pearson r between SAM valence and facial Happiness	+	Direct operationalisation of interoceptive accuracy
c4_facial_valence_accuracy	r between z-SAM valence and z-facial valence index	+	Most comprehensive single concordance measure
c4_change_detection_alignment	r between ΔSAM and ΔFV across 9 checkpoint differences	+	Detects emotion *changes*, not just static level
c4_block_stage_awareness	r between SAM drops at Block checkpoints and negative-AU increase	+	Aware of Block-related distress in real time
c4_positive_reporting_bias	Mean (z-SAM − z-facial valence)	−	Systematic over-reporting = repressive coping
c4_absolute_calibration_error	Negated mean |z-SAM − z-facial|	+	Less error = more accurate self-monitoring

Table A.6 continued

Feature	Computation	Dir.	EI Rationale
`c4_arousal_optimality`	Negated $\lvert\text{mean_arousal} - 3\rvert$	+	Moderate arousal optimal per Yerkes-Dodson
`c4_arousal_midrange_prop.`	Proportion of SAM ratings in $[2, 4]$	+	Consistent moderate arousal = sustained self-regulation
`c4_arousal_stability`	Negated SD of SAM arousal ratings	+	Stable arousal = maintained self-control
`c4_arousal_trend`	Slope of SAM arousal across 10 checkpoints	+	Rising trend = growing productive engagement
`c4_pride_production`	Mean Pride composite during Production stages	+	Appropriate achievement emotion during output
`c4_temporal_coherence`	Max cross-correlation SAM–FV at lags $0, \pm 1$	+	Gold-standard method for emotion coherence
`c4_valence_arousal_discrim.`	$1 - \lvert r(\text{SAM valence, SAM arousal})\rvert$	+	Distinguishing pleasantness from activation
`c4_SAM_AU_coherence`	Mean r(SAM valence, Happiness−Sadness) + r(arousal, Stress+Fear)	+	Coherent self-awareness enables accurate deployment
`c4_arousal_task_match`	Spearman ρ between SAM arousal and stage complexity scores	+	Arousal tracking cognitive demand
`c4_arousal_curvature`	Second derivative of SAM trajectory	±	Non-linear arousal response to task demands

A.3.5 D5 Emotion-Cognition Alignment Features

D5 operationalises Mayer-Salovey Branch 2 through the principle that high-EI individuals deploy stage-appropriate emotions: mild positive affect facilitates divergent Ideation; analytical focus (AU04 + AU07) drives committed Production; flow coherence sustains extended productive work.

Table A.7: D5 (Emotion-Cognition Alignment) feature specifications.

Feature	Computation	Dir.	EI Rationale
`c5_ideation_positive_affect`	Mean Happiness z-score during Ideation	+	Positive affect broadens divergent thinking

Table A.7 continued

Feature	Computation	Dir.	EI Rationale
`c5_production_concentration`	Mean (AU04+AU07) z-score during Production	+	Brow + lid focus = cognitive effort signal
`c5_evaluation_critical_eng.`	Mean (AU04+AU14) during Evaluation	+	AU14 dimpler predicted learning gains
`c5_stage_emotion_profile_fit`	Mean cosine similarity between observed and ideal AU profiles	+	Direct operationalisation of Branch 2 facilitation
`c5_emotion_stage_diff.`	η^2 of stage membership on each composite, averaged	+	Emotions co-vary with cognitive demands
`c5_cognitive_effort_deploy.`	(AU04 in Prod.+Eval.) − (AU04 in Prep.+Refine.)	+	More cognitive effort in high-demand stages
`c5_flow_proxy_production`	z[AU07]+z[Happiness]−z[Stress]−z[SD_AU] during Production	+	Attention + enjoyment + low variability
`c5_flow_stability`	Negated CV of flow proxy in 5-s windows	+	Sustained flow = stable emotion-cognition alignment
`c5_mastery_mode_ratio`	Mastery / (mastery + coping) windows	+	Boekaerts (2011) mastery pathway = task-focused
`c5_mastery_during_challenge`	Mastery ratio restricted to Prod.+Eval. stages	+	Mastery under highest cognitive demand
`c5_confusion_resolution_rate`	Proportion of confusion episodes resolving to focus	+	Productive confusion pathway
`c5_block_constructive_conf.`	(AU04+AU01) − Anger during Block	+	Confusion/interest over frustration in Block
`c5_transition_emot._prep.`	Negated Euclidean distance between stage n and $n+1$ profiles	+	Smooth emotional transitions between stages
`c5_pride_after_production`	Pride in first 30s of Evaluation − last 30s of Production	+	Achievement emotion at production-evaluation boundary
`c5_stress_appropriate_mod.`	Negated deviation from optimal stress per stage	+	Yerkes-Dodson arousal-complexity matching
`c5_arousal_SAM_task_match`	Spearman ρ between SAM arousal and stage complexity	+	Arousal following cognitive demand contour

Table A.7 continued

Feature	Computation	Dir.	EI Rationale
c5_positive_creative_vs_focused	Happiness (Ideation − Production) + (AU04+AU07) (Production − Ideation)	+	Crossover: creative positivity + production focus
c5_neg._affect_block_spec.	Negative composite during Block − non-Block	+	Negative affect concentrated where cognitively useful
c5_emotion_complexity_index	Mean composites per 10-s window exceeding threshold	+	Emotion schemas engage dynamically with cognition
c5_SAM_AU_coherence	Checkpoint correlations SAM-valence/Happiness and SAM-arousal/Stress	+	Coherent self-awareness enables deployment
c5_engagement_recovery_speed	Negated time until engagement indicators return after neg.-affect ep.	+	Rapid re-routing coping → mastery
c5_interest_during_ideation	Mean (AU01+AU02+AU07) during Ideation − during Production	+	Interest/curiosity is the exploratory emotion



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